

Efficient Testing and Design Methods for Underground Fiber Reinforced Concrete Structures

Barzin Mobasher, PhD, PE, FACI, FRILEM
Professor

School of Sustainable Engineering and the Built Environment
Arizona State University, Tempe, AZ 85287-5306

barzin@asu.edu

ORCID: <https://orcid.org/0000-0002-7580-2855>

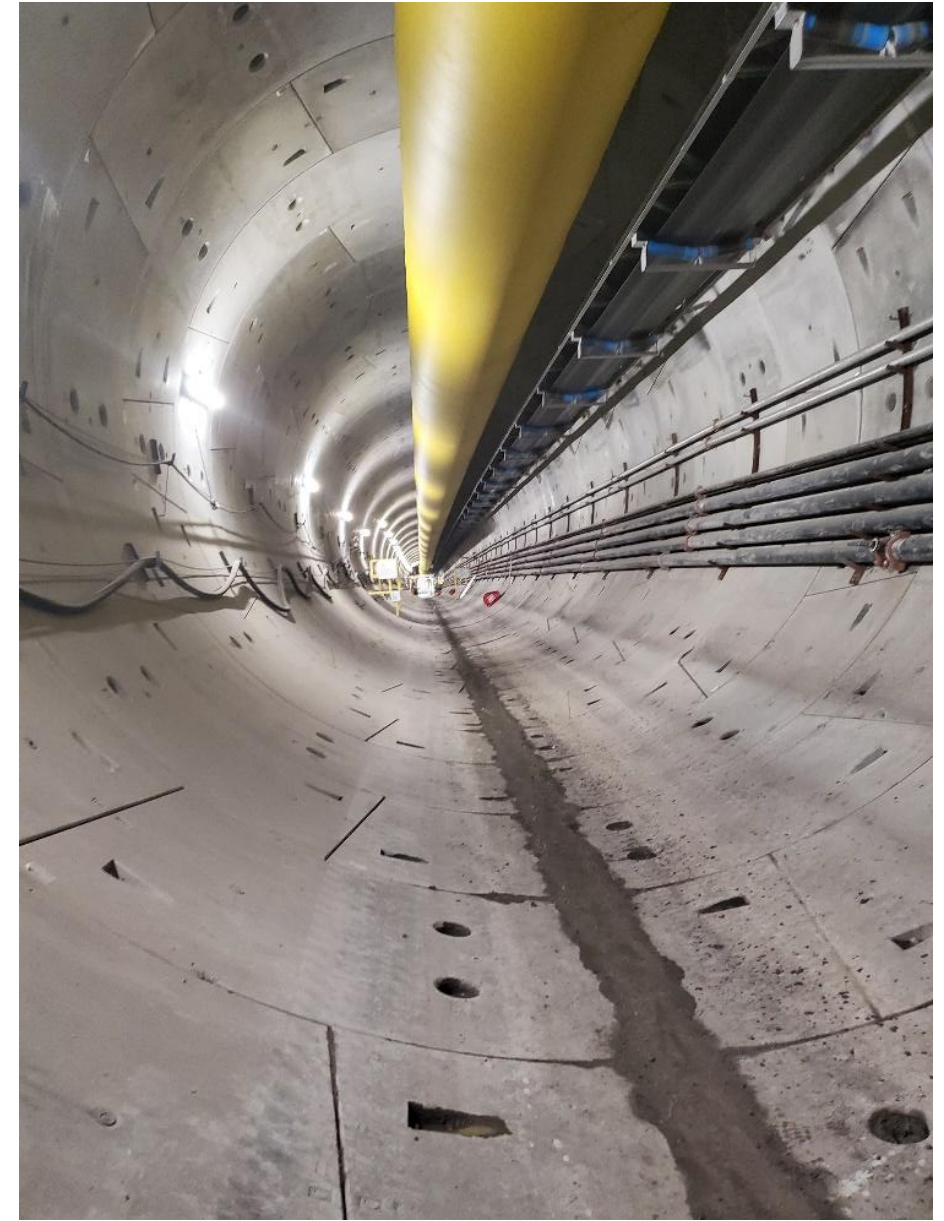
RETC Short Course

Advances in Concrete Design and Construction for Underground Structures
Dallas, Texas, June 8th , 2025

Precast Tunnel Sections

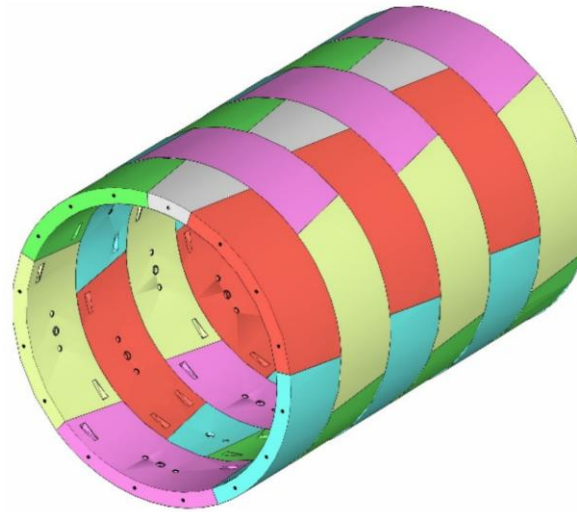
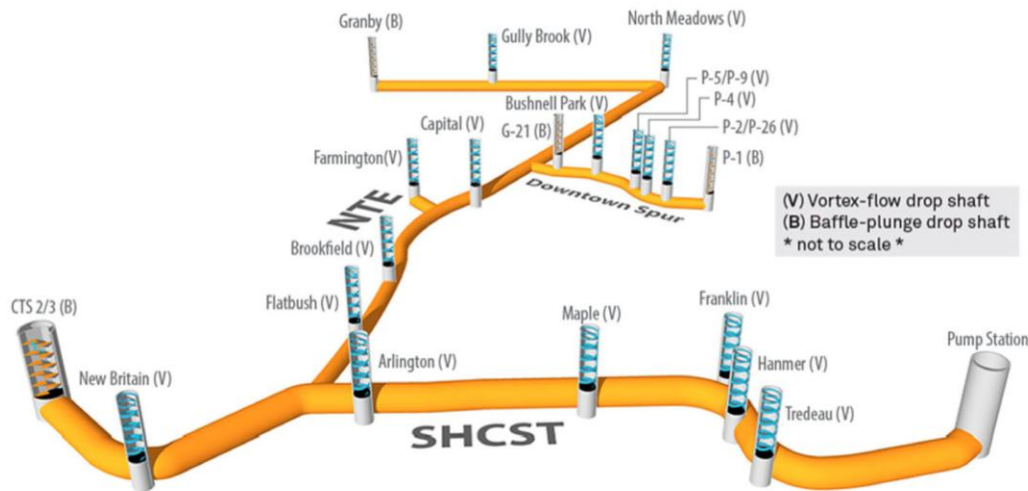
VERYA NASRI, DAVID R. KLUG, BRIAN FULCHER
AND JAMES A. MORRISON

HANDBOOK OF PRECAST SEGMENTAL TUNNEL LINING SYSTEMS



Sustainability of Concrete Structures

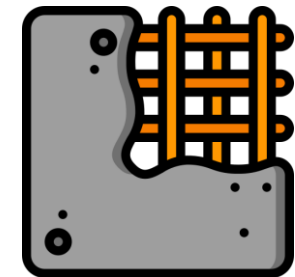
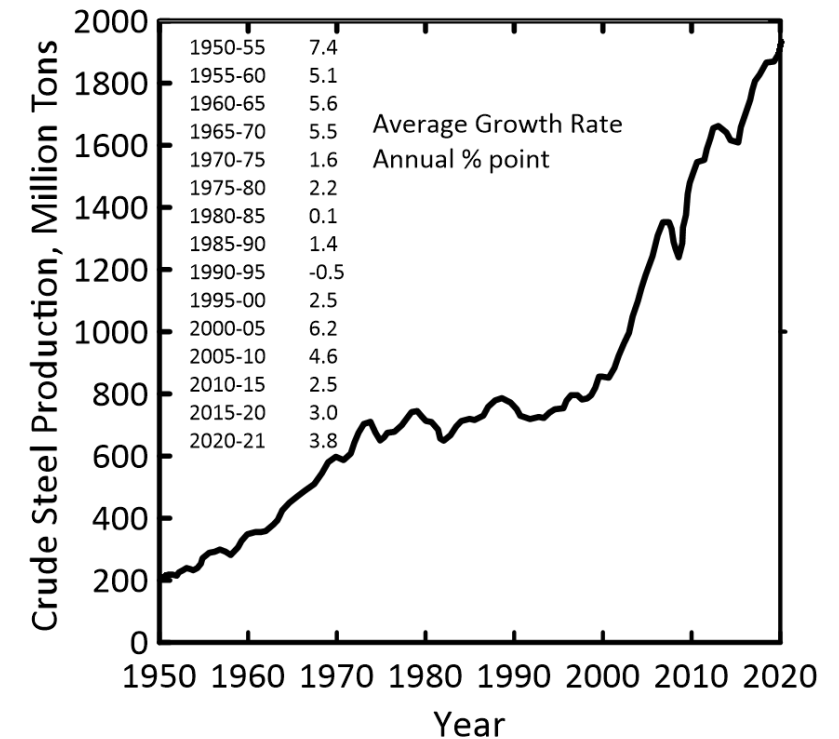
- Compared to all countries, concrete industry is the third-largest CO₂ emitter (4-8% of the world's emission)
- Fourth in ranking as a source of greenhouse emission after coal, oil and gas.
- Annual use of about 4 billion tons of cement creates 3 B tons of CO₂
- Uses 10% of the world's industrial water, affecting drinking and irrigation.
- 75% of water consumption is in drought and water-stressed regions; adding to the heat-island effect
- Carbon capture by storage or mineralization has yet to show any promise, (\$27B spent since 2017)
- **Very little has been done about efficient structural design of concrete**



South Hartford Conveyance Storage Tunnel (SHCST), flagship infrastructure project in Connecticut managing combined sewer overflows, 6.6 km, 6 m in diameter

World Production of Steel in the past 70 years

- Crude steel production increased from 800 to 2000 MT/Y since 1990
- 50% of production, 1000 MT/Y is by China, 10x of India, 2nd producer.
- For concrete consumption of 10Bt, rebar will be 522 MT
- 7.5 lbs of steel rebar per ft³ of concrete
- 100% recyclable, raw feed supplies of ore will exhaust due to demand.



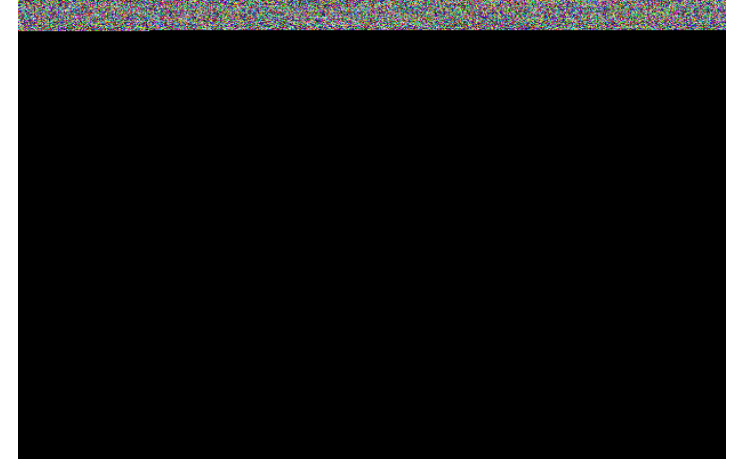
Fully Automated Procedures for robotic handling and installation of the Tunnel sections



Precast Tunnel Lining using Fiber Reinforcement

Serviceability Based Analysis, Design, & Testing of Hybrid Structural Sections

- Reduce steel reinforcement, section thickness
- Use Polymeric fibers for precast tunnel lining
- Testing, Modeling to promote innovative and sustainable alternatives.



Fundamentals of Concrete Technology

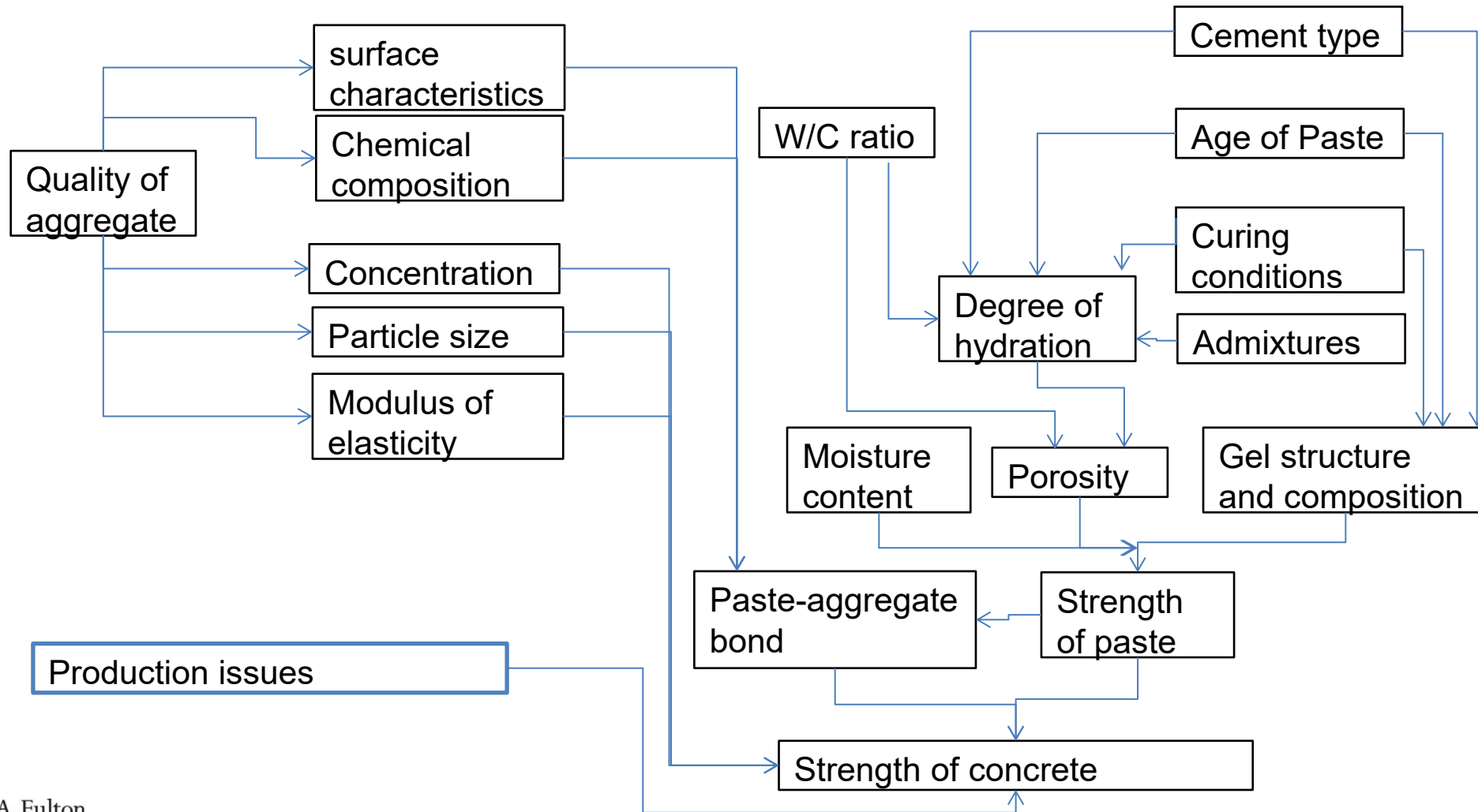
Fresh Concrete

- Constituents: Portland cement, aggregates, water, admixtures
- Mix Design
- Rheology & flow, consolidation, packing factors, slump, bleeding
- Production speed, placing, finishing.
- Curing, early age plastic shrinkage cracking
- Temperature effects

Hardened Concrete

- Factors affecting strength
- Choice of reinforcement
- Physical and chemical properties
- Mix design: Strength, permeability
- Durability, freeze-thaw, creep, shrinkage, permeability, corrosion, alkali-silica reaction, sulfate attack, carbonation, chloride induced corrosion of rebar.
- Specifications and quality control

Factors Affecting Strength of Concrete

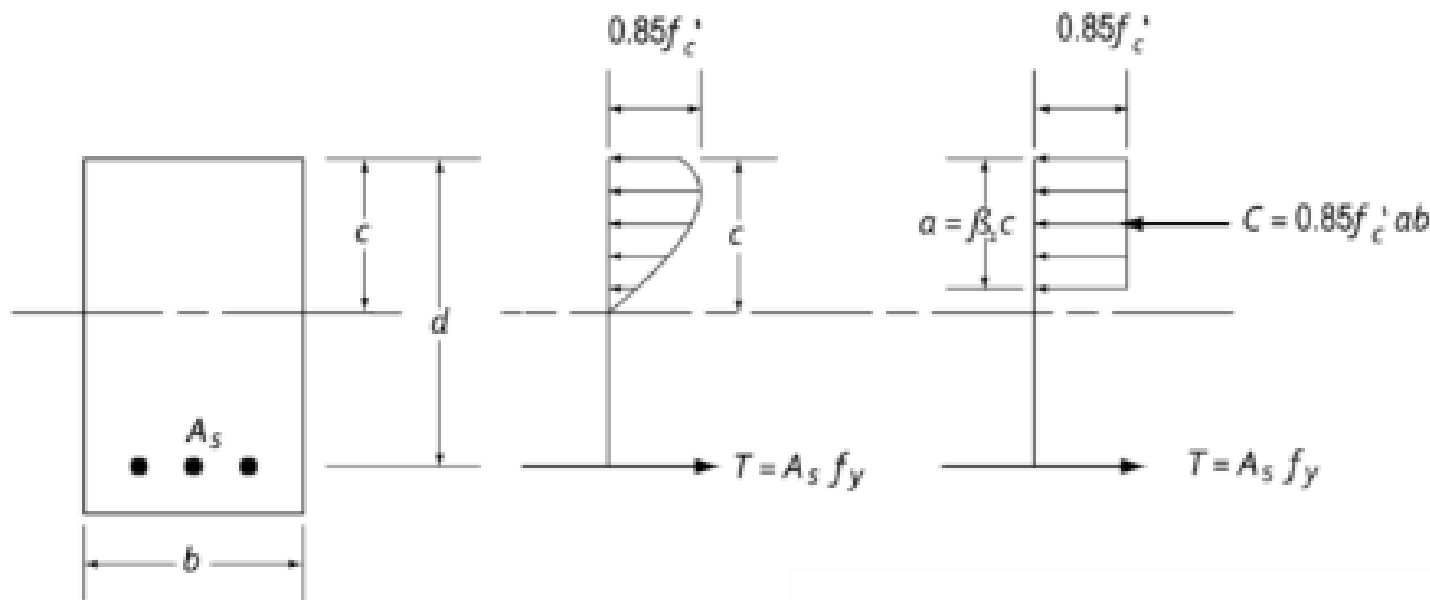


Rhetorical Discussions on Testing, Material Properties, and QC

- Why do the testing?
 - materials and systems, from mix design, to sample strength testing, to full-scale testing
- What do we measure and control the Material properties used in the design process?
 - Compressive and Flexural Strength have very little to do with controlling the outcome of design
- What tools can we use to get fundamental material parameters for structural design?
 - Properties that sever multiple purposes, for analysis, design, and QC, stress-strain properties
- How to improve QC/QA Processes using Statistical Process control?
 - Move from 1-sigma towards 6-sigma
- How to address the variability when the results look all over the place?
 - Avoid the borrowed empirical tools and relationships

Concepts for the structural design are more than 120 years old

- Structural Design of reinforced concrete, unchanged in 120+ years
- Guidelines developed in 1907, modified in 1930s, and codified in 1940s.
- Approximately 60-70% of concrete's volume that is subjected to tensile stress is still ignored and thus a waste of resource, providing no contribution other than providing a cover for the steel rebar.



$$M_u = \phi A_s f_y d \left(1 - 0.59 \frac{\rho f_y}{f'_c} \right)$$

Rectangular Stress Block
Charles S. Whitney, 1930s

Formulas.

Position of neutral axis,

$$k = \sqrt{2n \left(p + p' \frac{d'}{d} \right) + [n(p + p')]^2} - n(p + p') \quad (26)$$

Position of resultant of compressive stress, $C + C'$,

$$x = \frac{k + \frac{d'C'}{dC}}{1 + \frac{C'}{C}}; \text{ in which } \frac{C'}{C} = \frac{2p'n \left(k + \frac{d'}{d} \right)}{k^2} \quad (27)$$

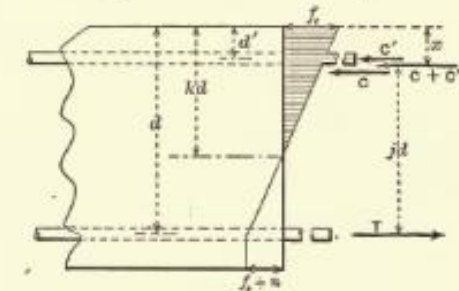


FIG. 67.

Arm of resisting couple,

$$j = \left(1 - \frac{x}{d} \right) \quad (28)$$

Moment of resistance,

$$M_s = f_s p j \cdot b d^2 \quad (29)$$

$$M_o = \frac{1}{2} f_c k (1 - \frac{1}{2} k) b d^2 + f'_s p' b d (d - d') \quad (30)$$

Fibre stresses,

$$f_s = \frac{M \div j d}{A} \quad (31)$$

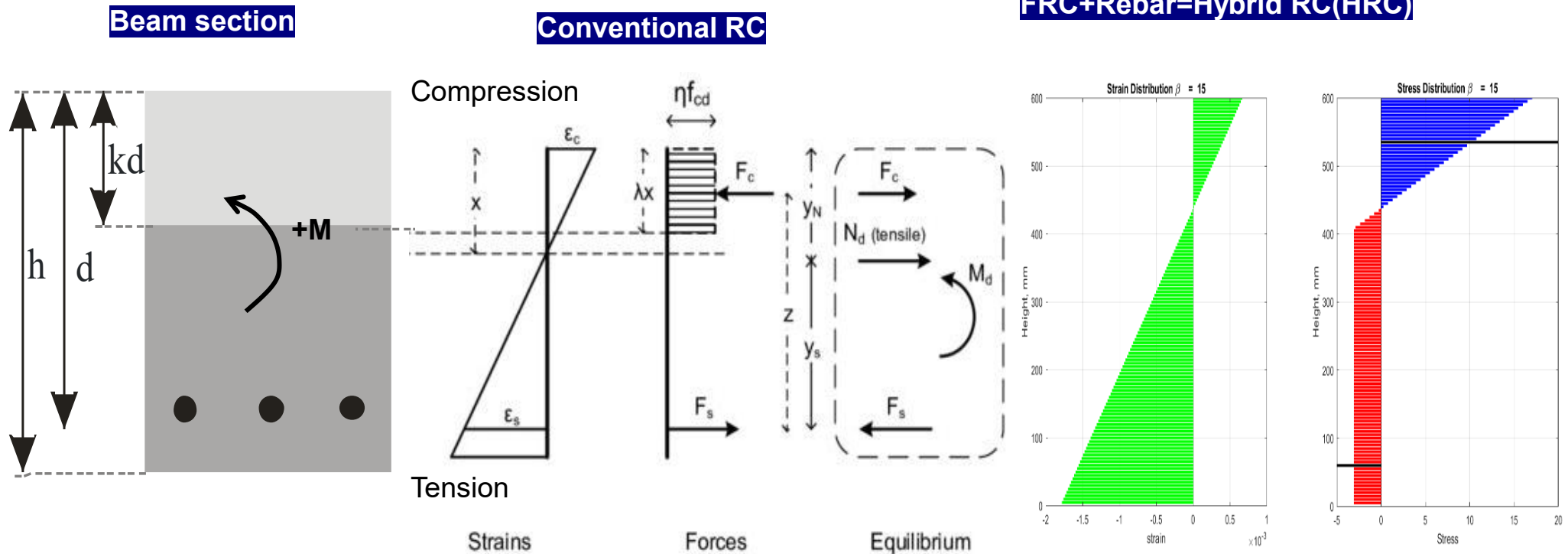
$$f_c = \frac{k}{n(1-k)} \cdot f_s \quad (32)$$

$$f'_s = \frac{n \left(k - \frac{d'}{d} \right)}{k} \cdot f_c = \frac{k d - d'}{d - k d} \cdot f_s \quad (33)$$

Ultimate Loads formula by Tuneaure 1907

Tensile contribution of FRC to the Hybrid Flexural Response

- Tensile strength of concrete (at 10% of its compressive strength), is ignored in structural calculations
- Steel reinforcement is expected to carry all the tensile loads immediately after concrete cracks
- Can't fully replace the rebars, but can enhance their contribution significantly
- Efficiency of total concrete volume is in the single digits.

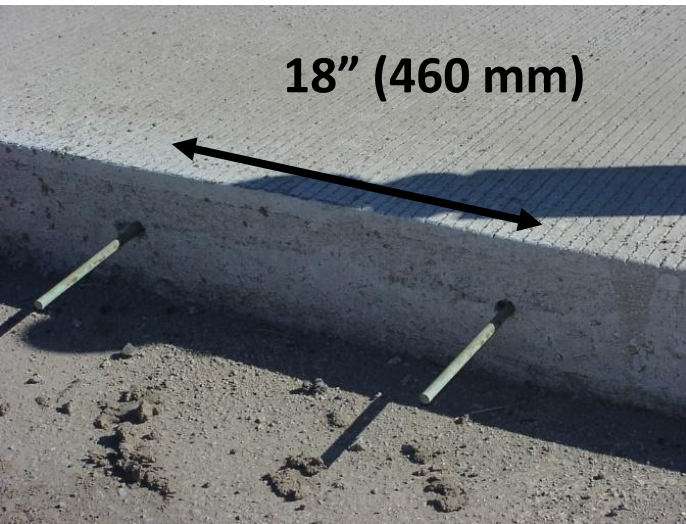


Soranakom, C., & Mobasher, B. (2008). Correlation of tensile and flexural responses of strain softening and strain hardening cement composites. *Cement and Concrete Composites*, 30(6), 465–477. <https://doi.org/10.1016/j.cemconcomp.2008.01.007>

ACI Committee 544.4R. (2018). ACI 544.4 R-18 -Guide to Design with Fiber-Reinforced Concrete.

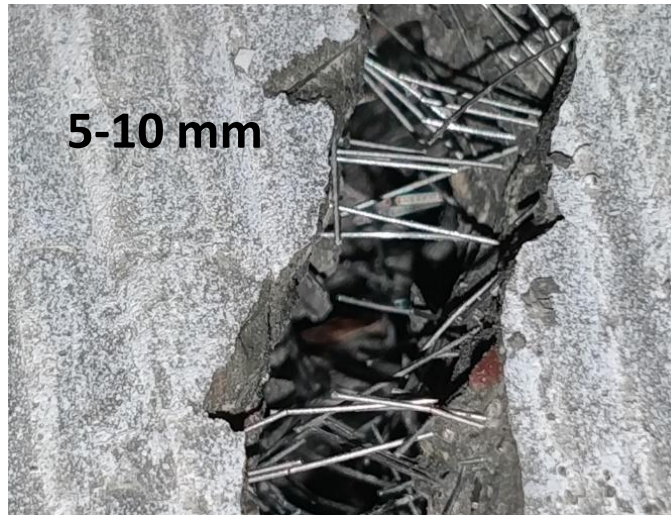
Role of Supplementary Reinforcement

Strengthening microstructure has a profound impact at the macro-structural strength & durability
Improved durability is a game changer in terms of Carbon footprint



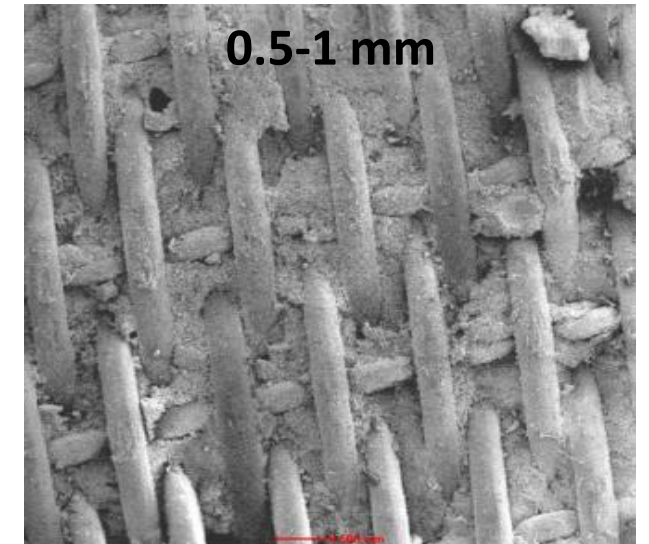
Ordinary reinforced concrete slab, 5-10" thick, standard rebars, 12"-18" apart

Reinforcing Bars: Macro-structure reinforcement



Fiber reinforced Concrete (FRC), at the meso-structure with $\frac{1}{2}$ " long fibers, $\frac{1}{2}$ " (10mm) millimeters apart

Fiber reinforced Concrete: Meso-structure reinforcement



Textile reinforced Concrete (TRC), at the micro-structure with long fibers, woven, only millimeters apart

Textile Reinforced Concrete: Micro-structure reinforcement

Delayed Activation of Reinforcement

ARIZONA STATE UNIVERSITY

Inhibited Localization

What are the potential mechanisms for toughening Concrete?

○ Mechanisms:

- Crack path tortuosity, bumps on the road, slows the speed of cracks
- Crack deflection
- Crack bridging, fiber pullout, yielding
- Delamination cracks
- Multiple cracking, Tension stiffening

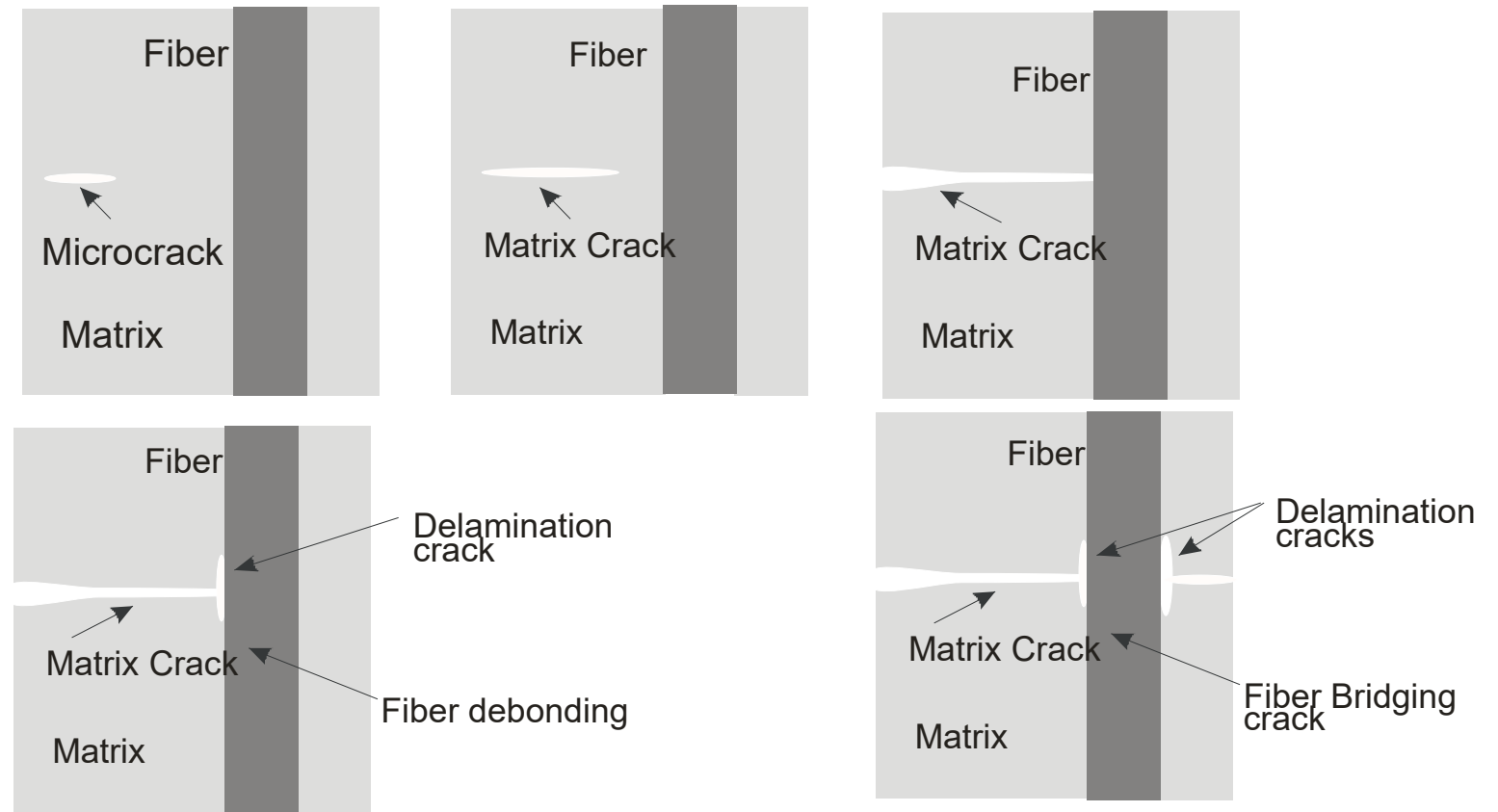
○ Results:

- Structural Redundancy in the Limit States
- Correlation of cracking with creation of new surfaces and energy dissipation
- Delay localization to increase the strength of concrete in tension
- Delays in crack propagation, reducing w/c , porosity, and improving interfaces
- Transverse Yarn anchorage by textiles

Sequence of cracking from initiation and growth, deflection, & Bridging perspective

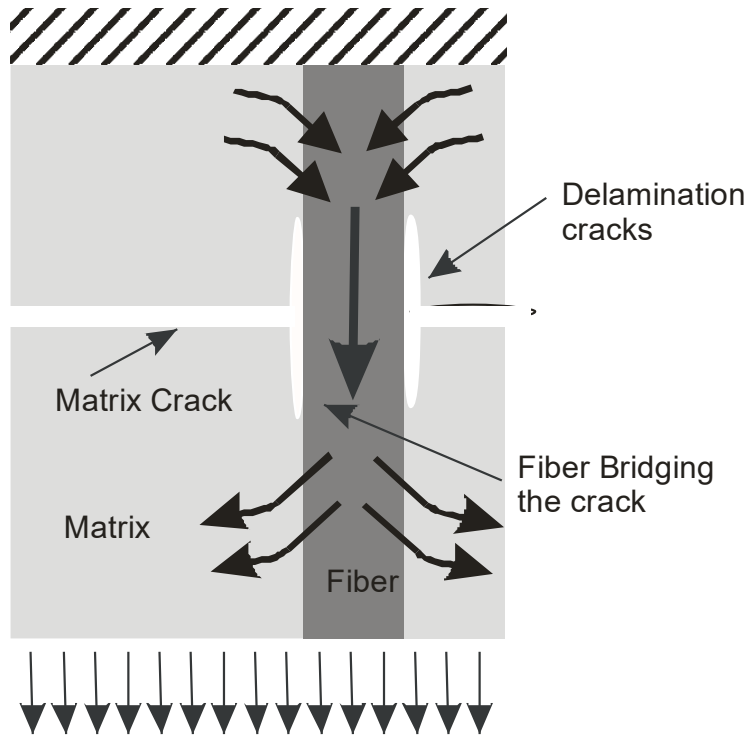
The cracking process starts from slow crack growth of a microcrack from a pore or aggregate interface, crack propagation, hindered by fibers, resulting in crack deflection, delamination, fiber bridging, pullout

1. Stable microcrack to macro crack growth
2. Crack coalescence and growth
3. Crack arrest by fibers
4. Fiber debonding and crack deflection
5. Interface debonding
6. Full fiber debonding growth
7. Fiber bridging

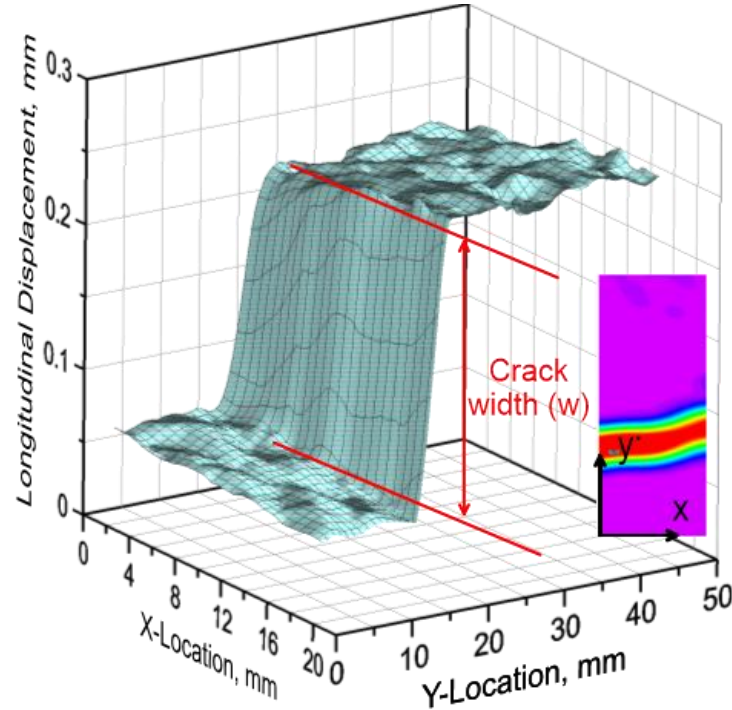


None of these mechanisms are present when the concrete is not reinforced at the microstructural level

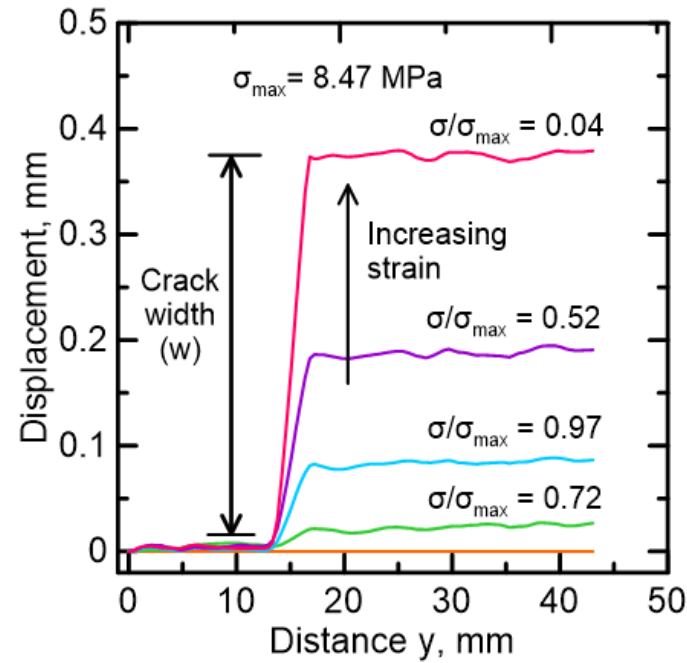
Tension Stiffening is the Fifth major mode of toughening



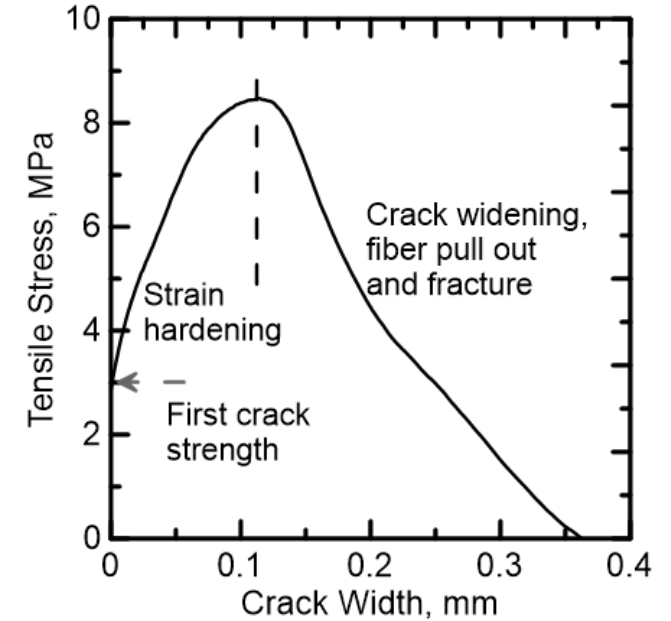
Crack Width Measurement for Strain Hardening Composite



Displacement Field



Displacement
Distribution Along
Specimen

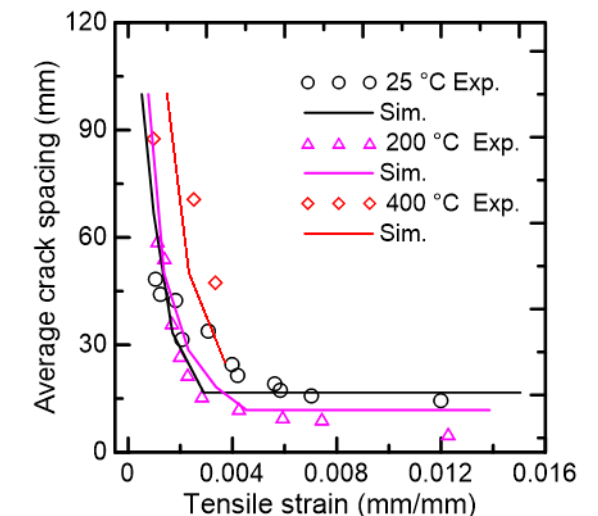
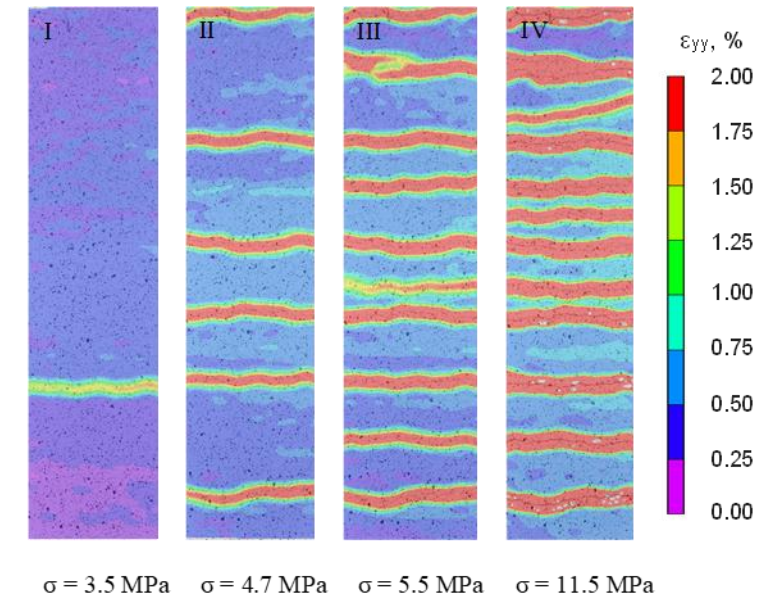
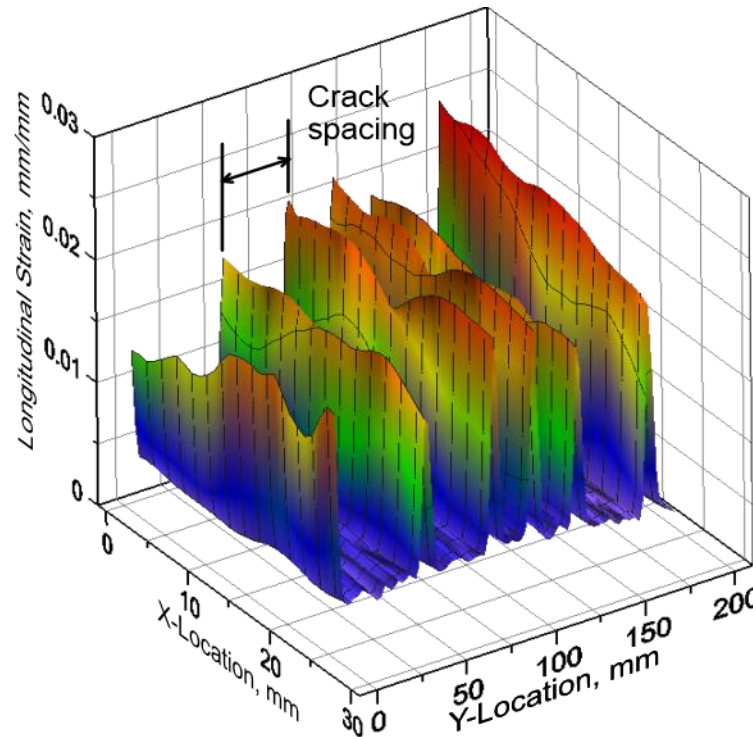


Stress-Crack Width
Relationship

- Non-contact measurement
- Quasi-static to high speed
- Single crack and multiple cracks

Ductility Induced enhanced behavior for structural composite applications

- Evolution of Crack Spacing in TRC
- Multiple cracking in tension
- Tension stiffening
- Development of parallel cracks
- Indication of toughening mechanisms
- Characteristic length in numerical modelling

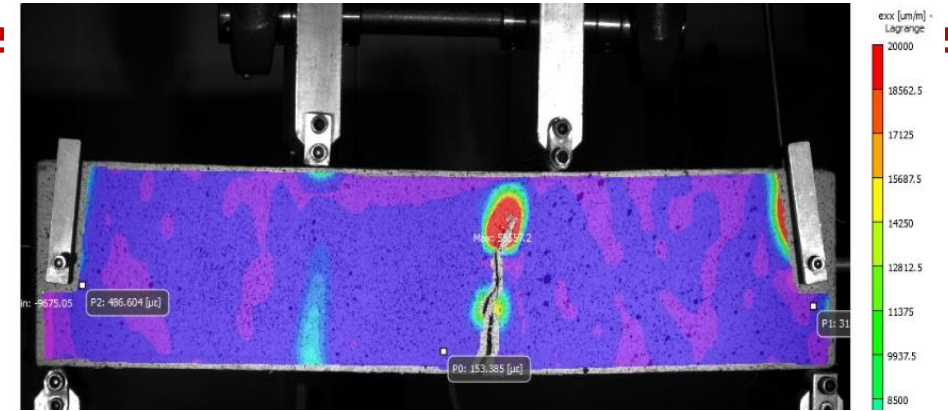
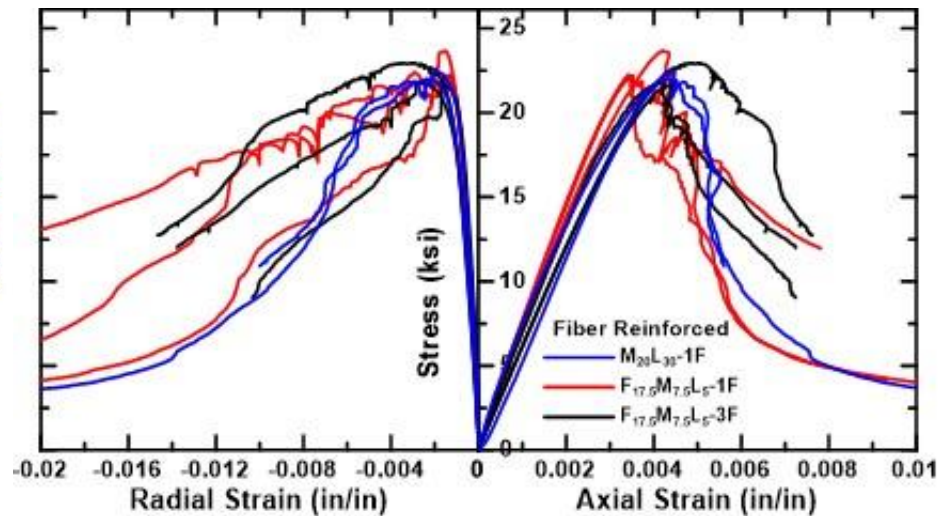
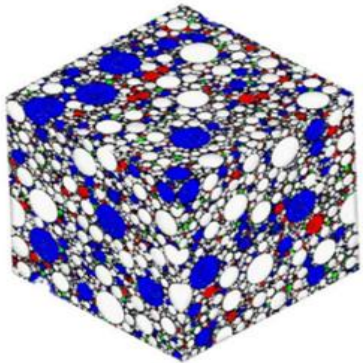


Rambo, D. A. S., Yao, Y., et al. (2017). Experimental investigation and modelling of the temperature effects on the tensile behavior of textile reinforced refractory concretes. Cem.

Concr. Compos. 75, 51-61.

UHPC Concrete Mixtures, Bridge Replacement Strategies MCDOT, Arizona, 2022

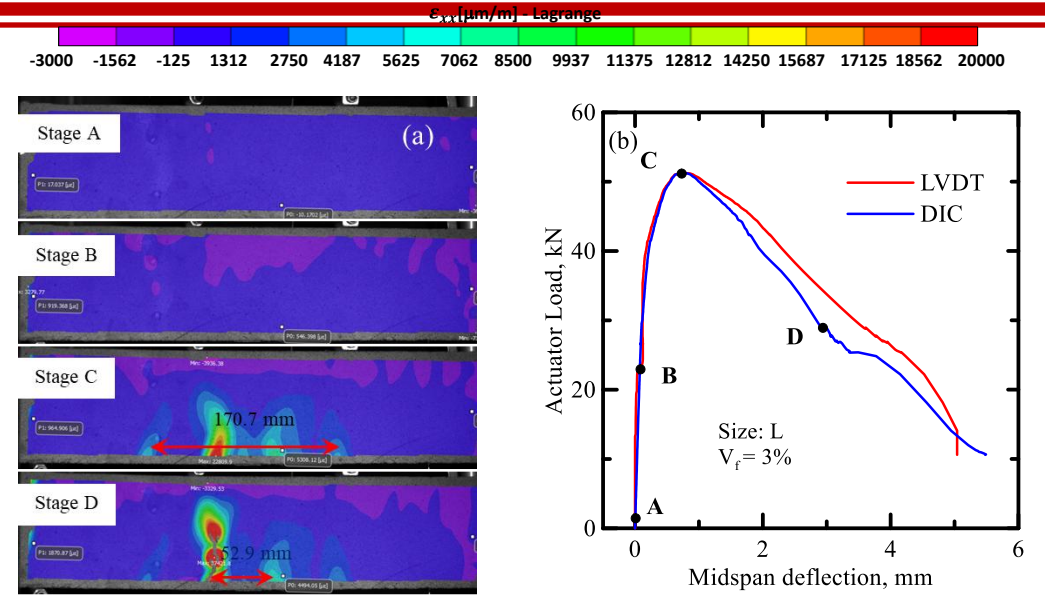
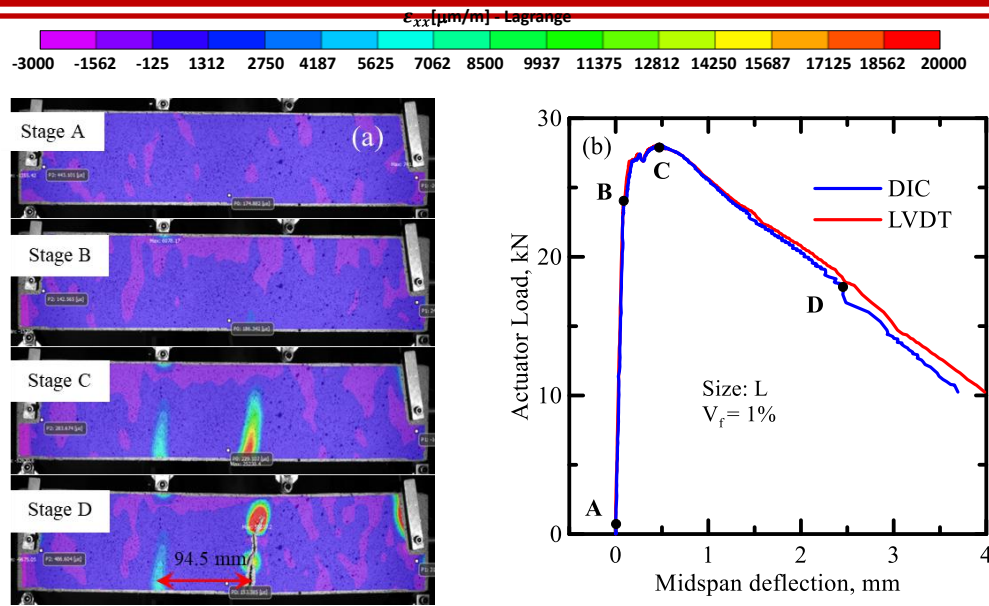
- How to replace an entire corroded Bridge deck in 10 days
- Life cycle cost and economic advantages of using UHPC systems in terms of structural applications.
- Precast UHPC elements reduce operational complications, labor-intensive tasks. From a design perspective, elastoplastic behavior improves durability, ductility, load redistribution, and weight reduction.
- Develop Sustainable Materials, using solid mechanics, material formulations, structural components, and systems.
- solutions for composite materials for transportation, water treatment facilities, pipes, tunnel lining, thin sections, Structural Shapes



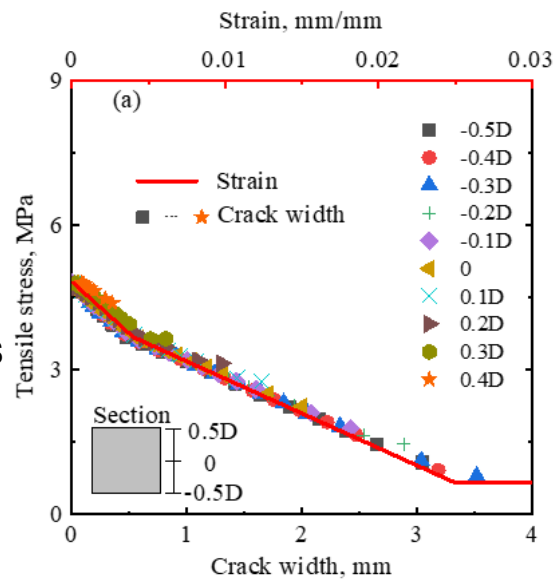
[University's open-source concrete formula offers speedy, cost-effective bridge repairs](https://www.asce.org/publications-and-news/civil-engineering-source/article/2025/05/21/concrete-collaboration-promising-for-speedy-cost-effective-bridge-repairs)

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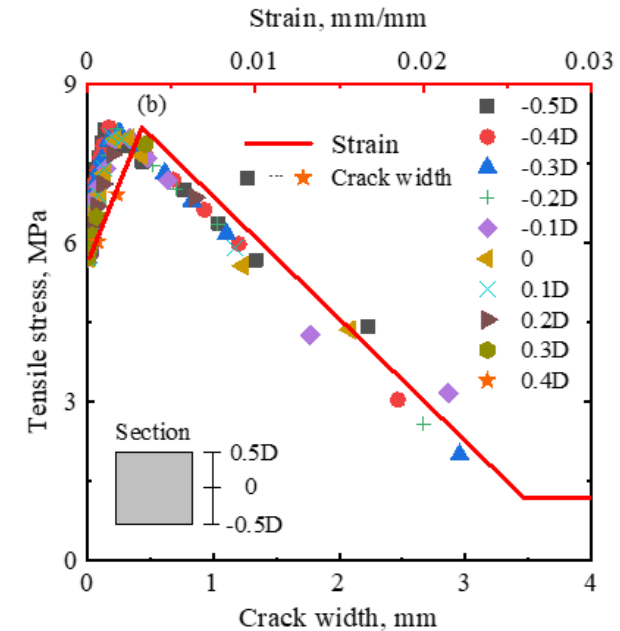
UHPC with 1% and 3% steel fibers, Stress Crack width Response



1% steel fibers

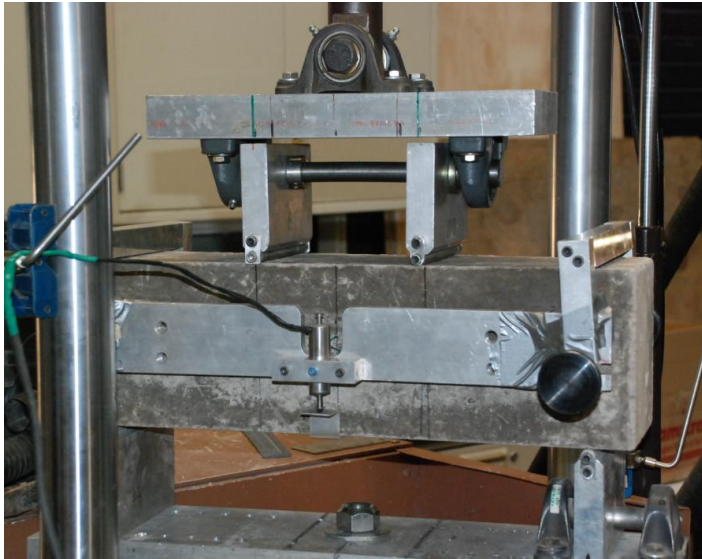


3% steel fibers



Specimen Level Testing Approaches

- FRC Material properties analyzed using standard closed-loop Testing at ASU
- Developing Closed-form solutions for Parameter Extraction using a parameterized approach
- A host of other characterization techniques



ASTM C1609



ASTM C1550



ASTM C39M

- Load-Deflection response
- Simulate L-D using parametric tension and compression models
- Backcalculation of other geometries using the bilinear/trilinear/quadrilinear approach

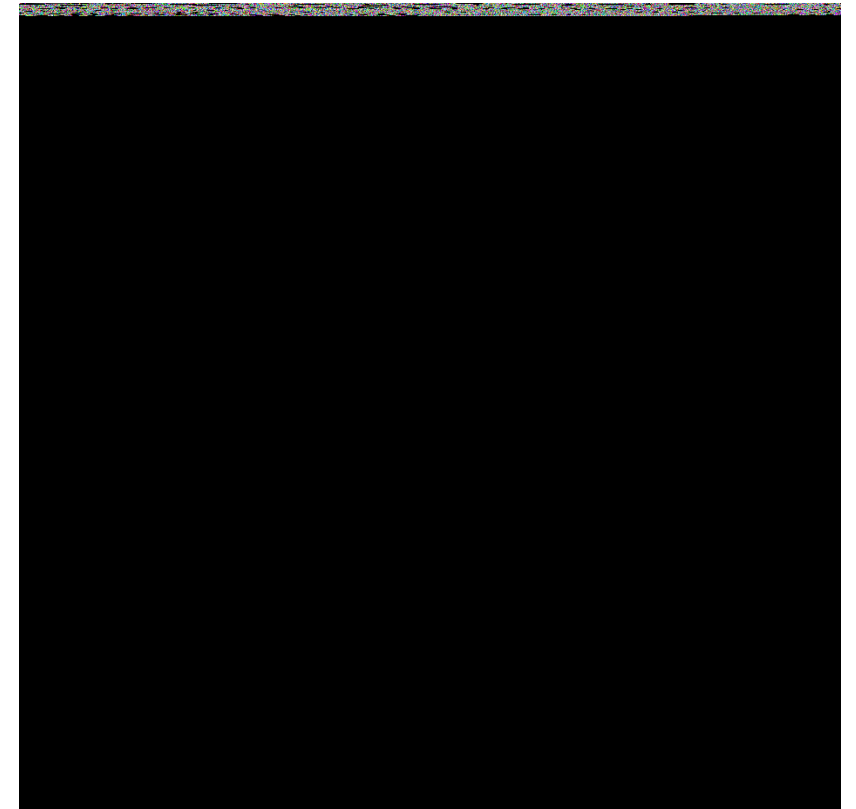
Resolution Copper – Magma Copper mine, Superior, Arizona

ASU evaluated the initial fiber-reinforced shotcrete support design for a 2000 m-deep shaft at Resolution Mine, Arizona.



Fiber-reinforced shotcrete for initial shaft sinking support

- Deep shaft (2189 m), 9 m dia, copper mine
- 400,000 tons copper per year for the next 40 years
- Three geological units
- A range of orthotropic stress conditions
- Several modes of instability: gravity driven, rockmass shear yielding, brittle failure
- The shotcrete system must achieve a high early strength and ductility within a short period (less than 24 hours).



Traditional Rock Bolt Mesh System is time consuming



Shotcrete Applications



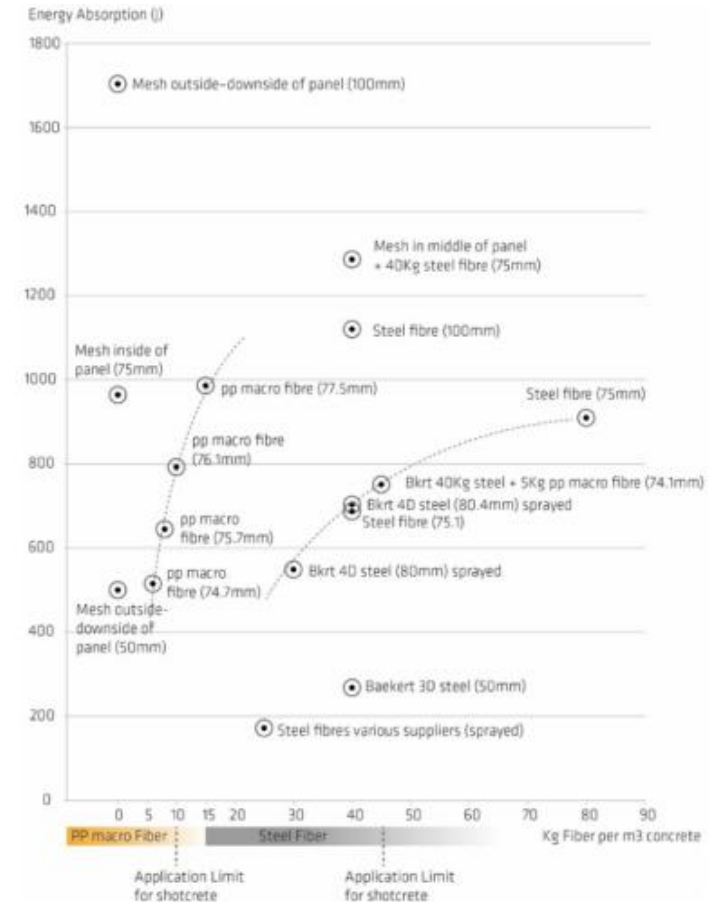
Analysis Methods

Selection of analysis methodology (Diederichs, 2007)

UCS/T or mi	GSI<55	GSI=55-65	GSI=65-80	GSI>80
<9	GSI	<i>GSI (TW)</i>	GSI	GSI
9-15	GSI	<i>GSI (KVS)</i>	GSI	<i>GSI/DISL</i>
15-20	GSI	<i>GSI/DISL</i>	<i>GSI/DISL (TAL)</i>	<i>DISL</i>
>20	GSI	<i>GSI/DISL</i>	<i>DISL</i>	<i>DISL</i>

Unit	Model of Instability
TAL:	Low stress – gravity driven instability Medium stress – local brittle failure
KVS:	GSI rockmass yielding and/or some local brittle failure
TW:	GSI rockmass yielding – raveling of conglomerate

Rock mass quality in geological strength index (GSI) chart



	Massive (RMR > 75)	Moderately Fractured (50 > RMR > 75)	Highly Fractured (RMR < 50)	
Low In-Situ Stress ($\sigma_1 / \sigma_c < 0.15$)	 Linear elastic response.	 Falling or sliding of blocks and wedges.	 Unravelling of blocks from the excavation surface.	Low Mining-Induced Stress $\sigma_{max}/\sigma_c < 0.4 \pm 0.1$
Intermediate In-Situ Stress ($0.15 > \sigma_1 / \sigma_c < 0.4$)	 Brittle failure adjacent to excavation boundary.	 Localized brittle failure of intact rock and movement of blocks.	 Localized brittle failure of intact rock and unravelling along discontinuities.	Intermediate Induced Stress $0.4 \pm 0.1 < \sigma_{max}/\sigma_c < 1.15 \pm 0.1$
High In-Situ Stress ($\sigma_1 / \sigma_c > 0.4$)	 Brittle failure around the excavation.	 Brittle failure of intact rock around the excavation and movement of blocks.	 Squeezing and swelling rocks: Elastic-plastic continuum.	High Mining-Induced Stress $\sigma_{max}/\sigma_c > 1.15 \pm 0.1$

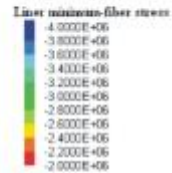
TAL: GSI=70, $\sigma_{max}/\sigma_c=0.6$

KVS: GSI=60, $\sigma_{max}/\sigma_c=1.5$

TW: GSI=60, $\sigma_{max}/\sigma_c=2.4$

3DEC Evaluation of Shotcrete

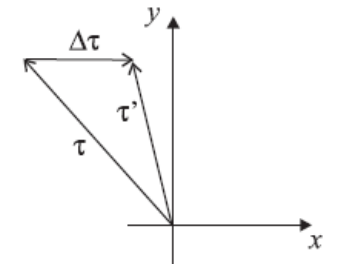
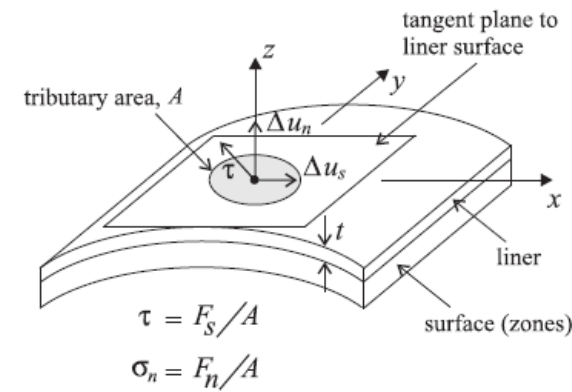
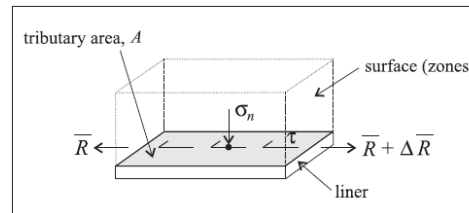
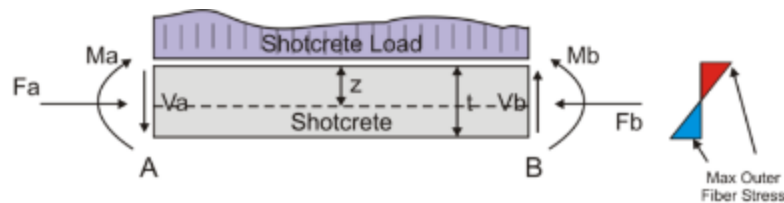
Tensile extreme fiber stress *exceeding* tensile strength of shotcrete



***** potential crack



3DEC-predicted liner stresses and potential cracking in TAL unit



$$\tau' = \tau + \Delta \tau = \tau + k_s \Delta u_s$$

$$\sigma_n' = \sigma_n + \Delta \sigma_n = \sigma_n + k_n \Delta u_n$$

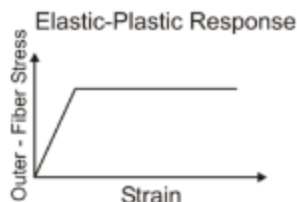
(a) shear stress, τ , and incremental relative shear and normal displacements, Δu_s and Δu_n , between liner and surface

(b) shear and normal stress update at each timestep

Combines Flexural Strength:

$$F/A \pm Mz/I \leq \sigma_{\text{ten, comp strength}}$$

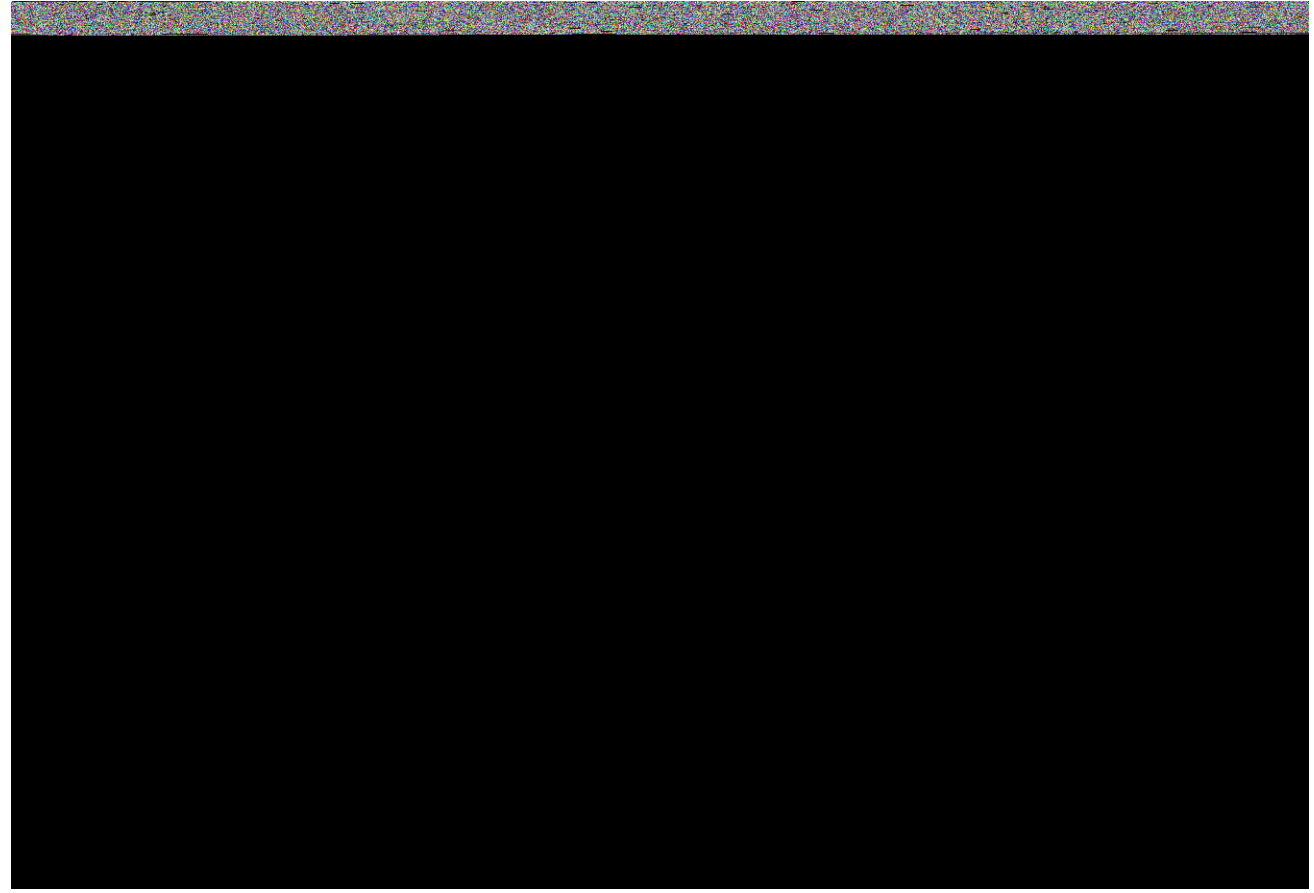
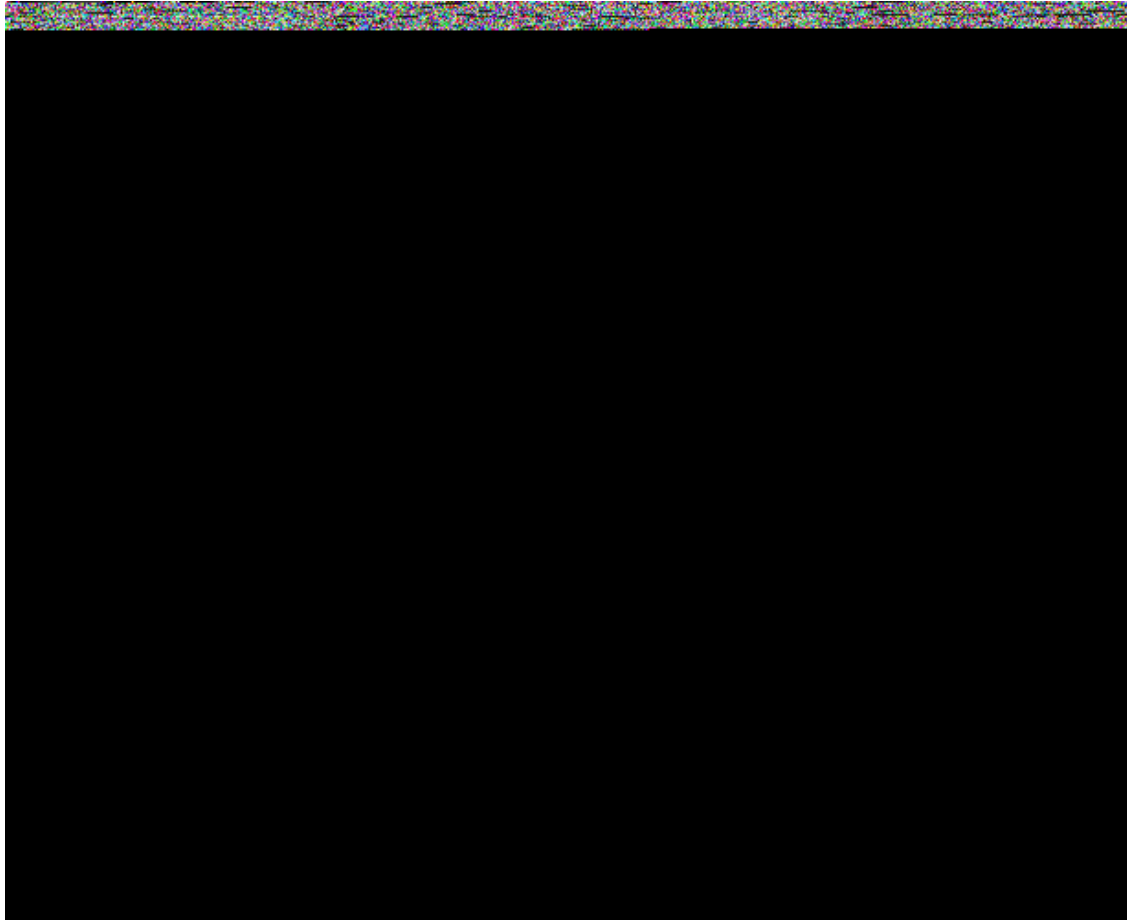
$$A = 1 \cdot t; I = 1 \cdot t^3 / 12$$



Extreme Fiber Stress

- Linear-elastic shotcrete liner installed in low stress, blocky ground.
- Early age properties use for shotcrete and rock-shotcrete interface along full length of model shaft.
- Extreme fiber stress used to evaluate potential shotcrete cracking.
- Tensile extreme fiber stress exceeding the tensile strength *combined* with tensile thrust at a given liner section indicates potential for a crack to form through the liner.
- Large regions of connected cracks would indicate potential for a block breaking through the shotcrete.

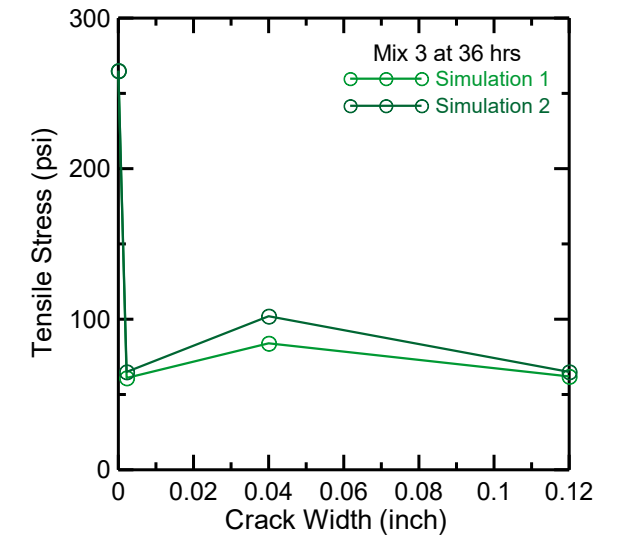
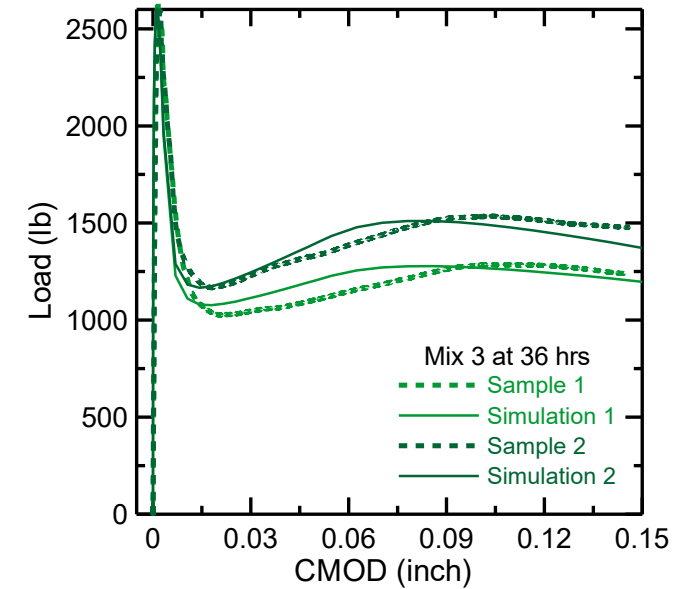
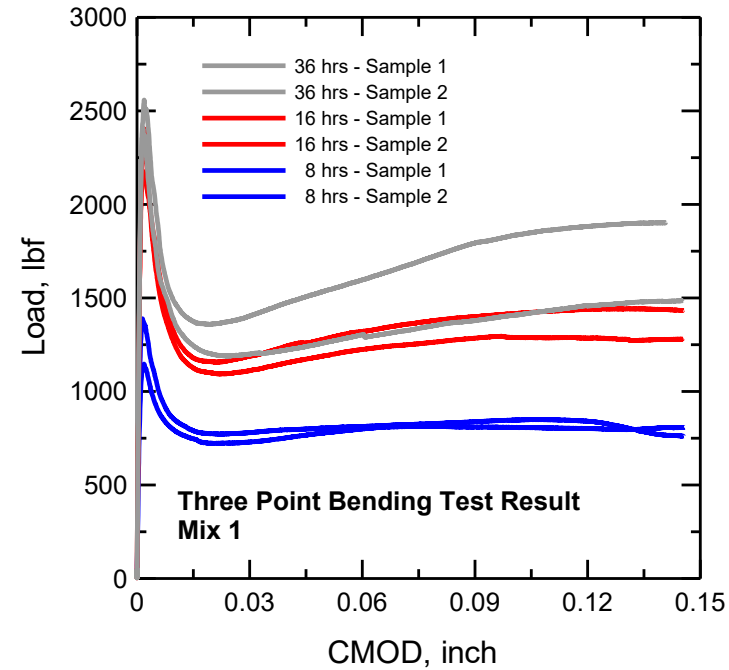
Modified Barton chart



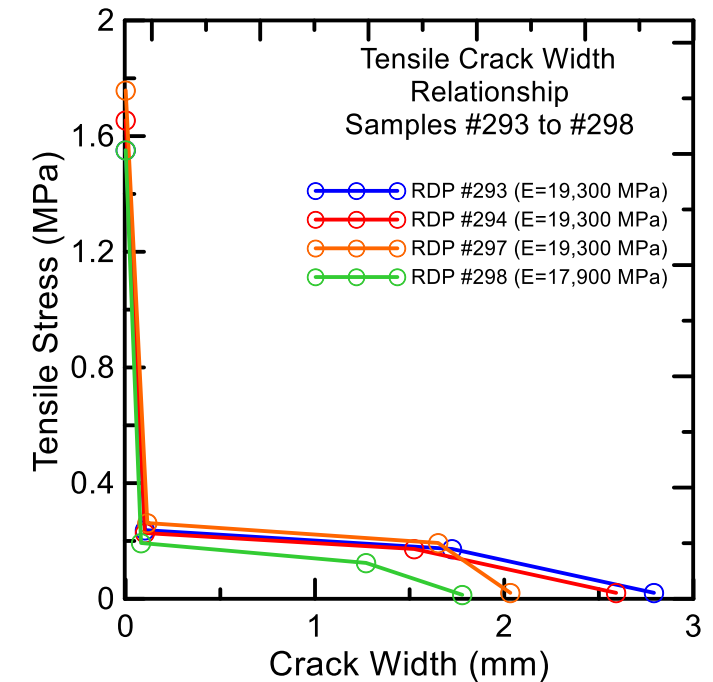
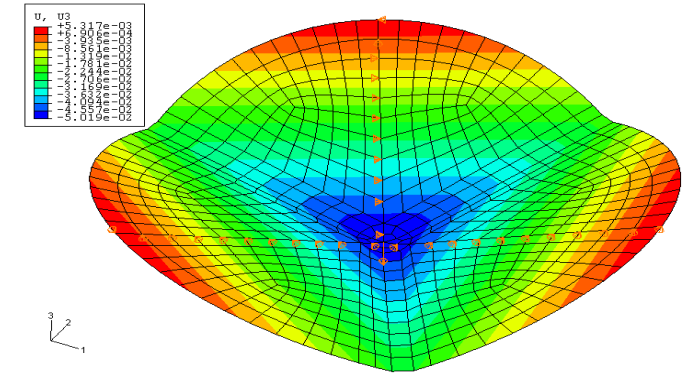
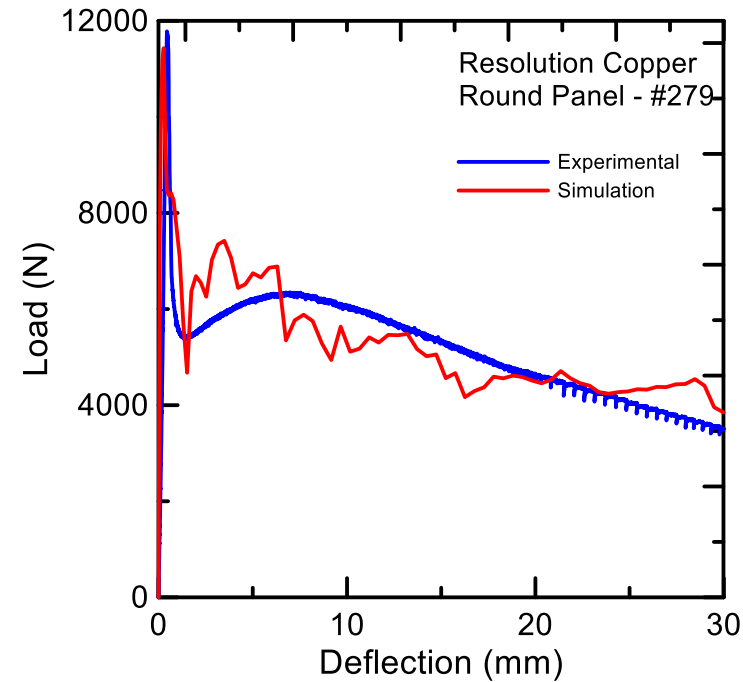
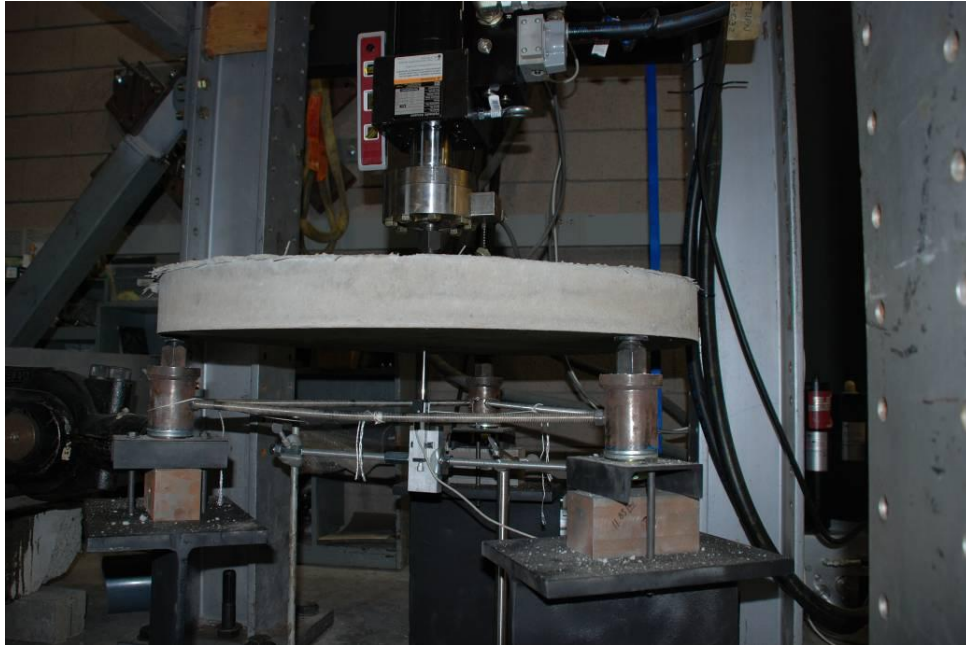
Effect of curing age on flexural response



10-12 lbs/yd³ of macro fibers

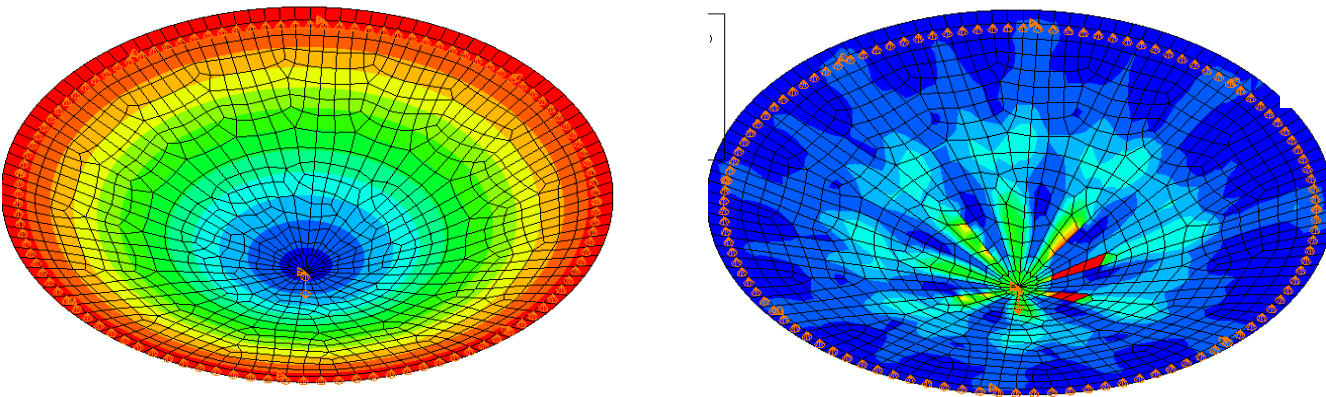
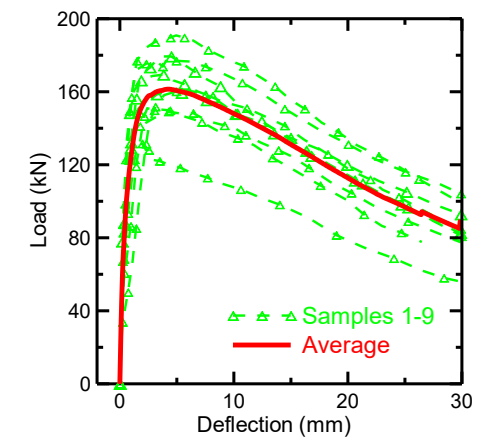
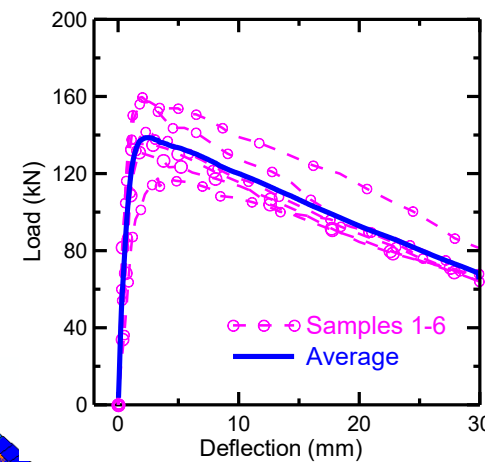
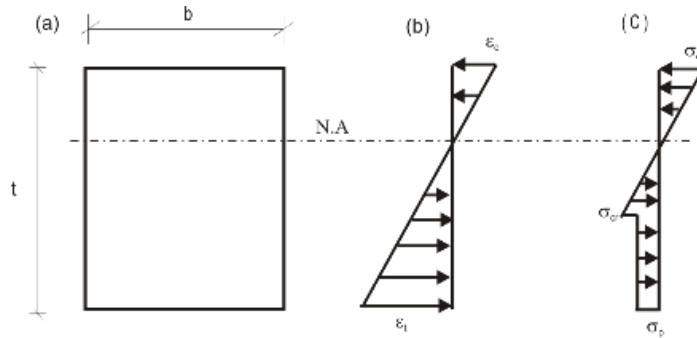
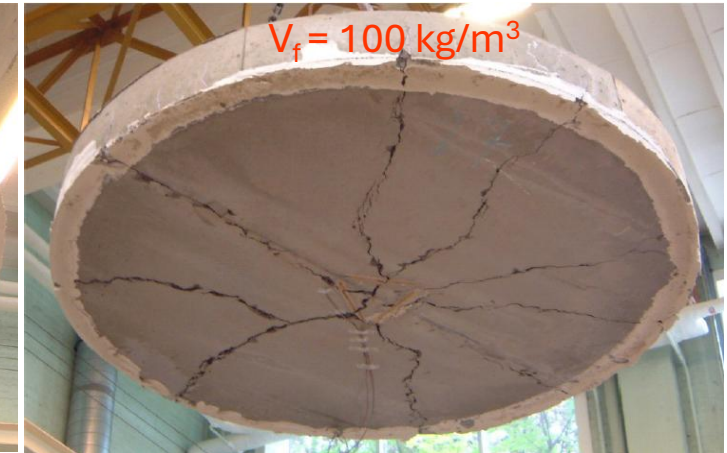
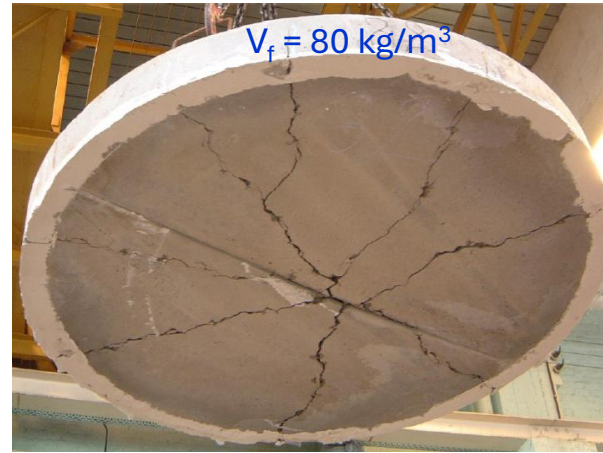


ASTM C1550-Round Panel 3P-support specimen



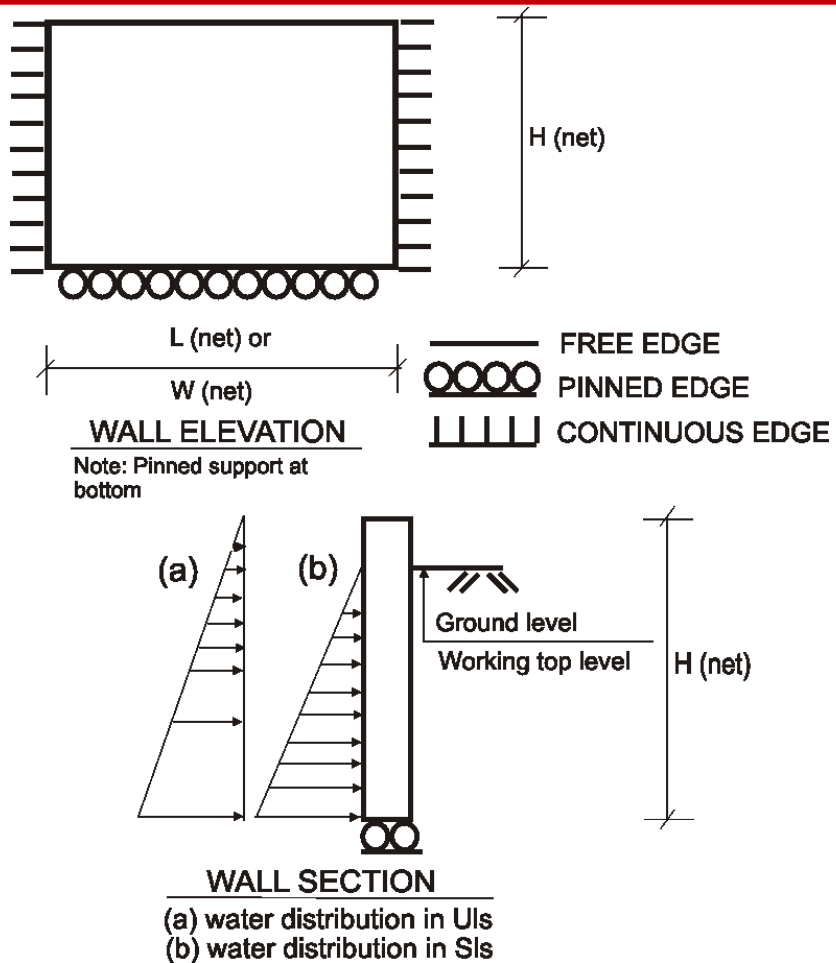
Typical Crack Patterns Deflection Hardening

clear diameter 1500 mm
thickness = 150 mm
stoke diameter = 150 mm



- The test reveals unsymmetrical multiple radial crack patterns
- In elastic range, deformation is axisymmetric imposed as boundary conditions to improve the efficiency of the model
- In plastic stage, strain energy density localizes in crack band regions, symmetry preserves

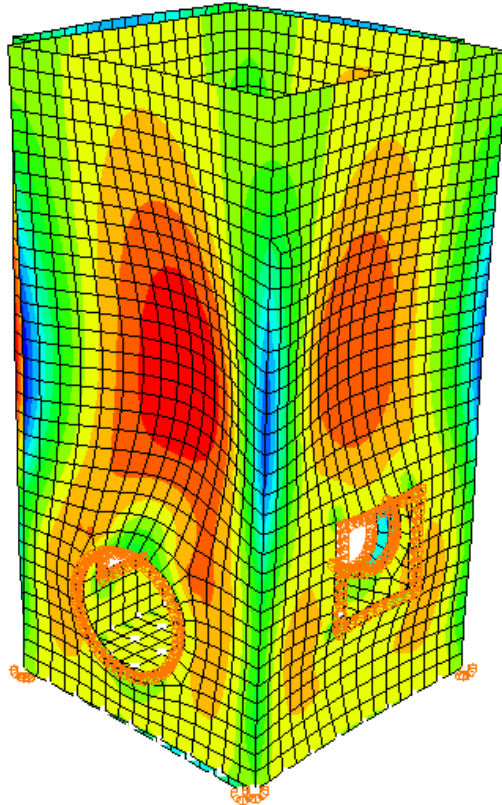
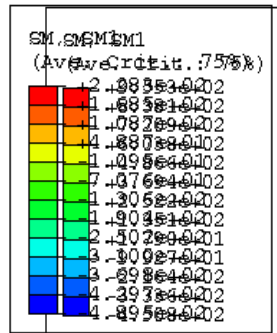
Analysis of Precast Wall Panels



Assume continuous wall, pin connection at the bottom and free at the top
Lateral water pressure in ultimate and serviceability limit states



Analysis, Design and Installation of precast water tank panels



■ Load Case2:

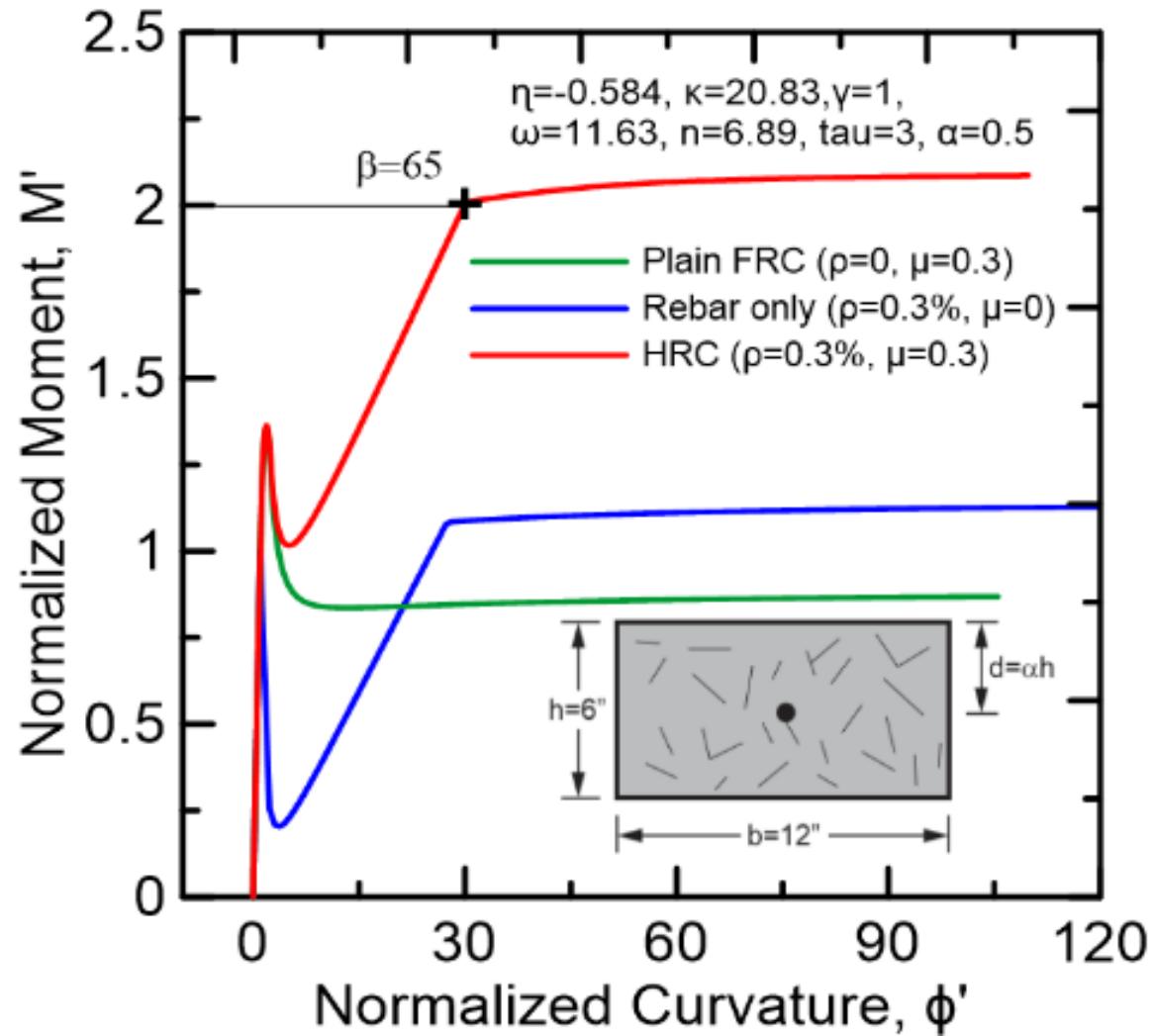
- 1.4 Self weight +
1.7 Earth pressure +
1.7 Uniform pressure due to surcharge
- Moment in short span direction SM1

Modeling and Validation of Experimental approach

- Hybrid Structural design: First principal tools to incorporate Supplementary Reinforcing Materials,
- Labor, Efficiency: Enhancing construction speed, design alternative checks
- Performance-based Metrics: Analysis, Design, and novel test methods
- Design for serviceability and durability: Different reinforcements operate at different strain ranges
- Optimization of generalized solutions envelop for new and repair/retrofit apps, Applicable to:
 - fibers, textiles, FRP, or ordinary steel reinforcement systems.
 - Structural elements meeting durability or serviceability limit states.



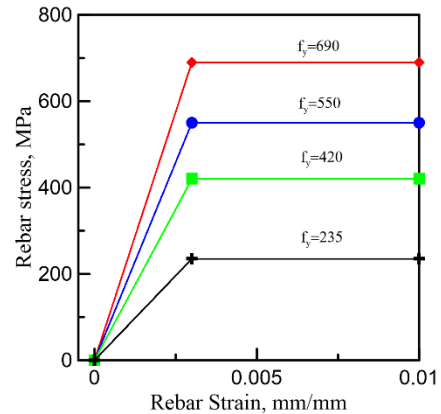
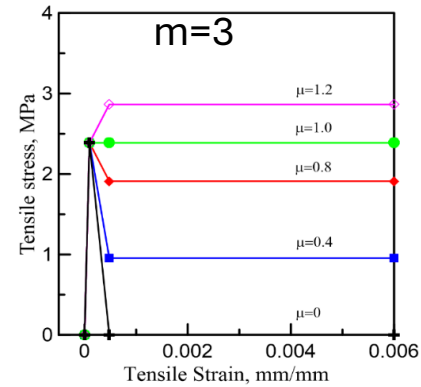
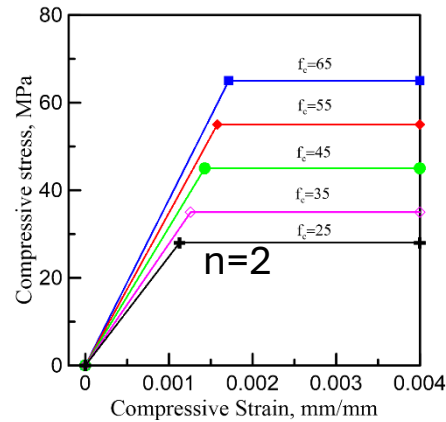
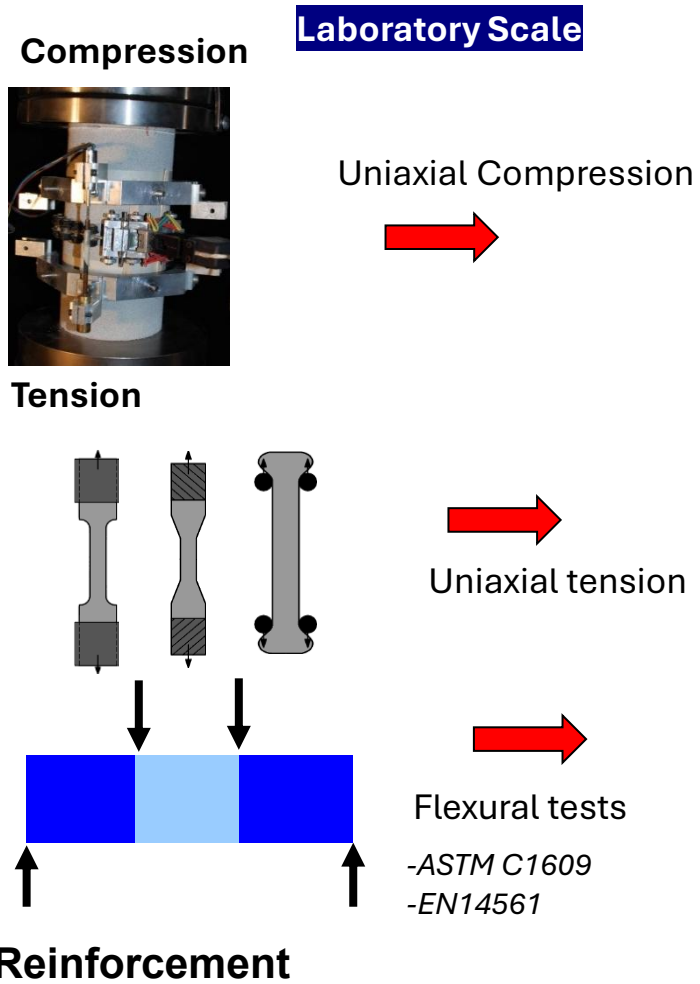
Contribution of Fibers to the reinforced concrete



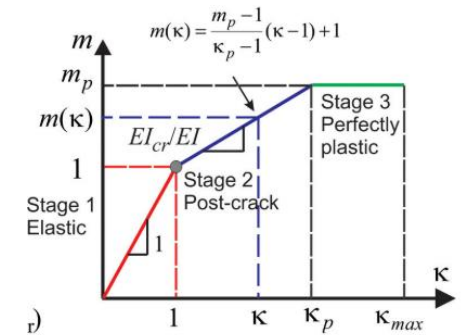
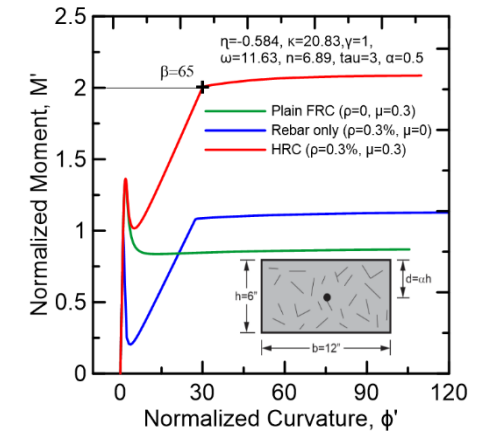
Modelling procedures

- Tension model
- Compression model
- Bond (interface) model
- Stress/strain distribution
- Moment-curvature relationship
- Normalized parameters
- Incorporations of equilibrium equations
- Moment and curvature distributions throughout the volume
- Integration, application of boundary conditions
- Data reduction and solutions for stress, strain, load, deflection distributions

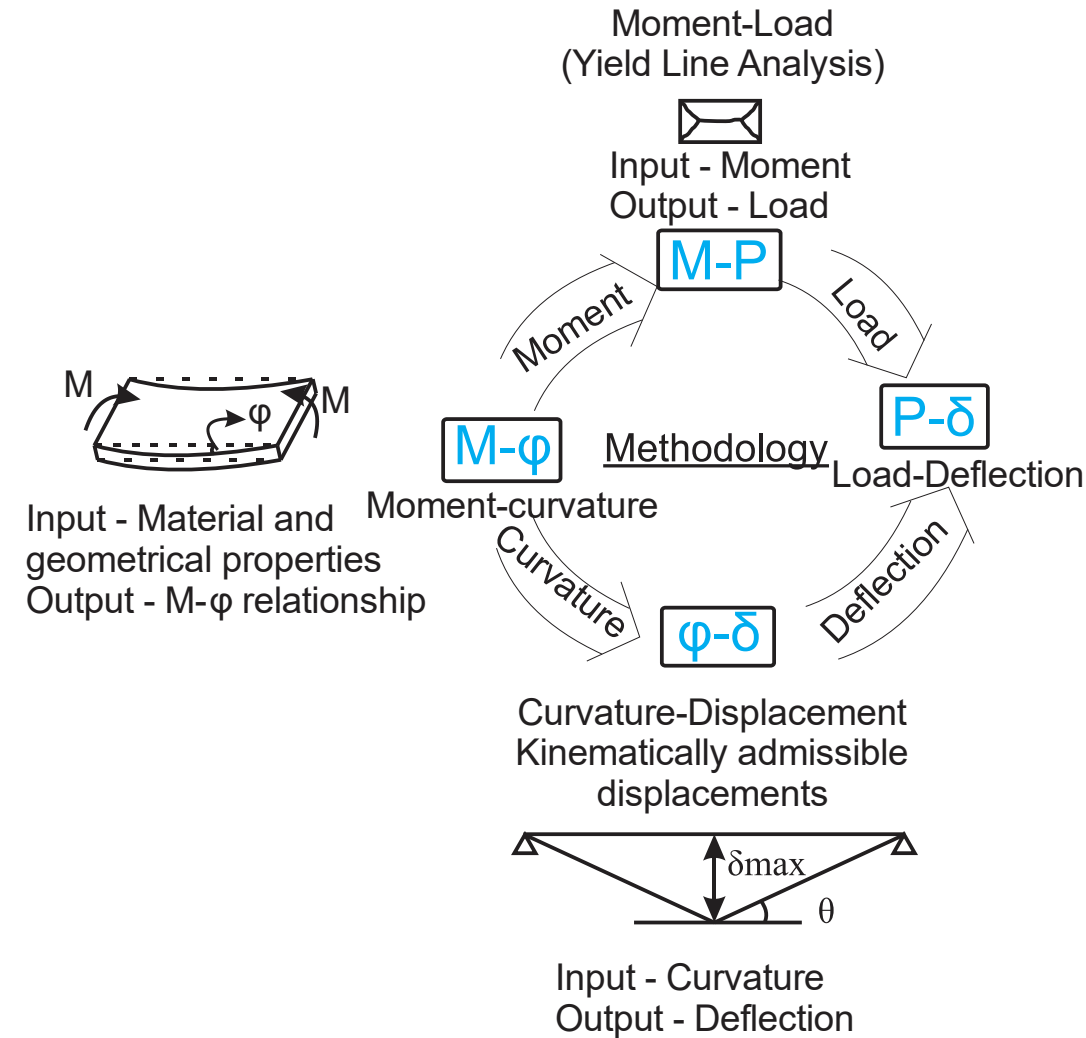
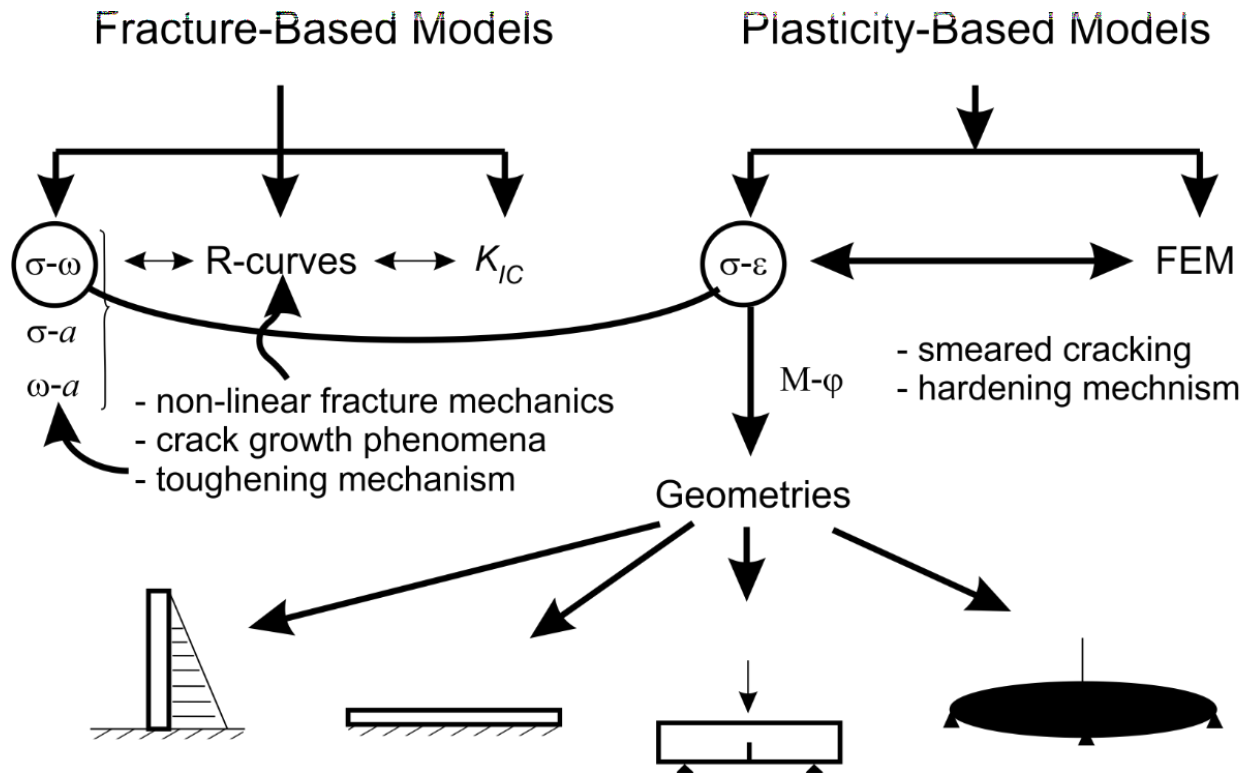
Mechanical Characterization and Design From laboratory to structure scale



Closed-Form Solutions

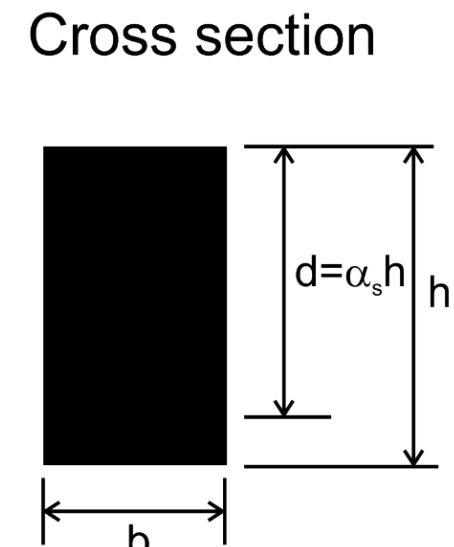
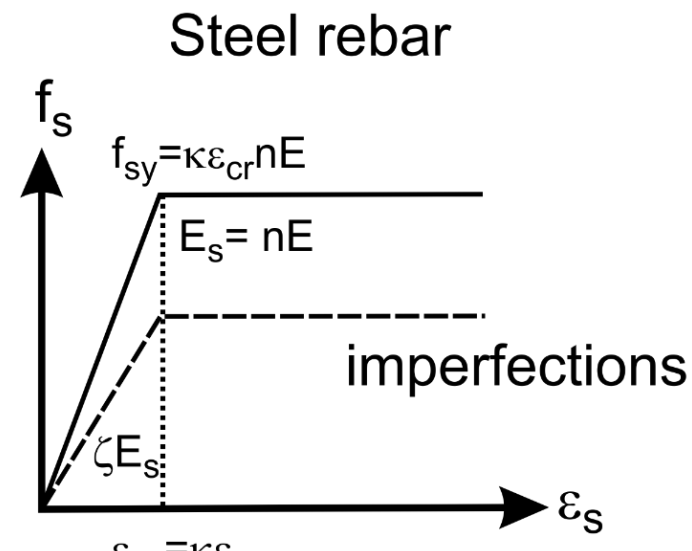
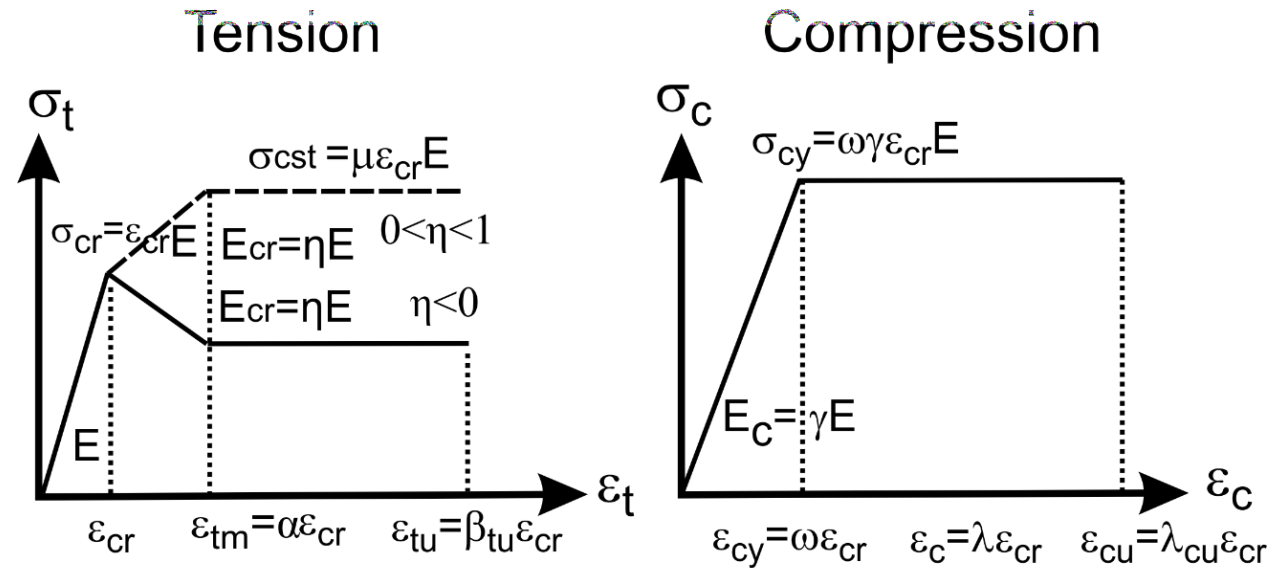


Fracture and plasticity models



Methodologies to account for the contribution of fibers

Parameterized Material Models



Moment-Curvature Relationship using the Strain Compatibility Analysis

Impose $0 \leq \varepsilon_t \leq \varepsilon_{tu}$

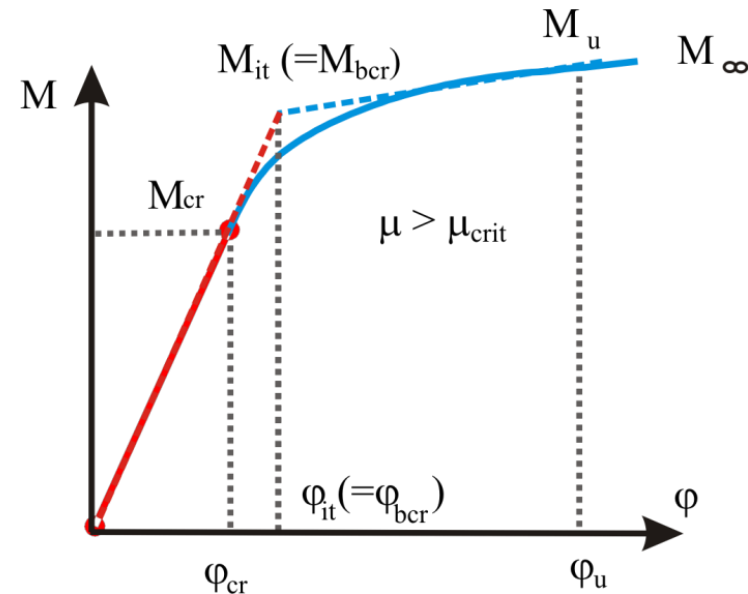
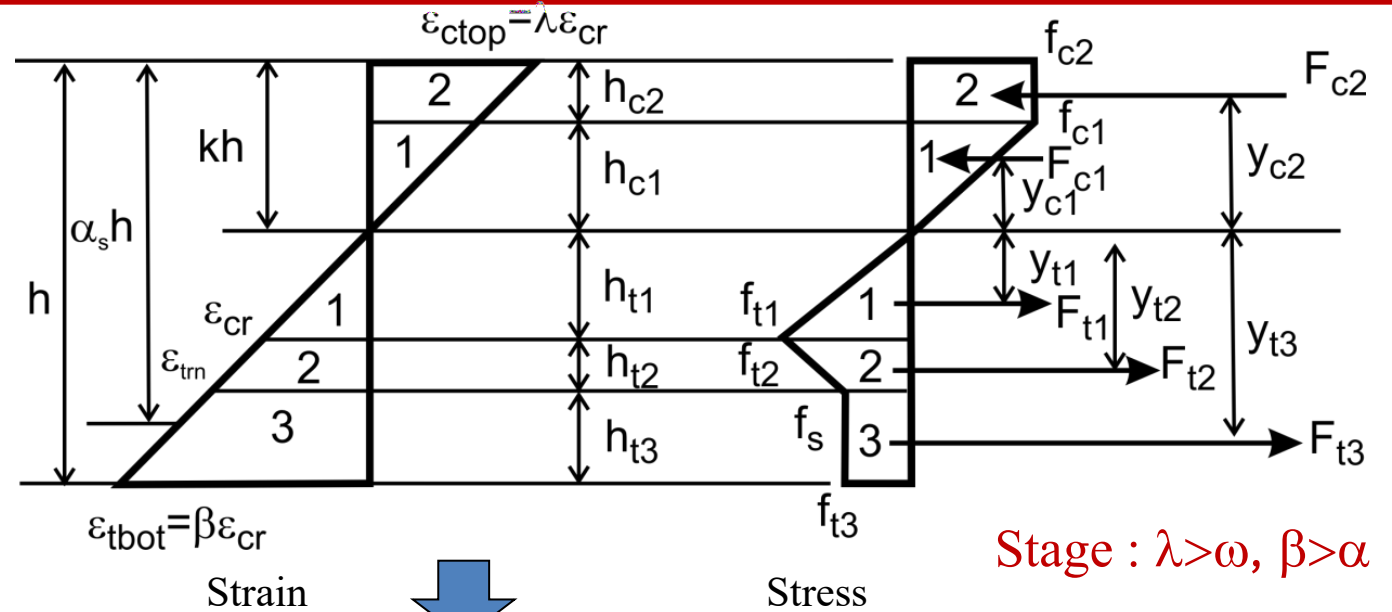
Strain Distribution
Stress Delivery

$\Sigma F = 0$, determine k

Moment:

$$M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$$

Curvature: $\phi = \varepsilon_c / kh$

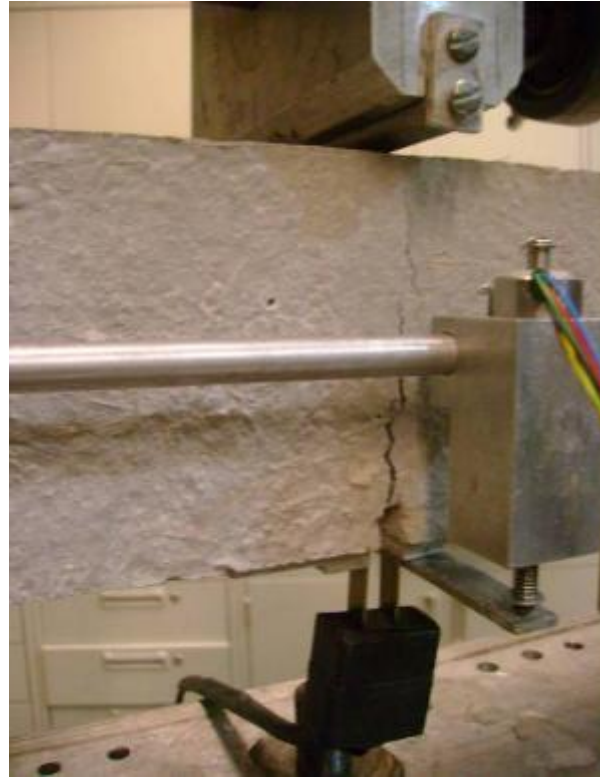


Simplified bilinear
moment-curvature

Three Point Bending, EN14651 Test notched beam



Closed-loop flexural test set up

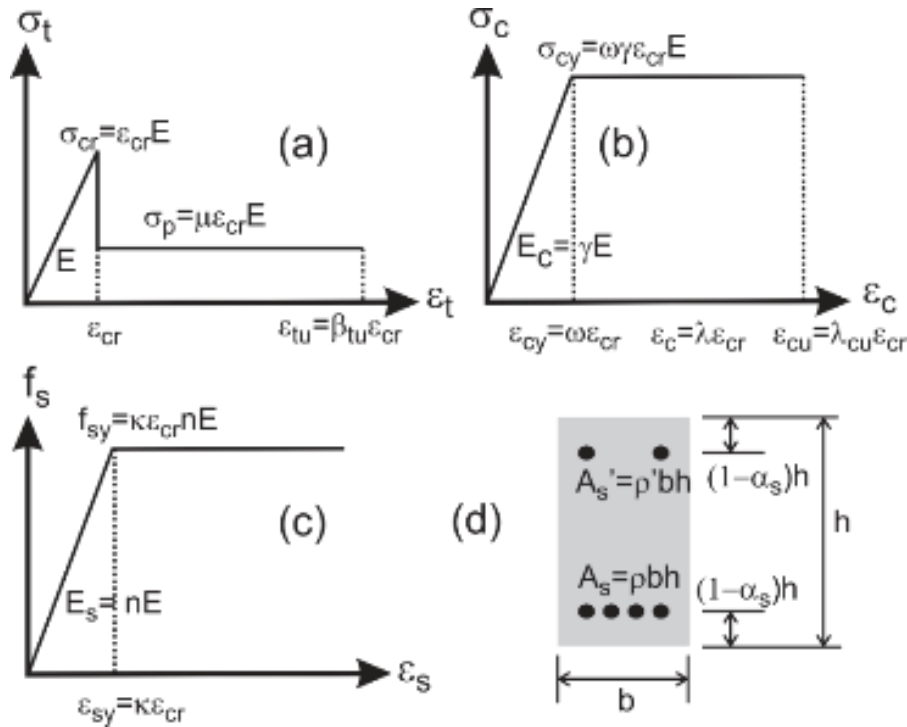


Crack pattern in a fiber reinforced specimen developed from the notch through concrete mass



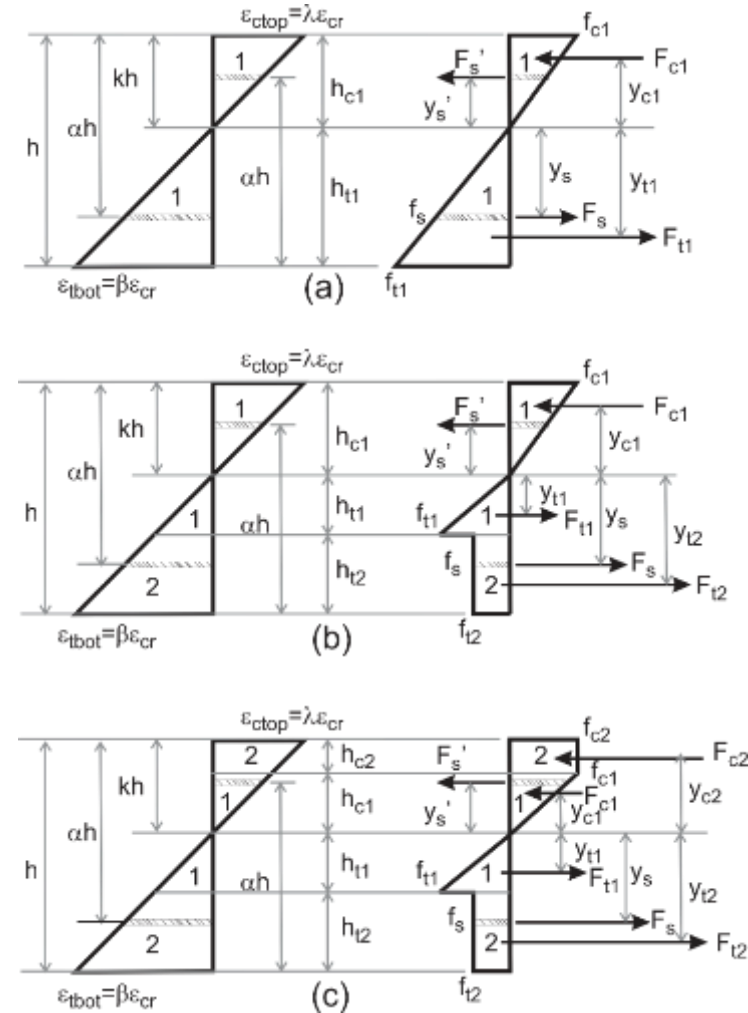
<https://www.gcts.com/>

Strain-Softening Material Models and Cross-Sectional Analysis for Hybrid systems



Material models

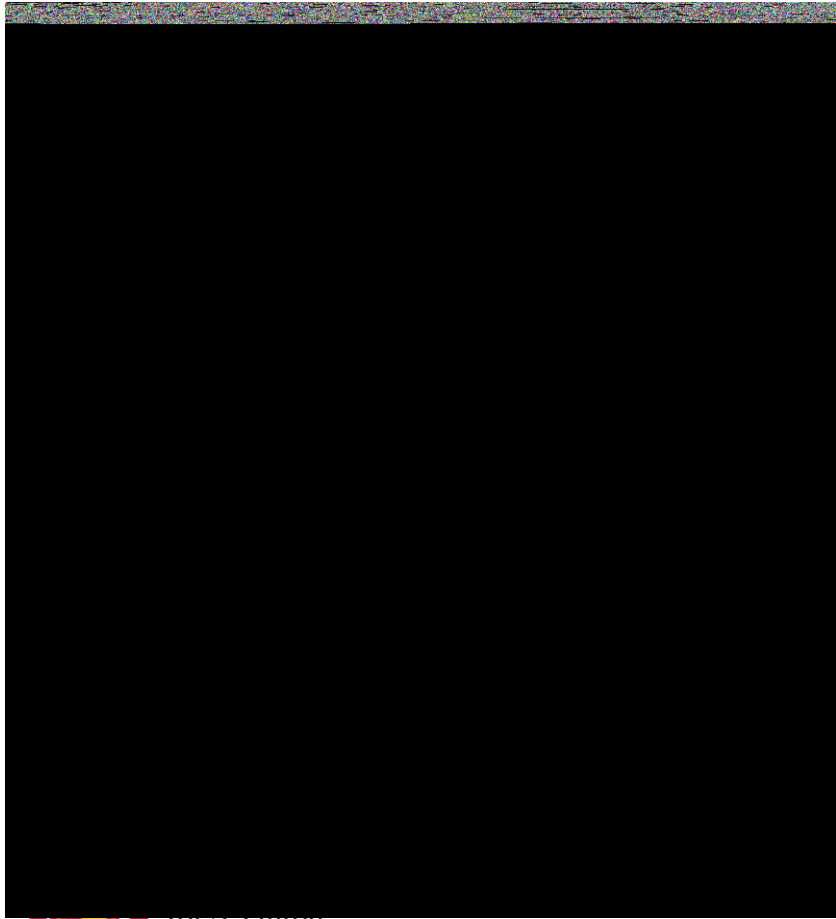
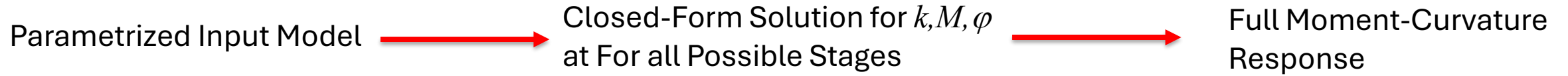
- Plane section remain plane
- Force equilibrium
- Summation of moment
- Moment-curvature response
- Load-deflection response



Stress diagrams

Flexural Modeling of Reinforced FRC/UHPC (Hybrid Approach)

- Closed-form based analytical Model for flexure simulation and Inverse analysis



Analytical solutions of neutral axis, moment, curvature and stiffness

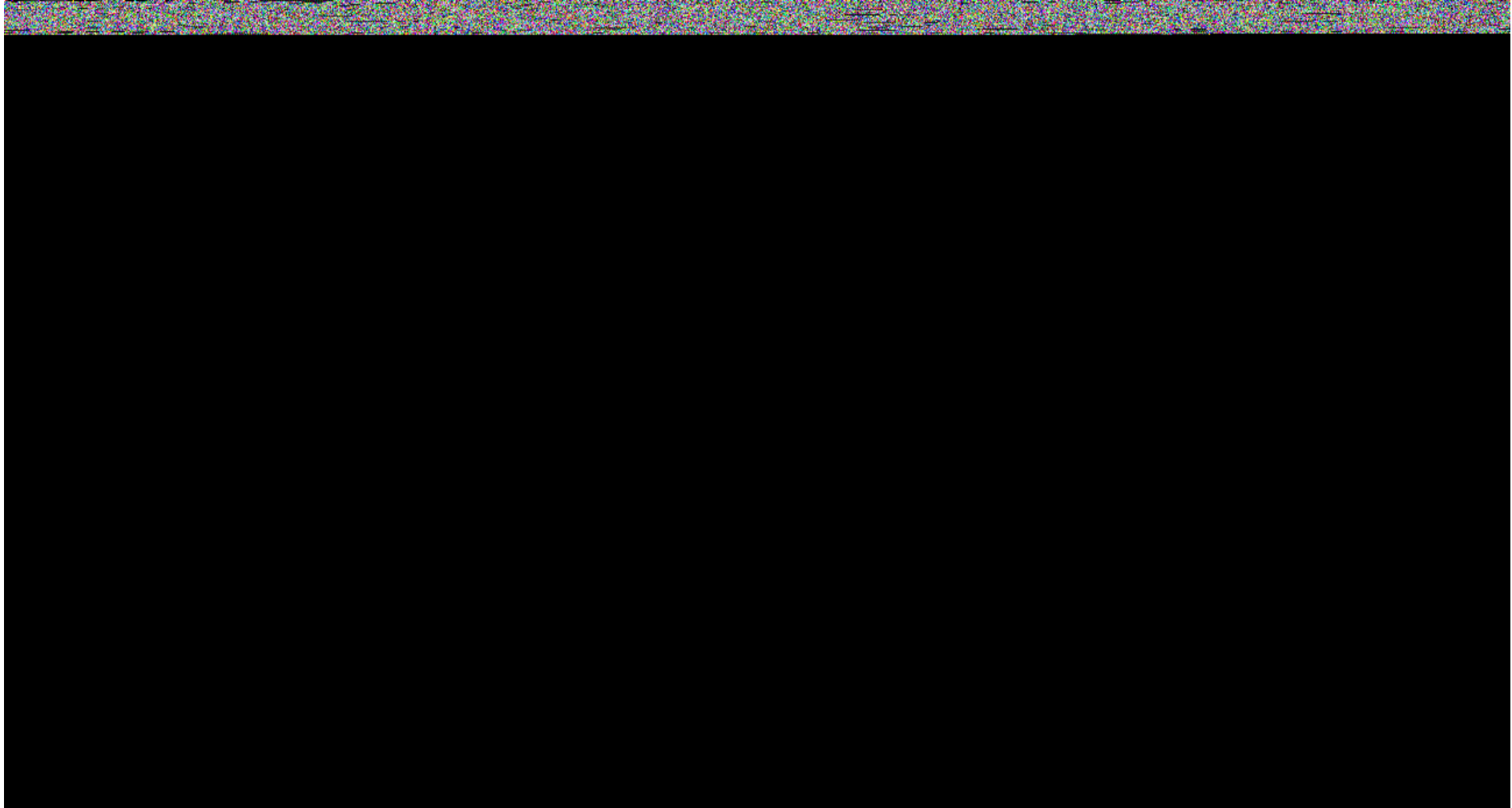
Bilinear Tension/compression/Reinforcement Hybrid Model

Stage	k	M'	ϕ'	K'
1	$k_1 = \frac{18\rho_g\alpha + 1}{18\rho_g + 2}$	$M'_1 = \frac{2\lambda}{k_1} [C_1k_1^3 + C_2k_1^2 + C_3k_1 + C_4]$	$\phi'_1 = \frac{\lambda}{2k_1}$	$K'_1 = \frac{M'_1}{\phi'_1}$
2.1	$k_{21} = \frac{\lambda}{B_1} (B_2 + \sqrt{B_3 + 2\alpha\rho_g n B_1})$	$M'_{21} = \frac{1}{\lambda^2 k_{21}} [C_5k_{21}^3 + C_6k_{21}^2 + C_7k_{21} + C_8]$	$\phi'_{21} = \frac{\lambda}{2k_{21}}$	$K'_{21} = \frac{M'_{21}}{\phi'_{21}}$
2.2	$k_{22} = \frac{B_4}{B_1}$	$M'_{22} = \frac{1}{\lambda^2} [C_5k_{22}^2 + C_9k_{22} + C_{10}]$	$\phi'_{22} = \frac{\lambda}{2k_{22}}$	$K'_{22} = \frac{M'_{22}}{\phi'_{22}}$
3.1	$k_{31} = \frac{\lambda}{B_5} (B_2 + \sqrt{B_3 + 2\alpha\rho_g n B_5})$	$M'_{31} = \frac{1}{\lambda^2 k_{31}} [C_{11}k_{31}^3 + C_6k_{31}^2 + C_7k_{31} + C_8]$	$\phi'_{31} = \frac{\lambda}{2k_{31}}$	$K'_{31} = \frac{M'_{31}}{\phi'_{31}}$
3.2	$k_{32} = \frac{B_4}{B_5}$	$M'_{32} = \frac{1}{\lambda^2} [C_{11}k_{32}^2 + C_9k_{32} + C_{10}]$	$\phi'_{32} = \frac{\lambda}{2k_{32}}$	$K'_{32} = \frac{M'_{32}}{\phi'_{32}}$

where, the coefficients are:

$$\begin{aligned}
 B_1 &= \lambda^2 + 2\mu(\lambda + 1) - 1; \quad B_2 = \mu - 9\rho_g\lambda; \quad B_3 = 9\rho_g(\rho_g 9\lambda^2 - 2\mu\lambda) + \mu^2; \quad B_4 = 2\lambda(9\rho_g\kappa + \mu); \quad B_5 = 20\lambda - 101 + 2\mu(\lambda + 1) \\
 C_1 &= 0; \quad C_2 = 27\rho_g + 3; \quad C_3 = -3 - 54\rho_g\alpha; \quad C_4 = 1 + 27\rho_g\alpha^2; \quad C_5 = 2\lambda^3 + 3\mu(\lambda^2 - 1) + 2; \quad C_6 = 6\lambda^2(9\lambda\rho_g - \mu) \\
 C_7 &= 3\lambda^2(\mu - 36\rho_g\alpha\lambda); \quad C_8 = 54\rho_g\alpha^2\lambda^3; \quad C_9 = -6\lambda^2(9\rho_g\kappa + \mu); \quad C_{10} = 3\lambda^2(18\rho_g\alpha\kappa + \mu); \quad C_{11} = 30\lambda^2 + 3\mu(\lambda^2 - 1) - 998
 \end{aligned}$$

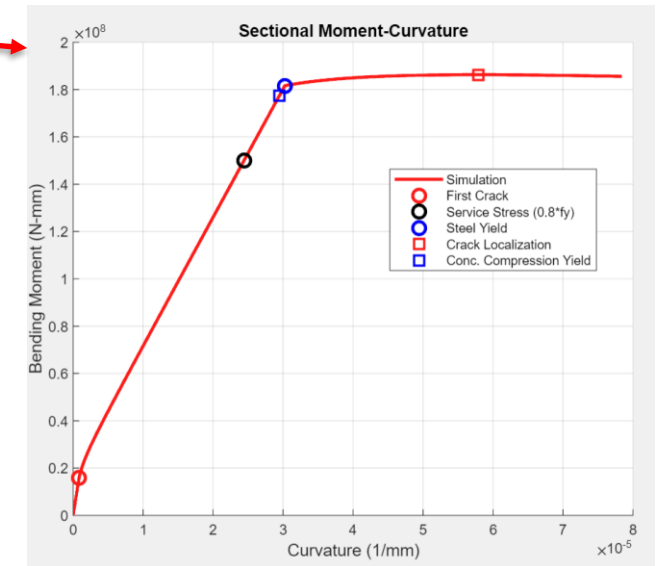
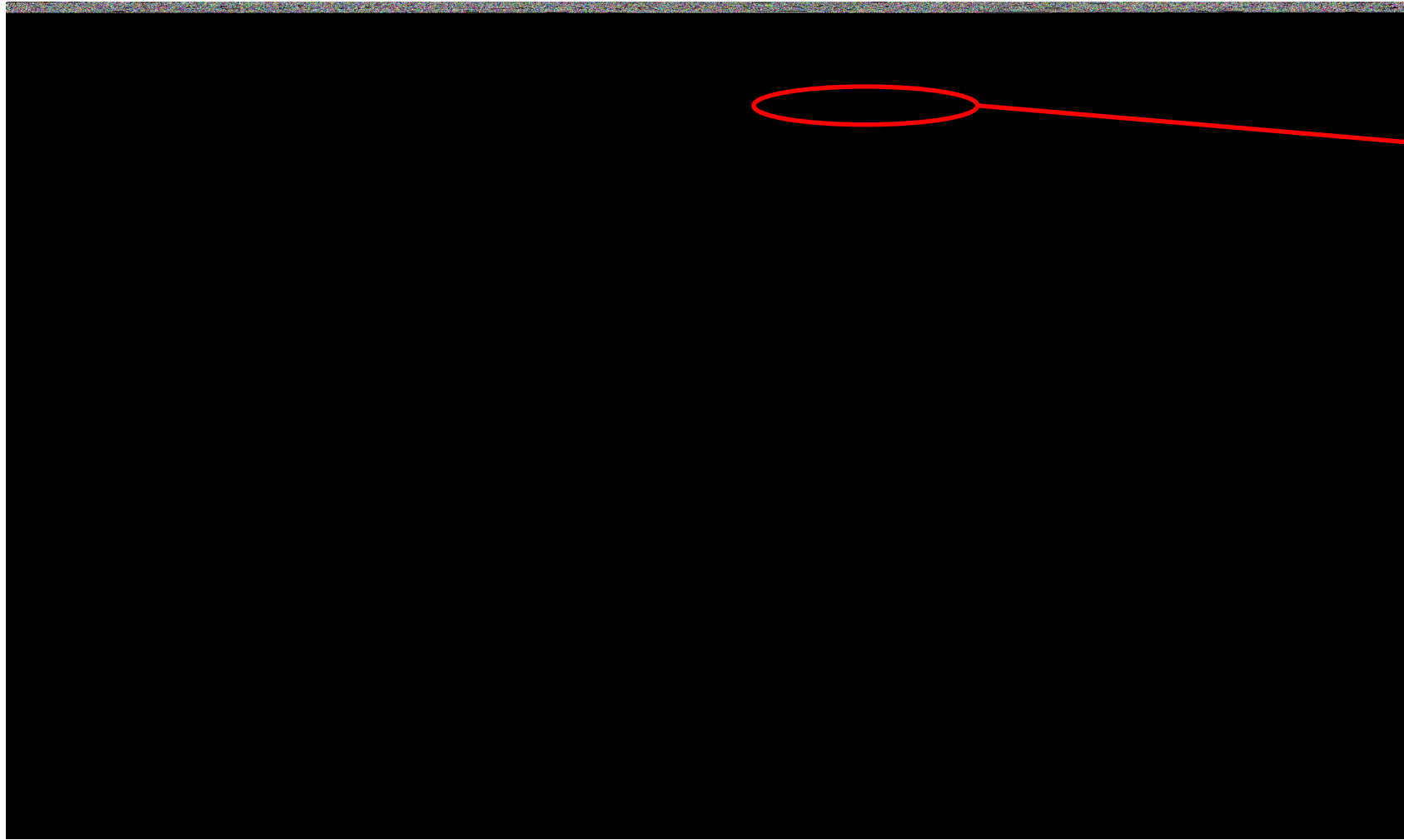
Envelope Moment-Curvature



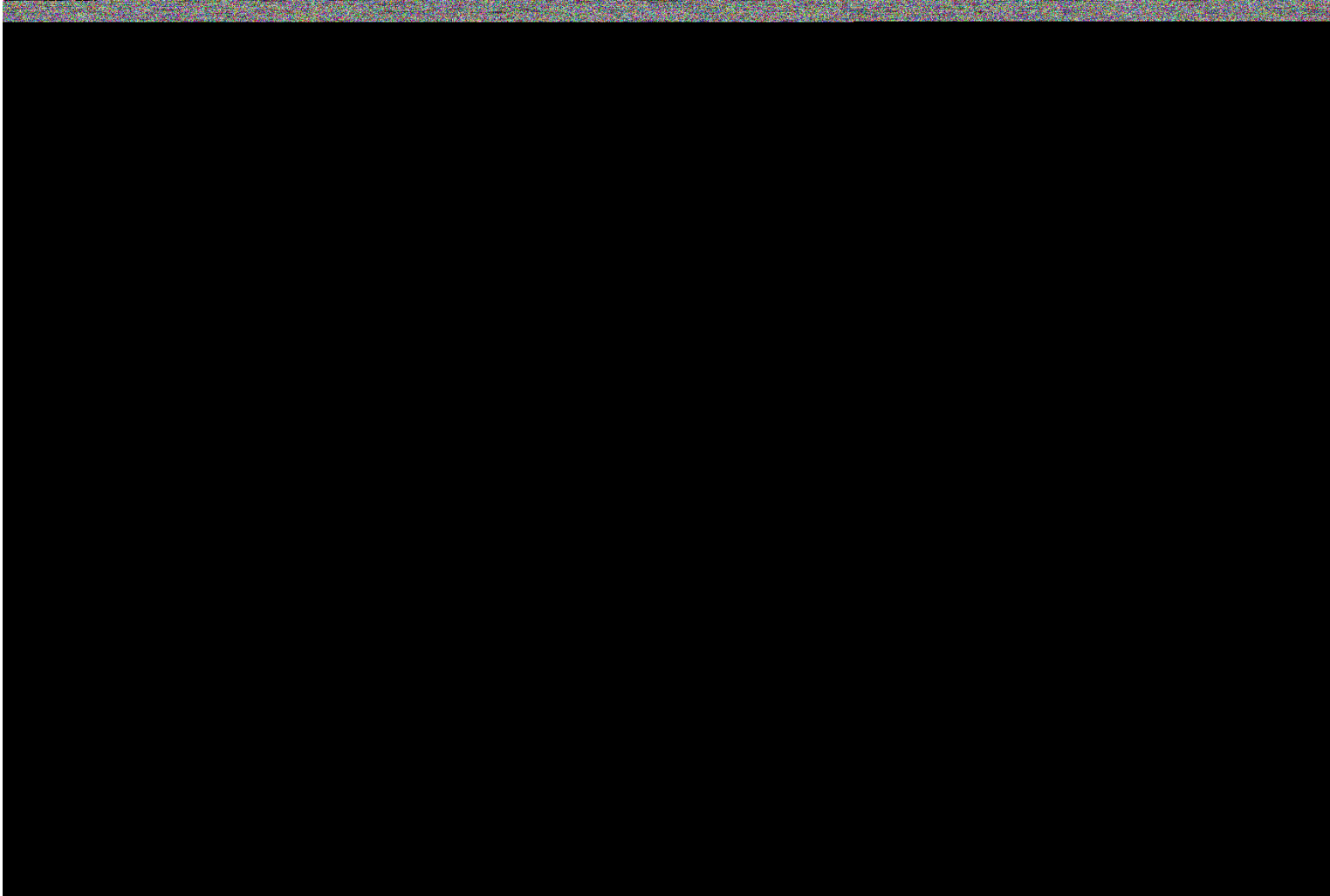
Inverse Analysis for precise characterization

-Quadrilinear Tension model

Design software for Plain FRC, UHPC using parametric Modeling



Load Frame for full scale testing of Precast Tunnel Segment

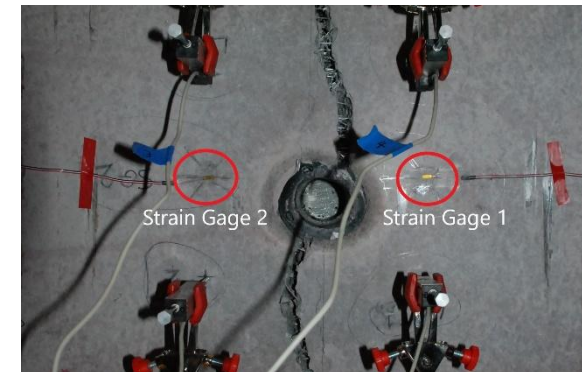
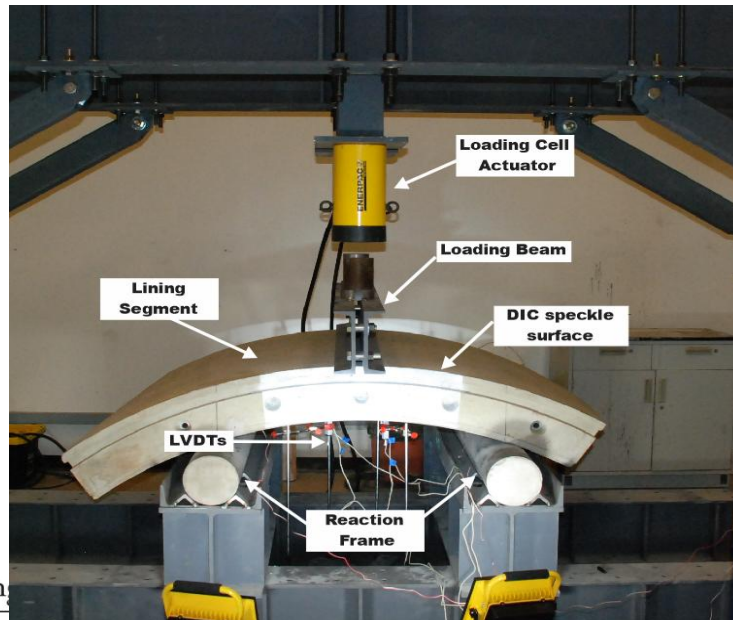


Hydraulic rams with 900 kN max. loading capacity, each

Analysis of PCTLs through full-scale testing

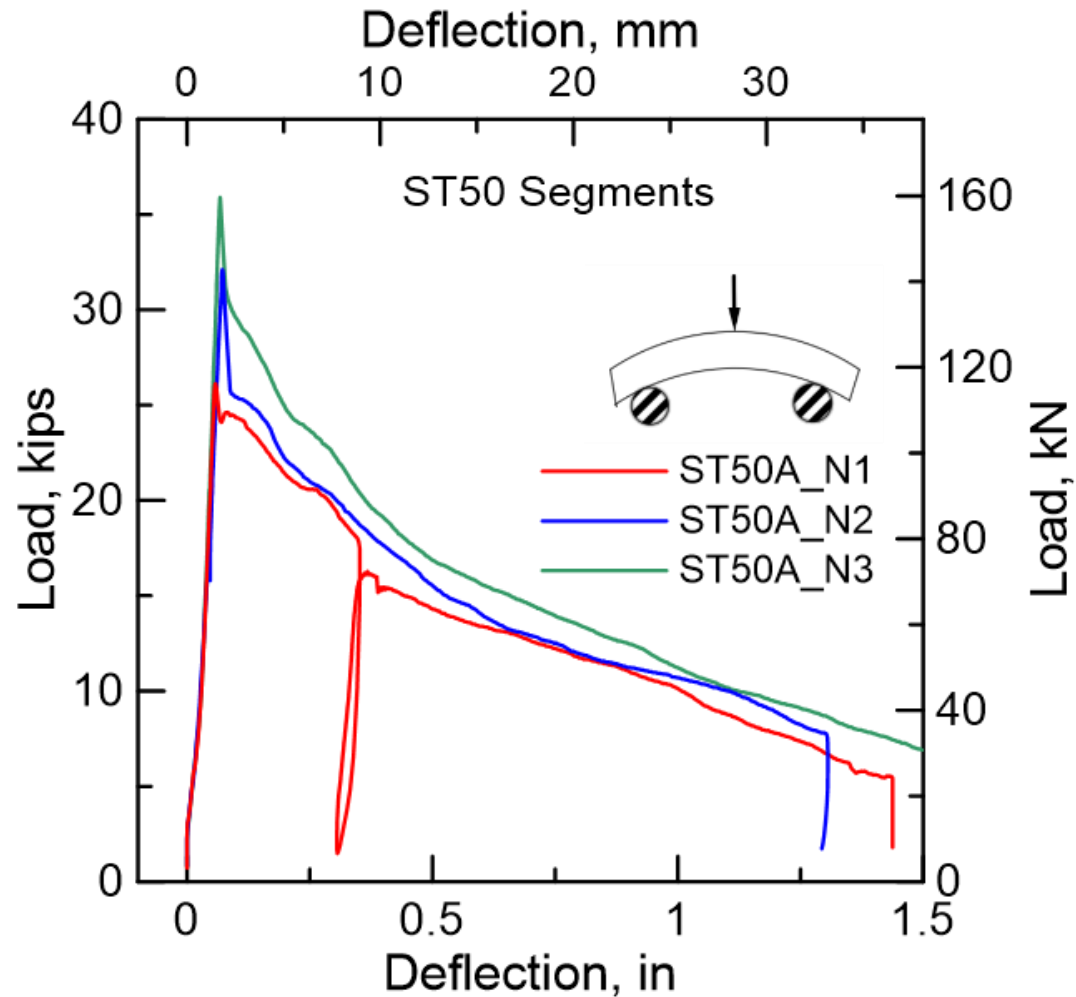
Total Instrumentation in Flexure Testing to

- Deflection (LVDT x4)
- Crack-Width (String Sensor x4)
- Strain in Tension and Compression (Strain gage x4)
- Digital Image Correlation (DIC)



Strain gage in Tension zone, Bottom view

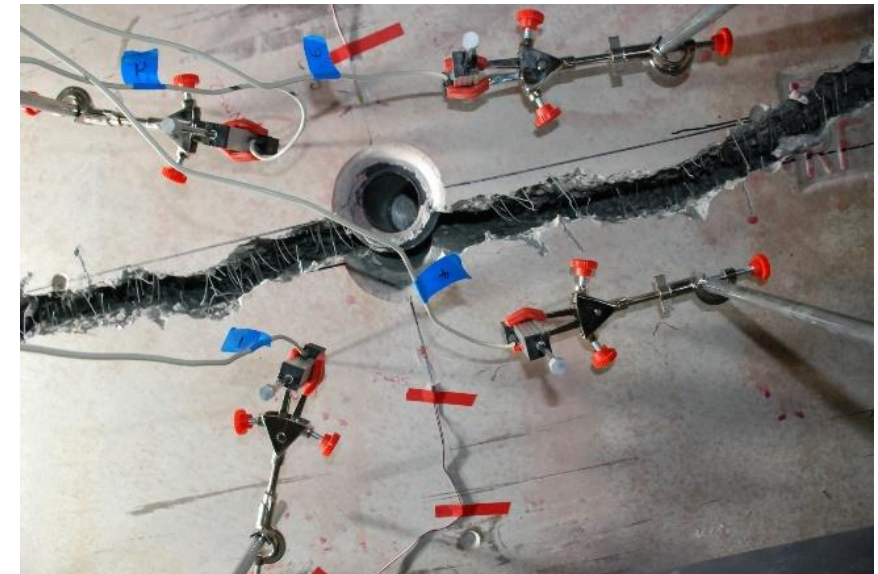
Load Deflection Response, SFRC, Full scale Tunnel Lining, 38kg/m³



N1, N2, N3 are three replicate Samples, note the variation



Cracked SFRC segment



Full scale Edge Compression Testing of Precast Tunnel Segment

- Point Load Compression Test Setup
- Max. Loading capacity of 400 kips (1780 kN)



Load Frame for compression testing

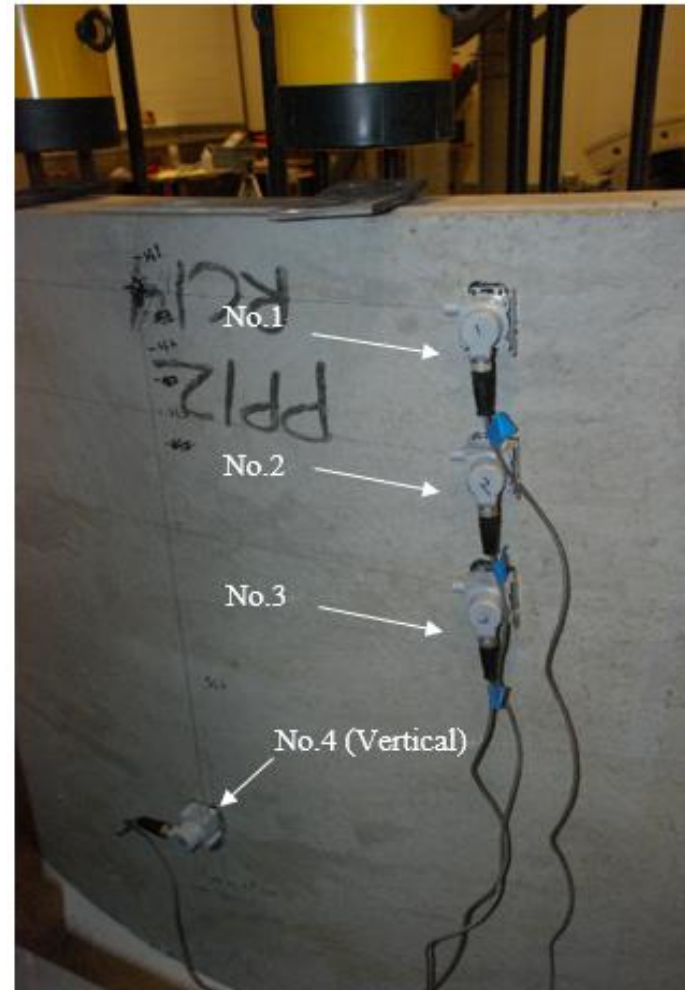
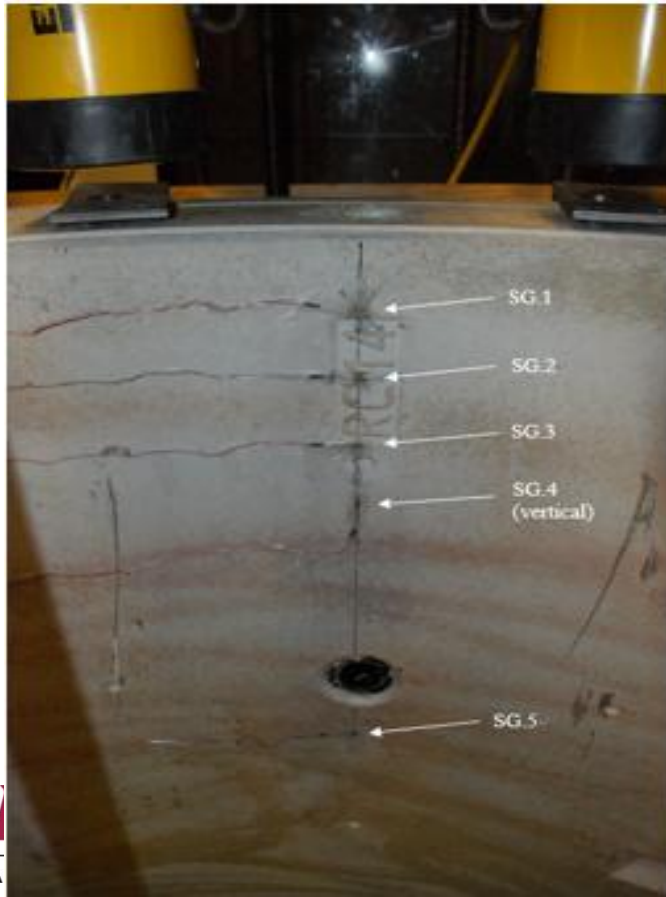


Hydraulic rams with 200 kips max. loading capacity, each

Instrumentation of Edge Compression testing

Total Instrumentation in Compression Testing to measure:

- Displacement (String Sensor x4)
- Strain (Strain gage x5)

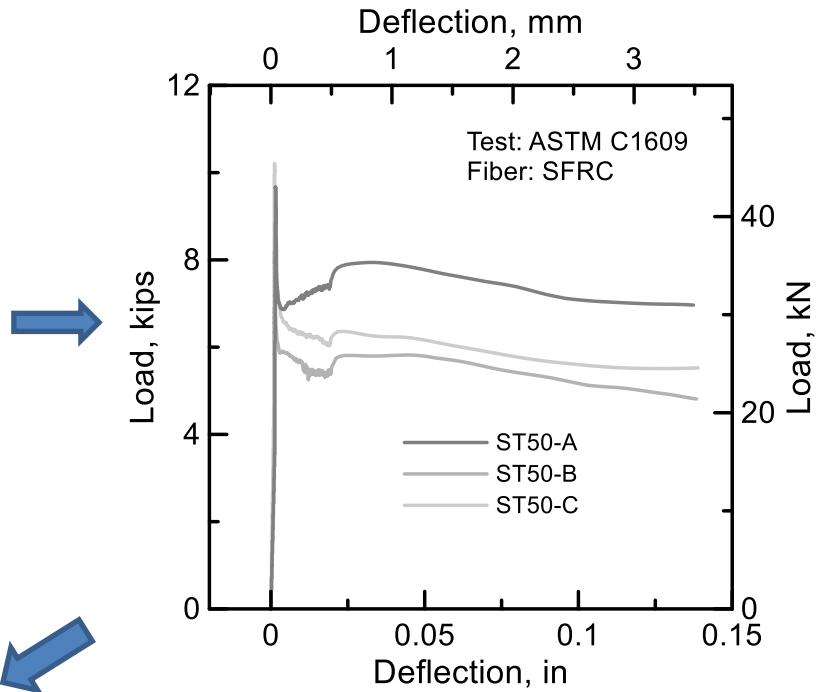
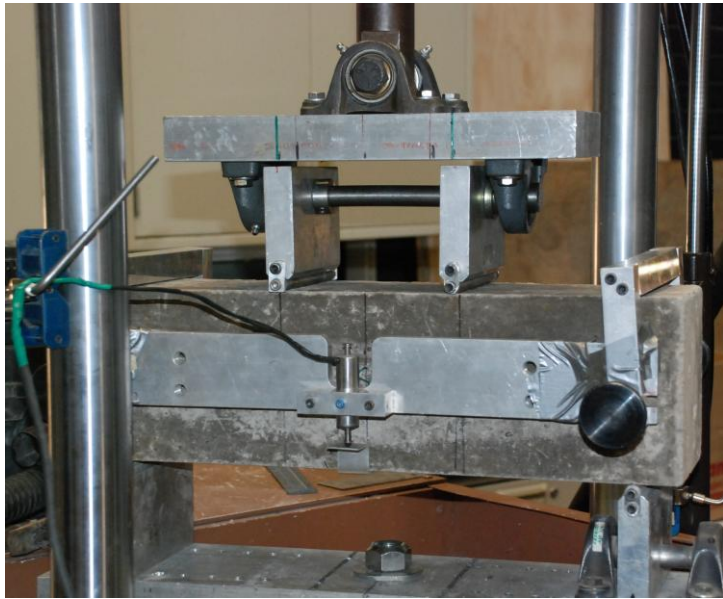


Tensile Cracking during the Edge compression tests

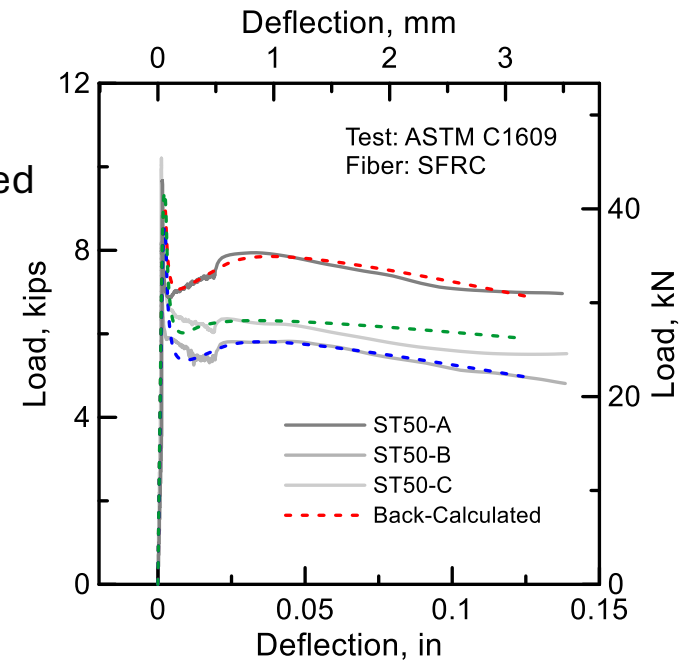
Tracing Cracks at every loading Stage



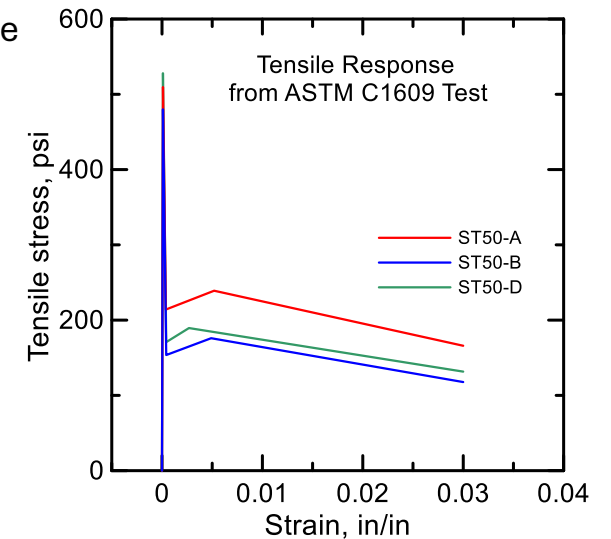
Quadrilinear Modeling for Simulating the response of tunnel segments based on ASTM C1609 Tests



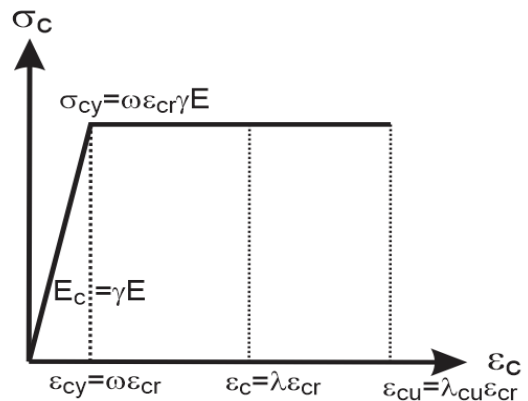
Model
Simulated
Data



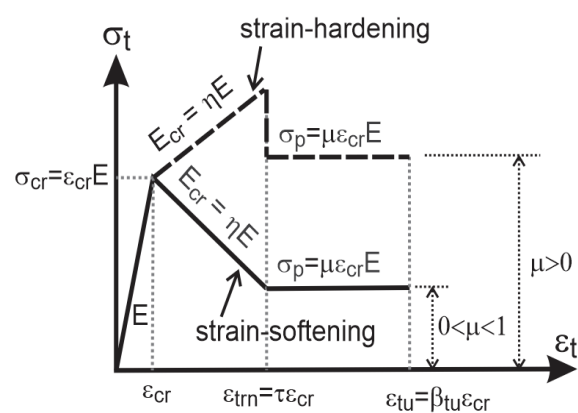
Constitutive
Response



(a) Compression Model

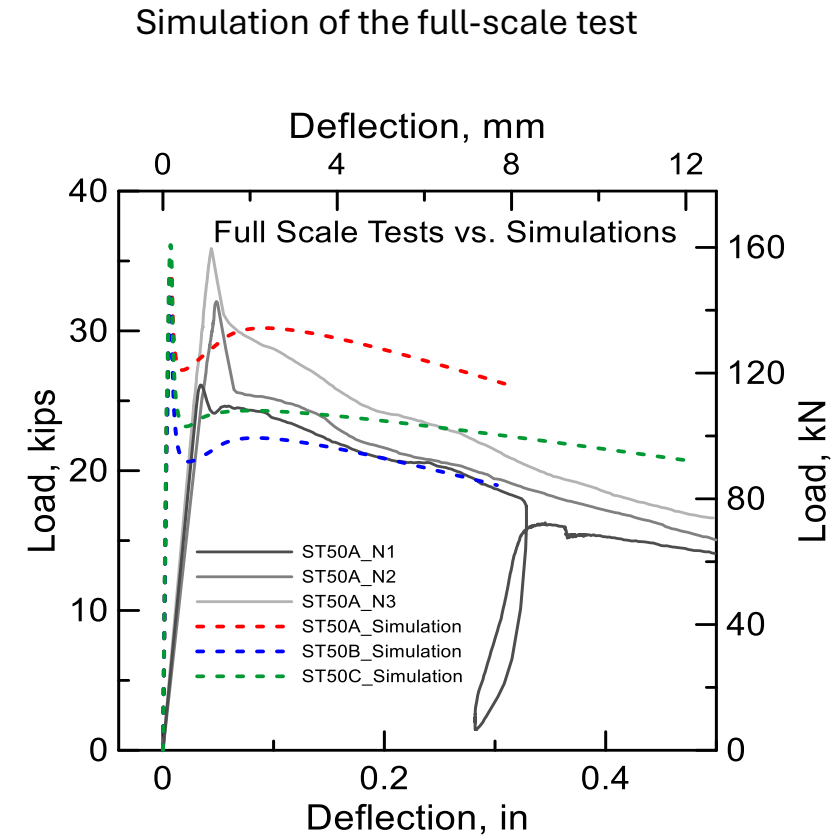
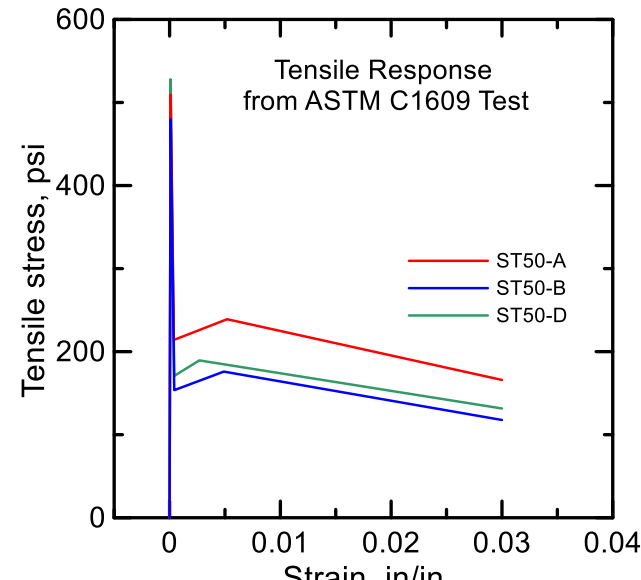
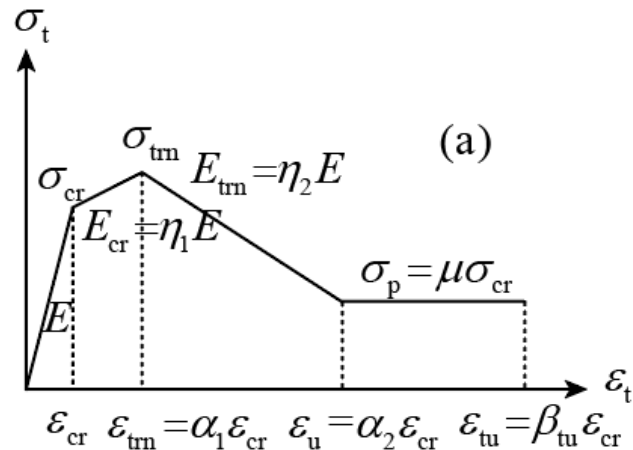
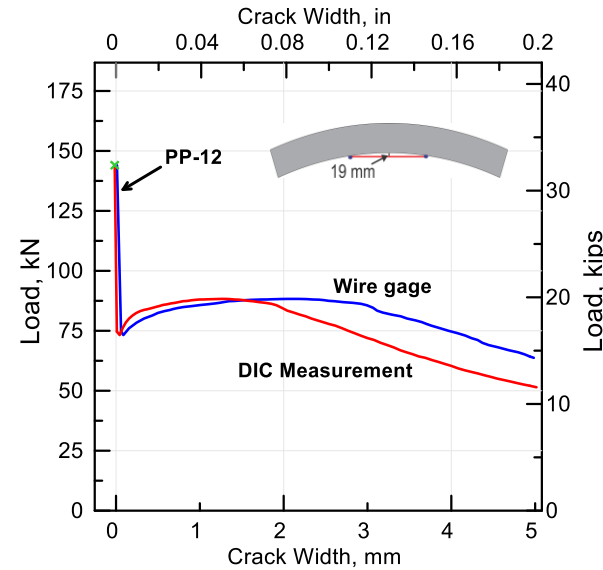
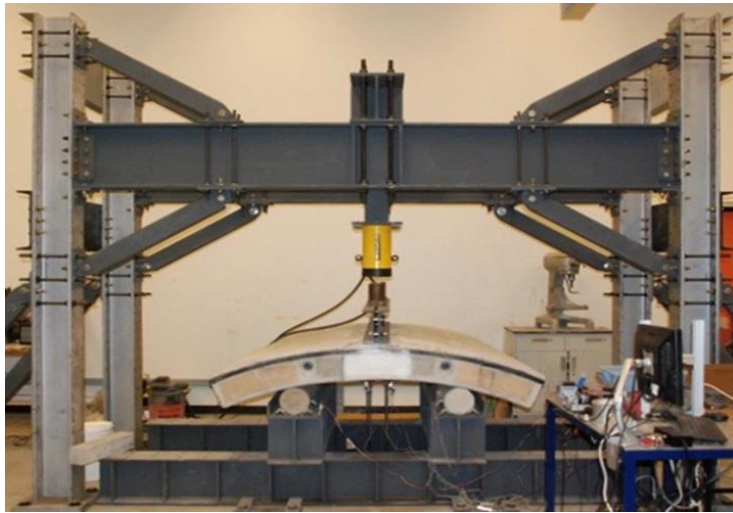


(b) Tension Model



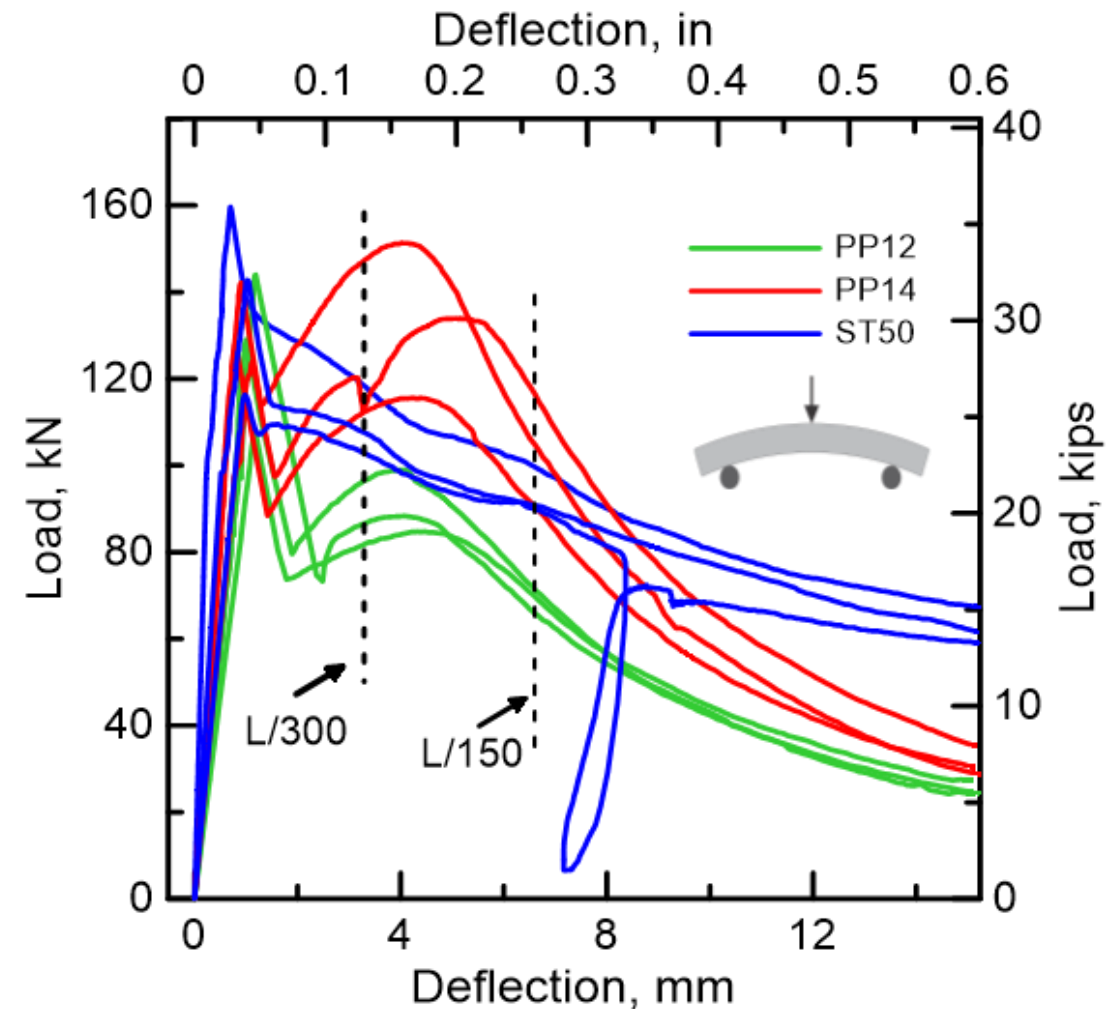
Specimen Level testing

Quadrilinear Modeling for Simulating response of tunnel segments based on ASTM C1609 Tests



Precast Tunnel Lining Full Scale testing validation, ACI 544-7R

- The question is no longer rebar vs. fibers, but what type of fiber at what volume fraction?
- Increased energy absorption, fatigue life, seismic, impact conditions, edge cracking
- Ductility, Durability, Sustainability

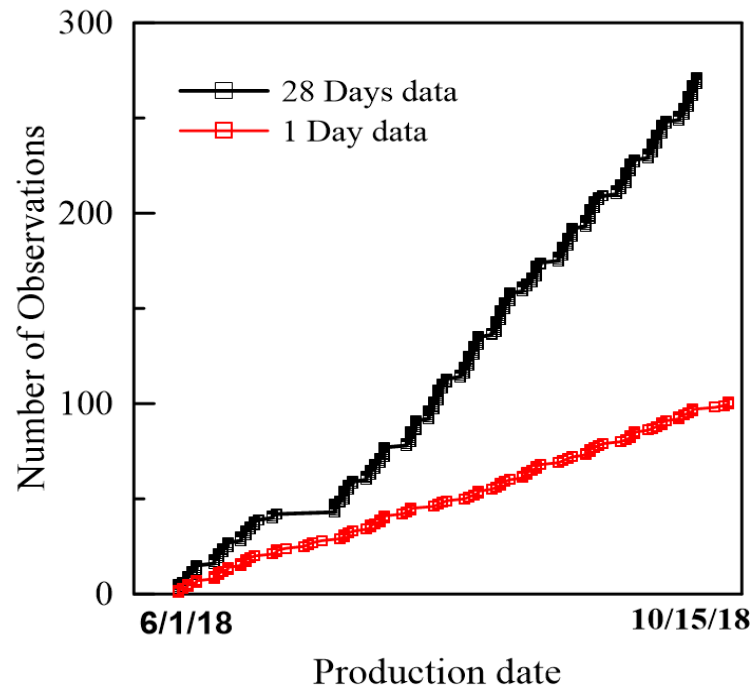


Patel, D., Plesudjai, C., Bui V., Pridemore, P., Schaef, S. and Mobasher, B. 2023. "Mechanical Response of Precast Tunnel Segments with Steel and Synthetic Macro-Fibers." *Cement and Concrete Composites* 144 (November): 105303.

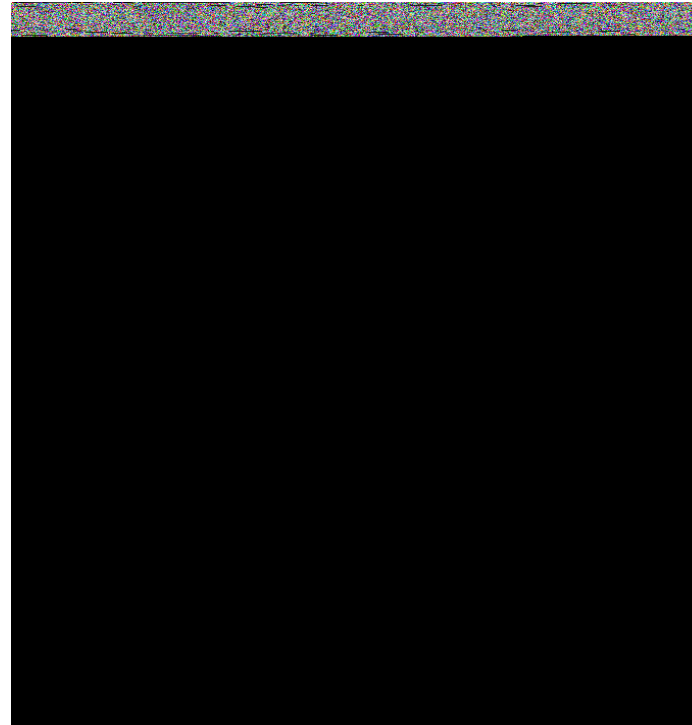
<https://doi.org/10.1016/j.cemconcomp.2023.105303>.

Statistical Process Control Of Residual Strength measures

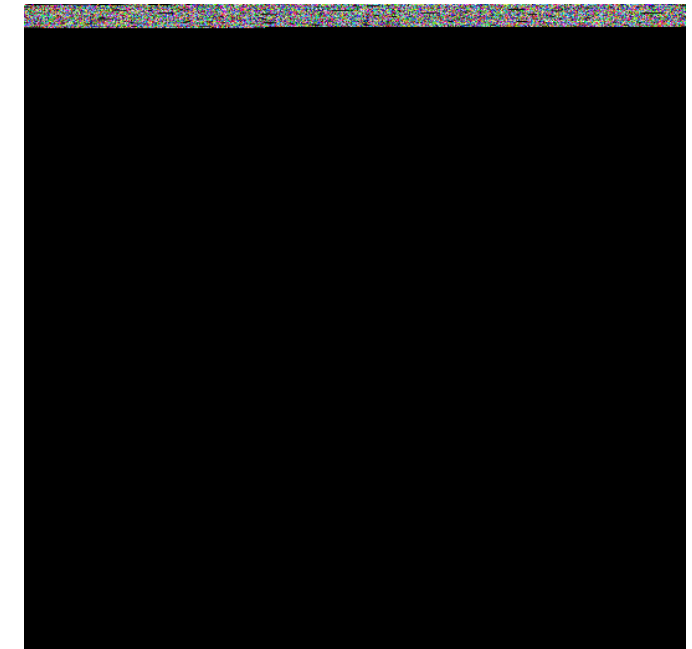
- Analysis of QC data to Improve production efficiency of PCTLs



Sample collection data points



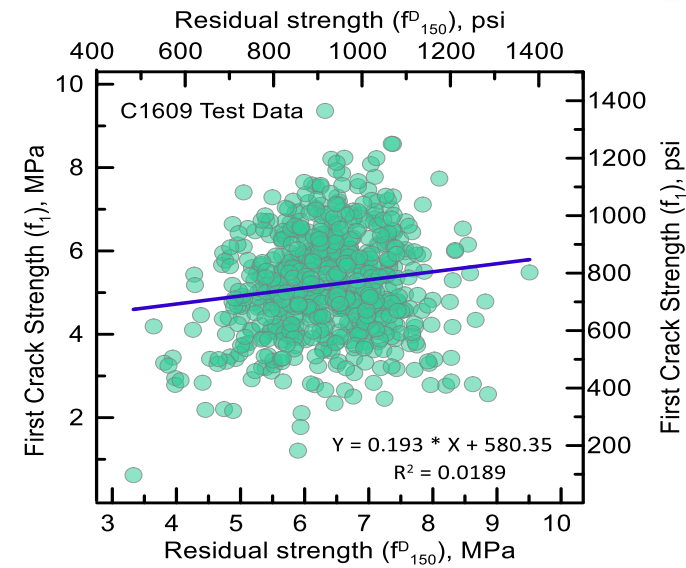
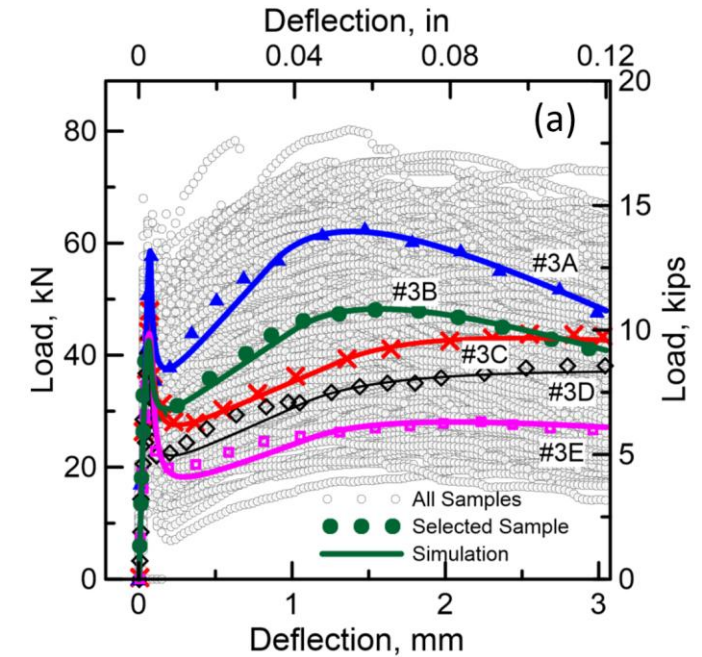
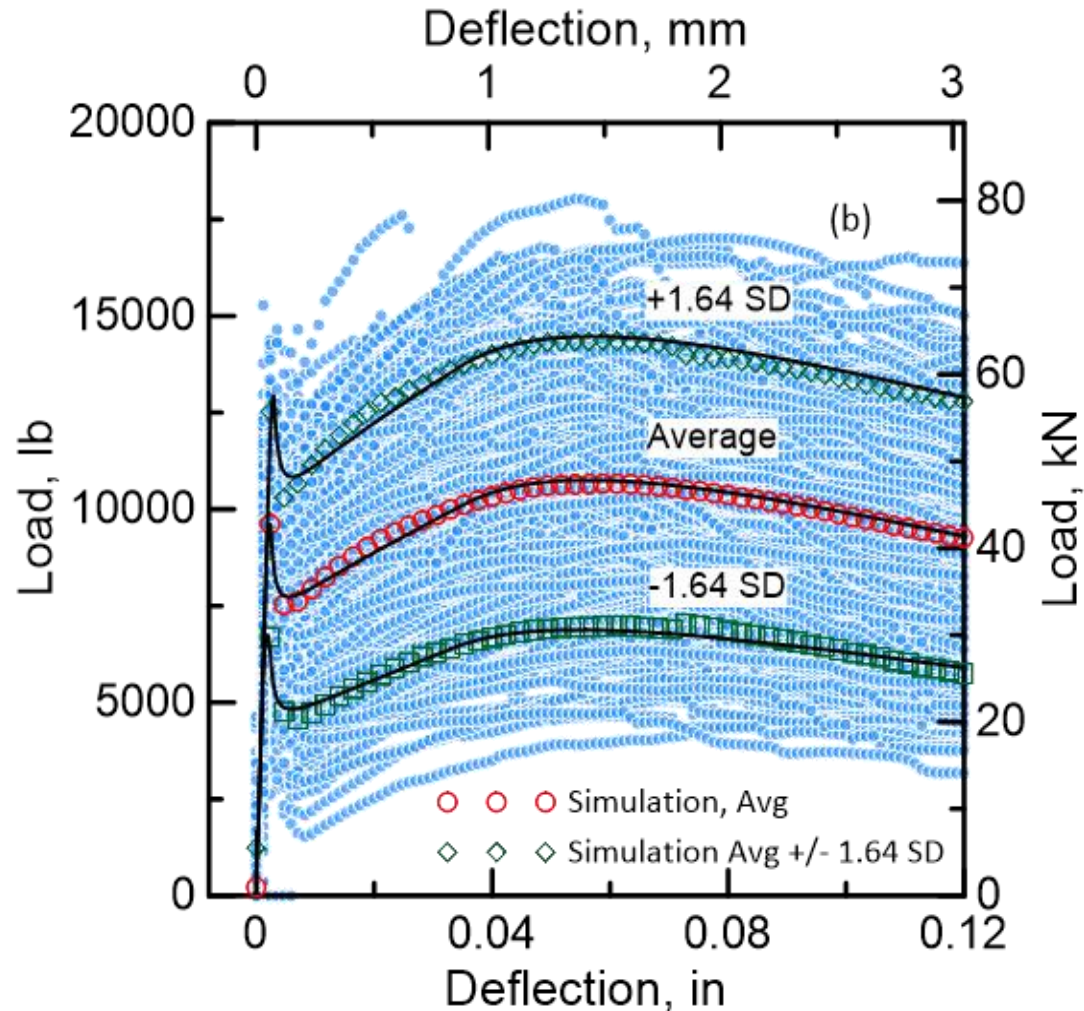
Run Chart for residual strength as a function of Production schedule



Stress vs. deflection plot of 270 C1609 test data

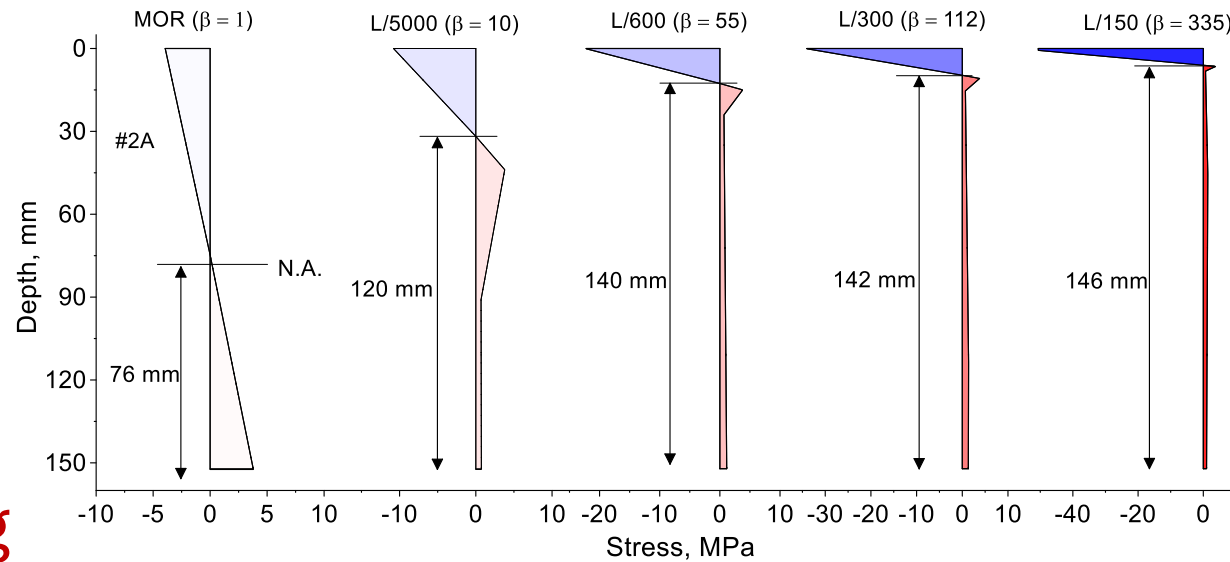
Flexural Testing used for precast Tunnel Linings QC/QA Procedures

- Evaluation of more than 9 months of QC/QA beam samples, 1200+ flexural C1609 beams
- Address the variations using Statistical Process Control tools

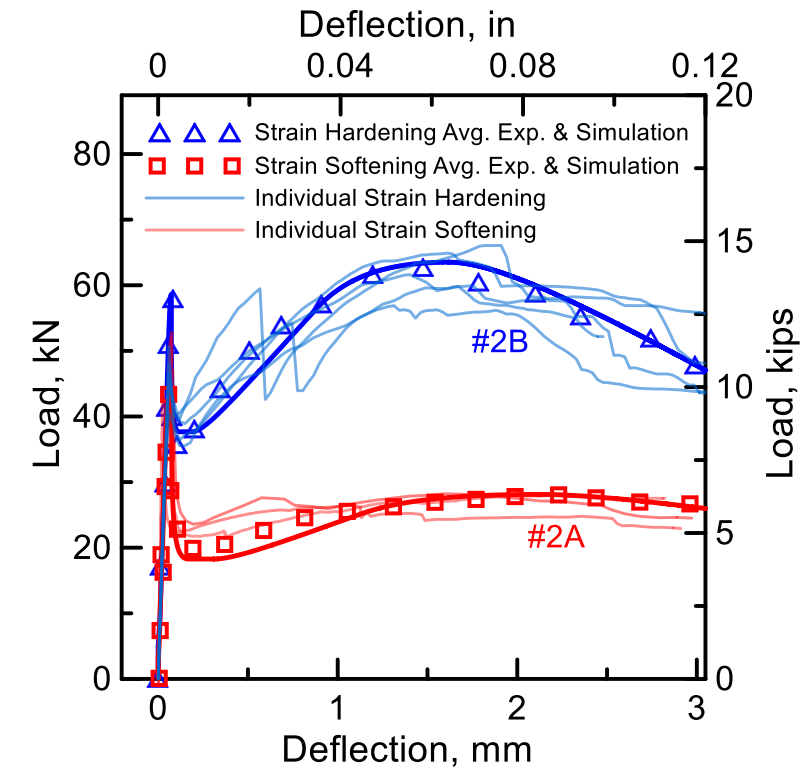
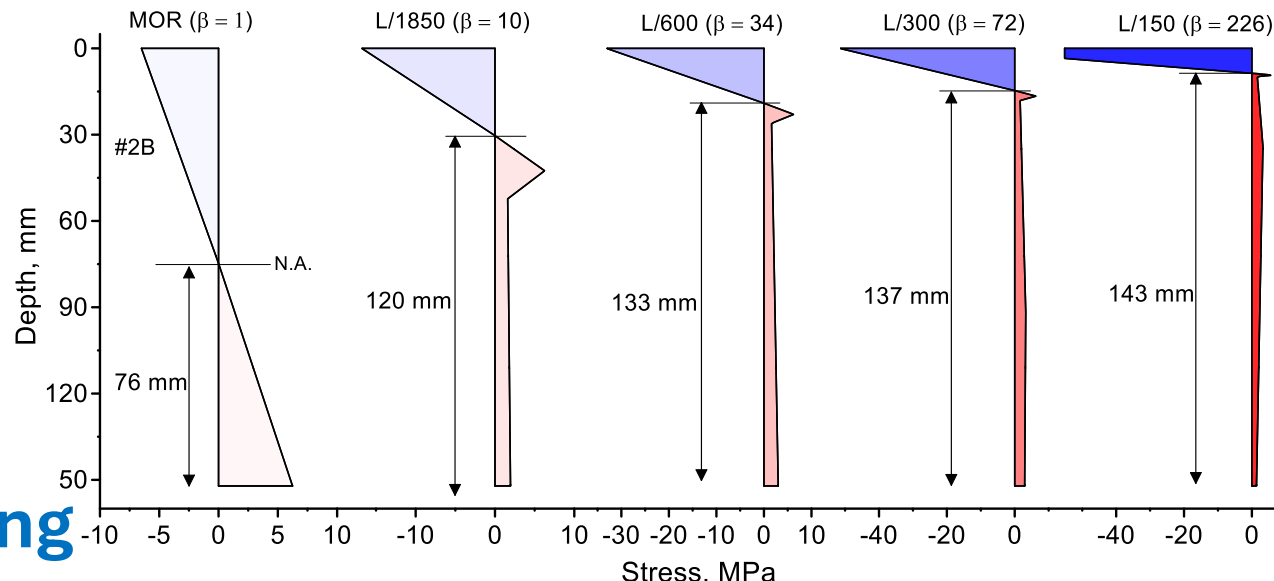


Evolution of Stress Distribution across the thickness

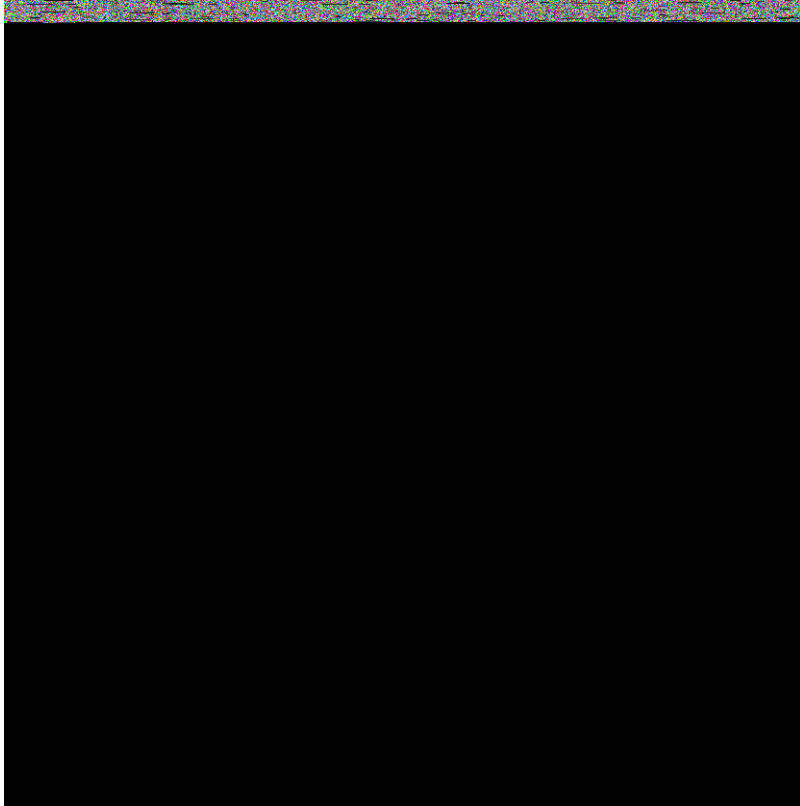
2A-Softening



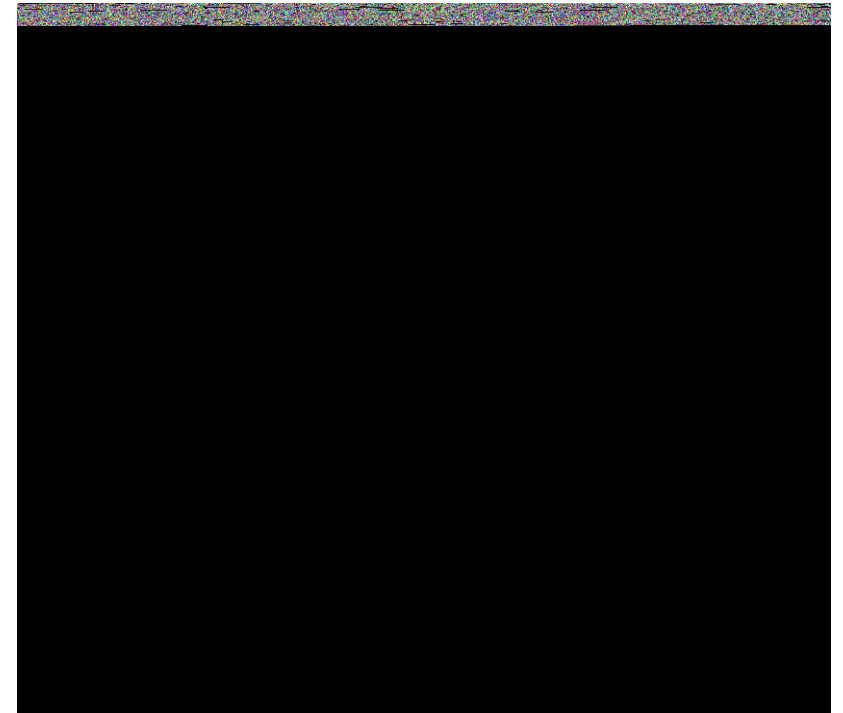
2B-Hardening



Statistical measures of distribution and correlation

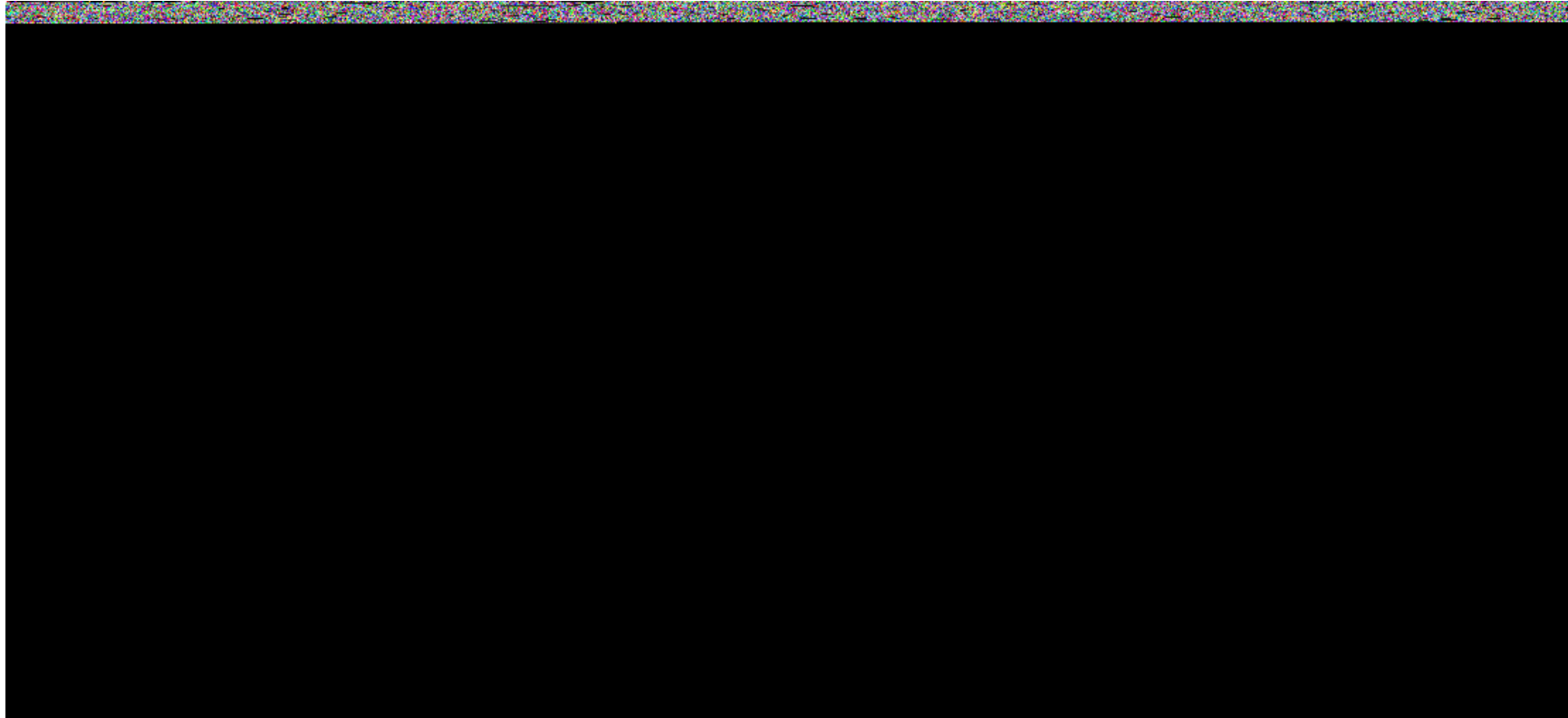


Fit Exp. Data



Backcalculated Tension
Model

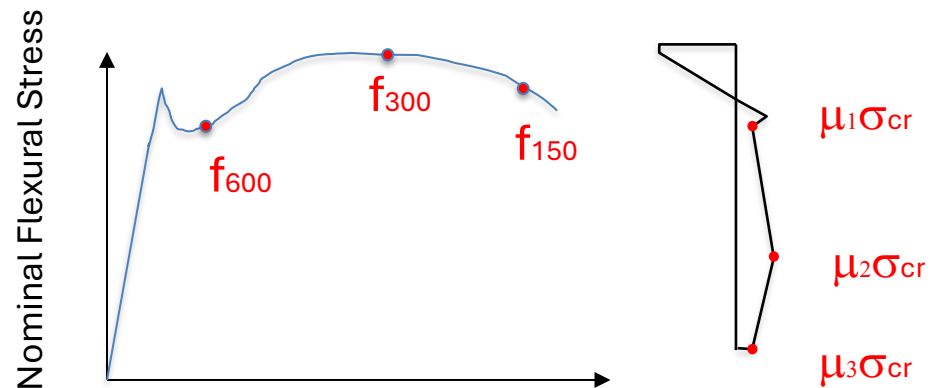
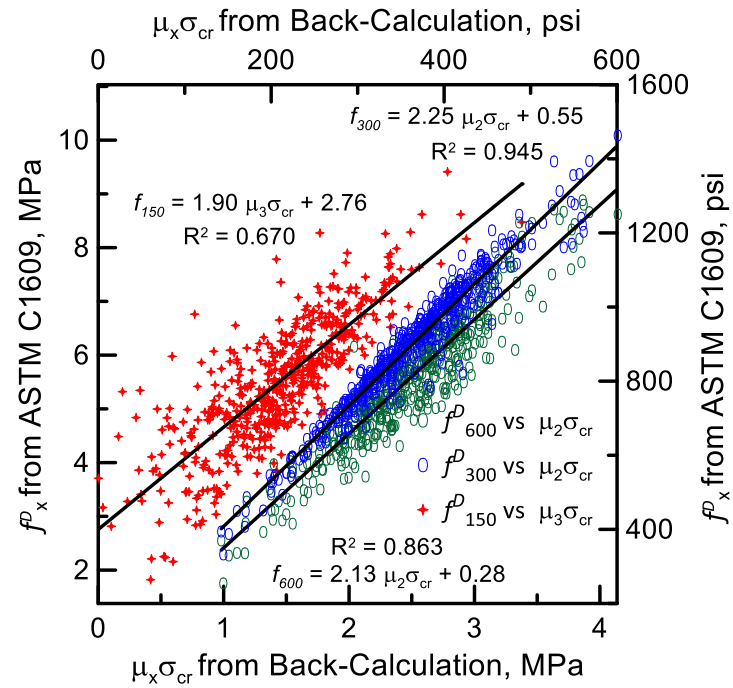
Definition of Average Tensile Stress for Simplified Design



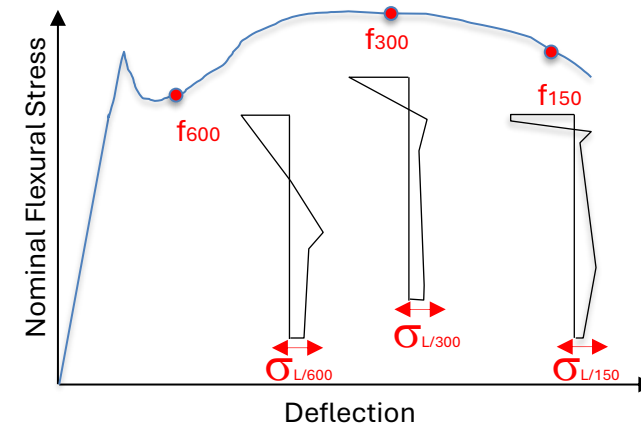
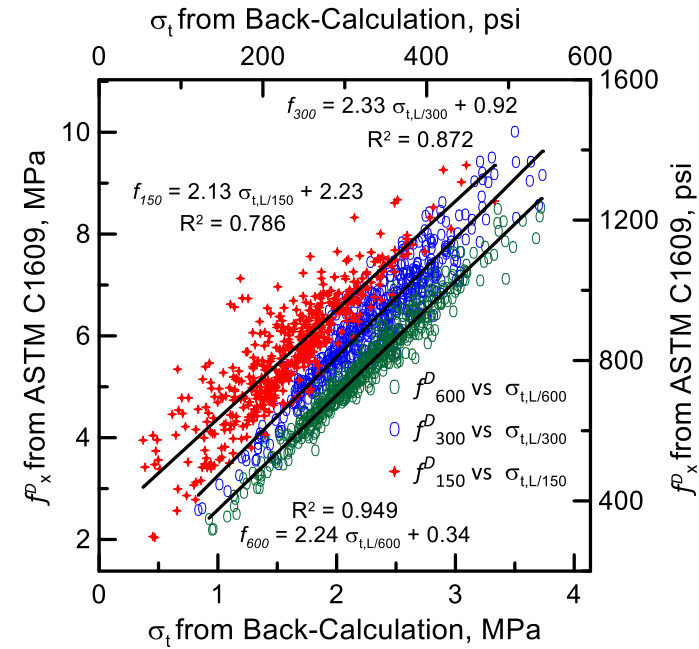
Total Force in T/C $F_C = F_T = \sum_0^{(1-k)h} F_t(\varepsilon) = b \int_0^{(1-k)h} \sigma(\varepsilon) dy = bh \int_0^{\beta_i} \sigma_t(\beta_i) \left(\frac{(1-k)}{\beta_i} \right) d\beta_i$

Avg. Force in T/C $\bar{\sigma}_t = \frac{F_T}{A_t} = \frac{F_T}{(1-k)hb}$

Correlation of experimental and simulation-based stresses at various Deflections



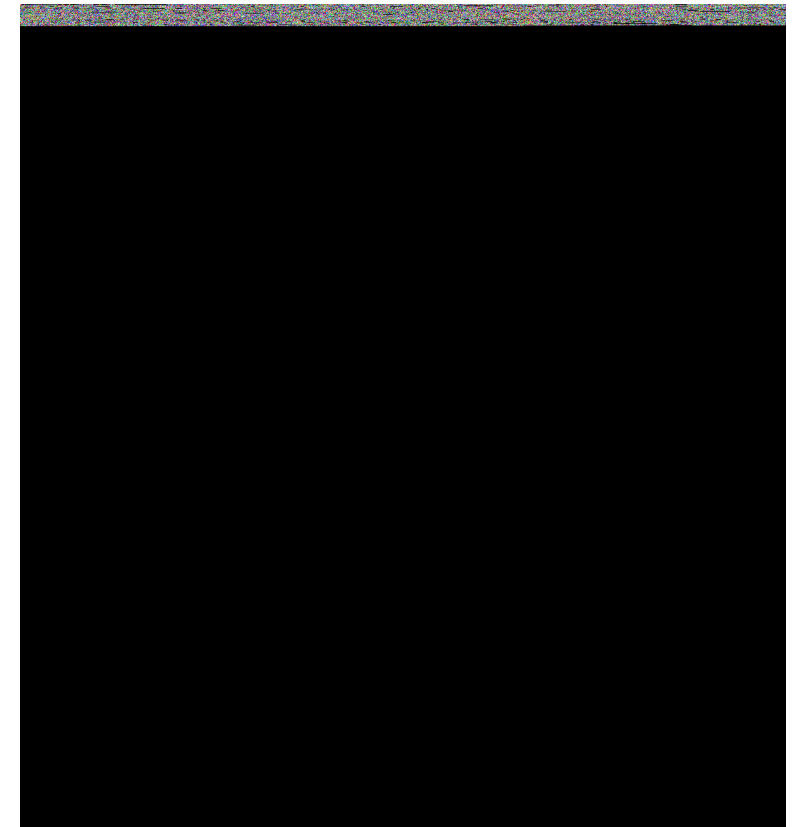
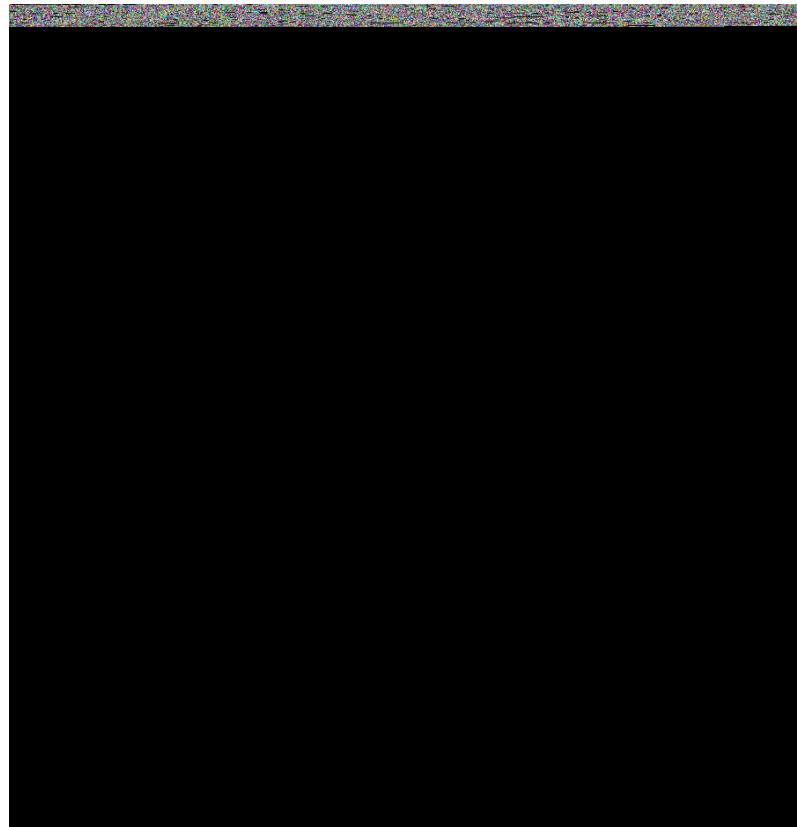
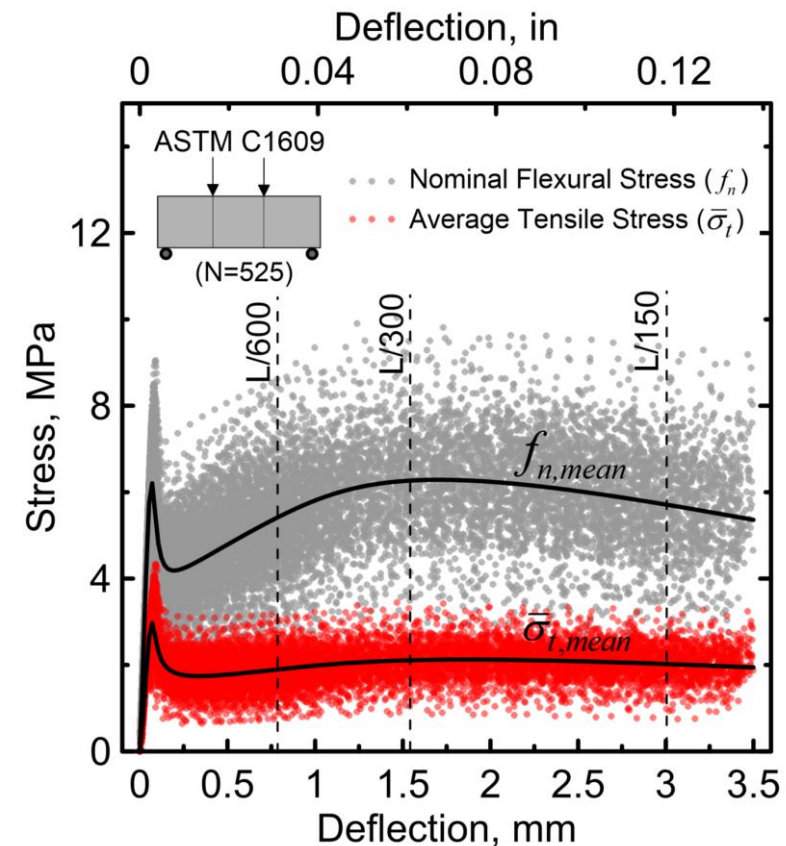
ASTMC1609 Parameters vs Tension Model Parameters



ASTMC1609 Parameters vs stress at extreme fiber

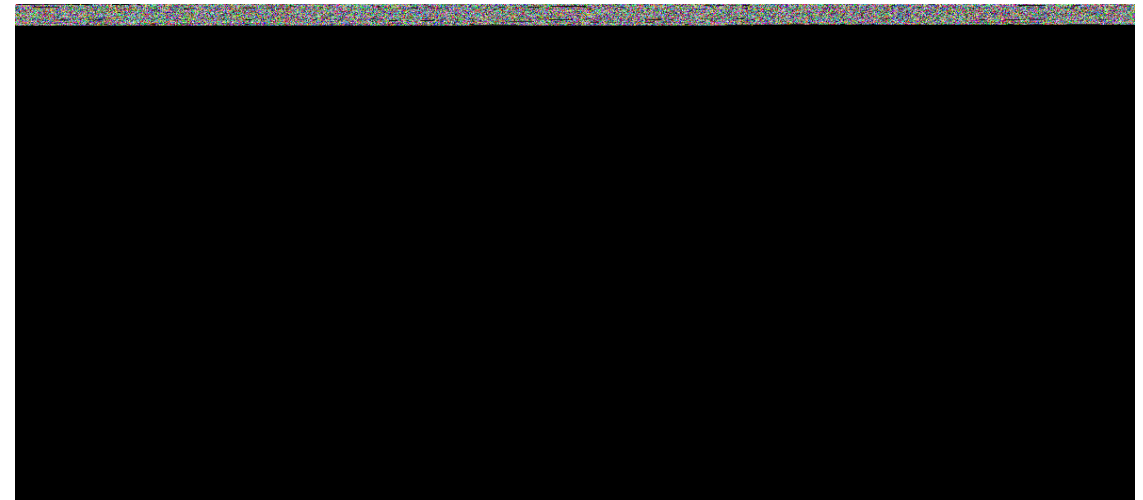
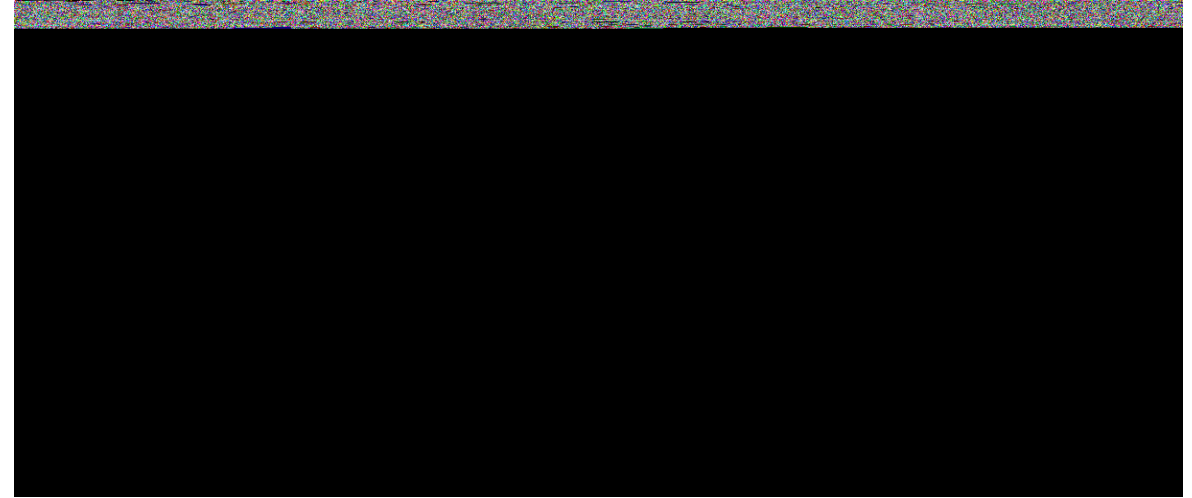
Average tensile stress calculation for simplified design

- The Nominal stress ratio and the neutral axis are relatively consistent compared to the nominal flexural stress which has a high scatter.
- The empirical factors for flexure are not constant and compound the definition of characteristic empirical factors.
- The post-peak response of different fiber types and amounts are fundamentally different.

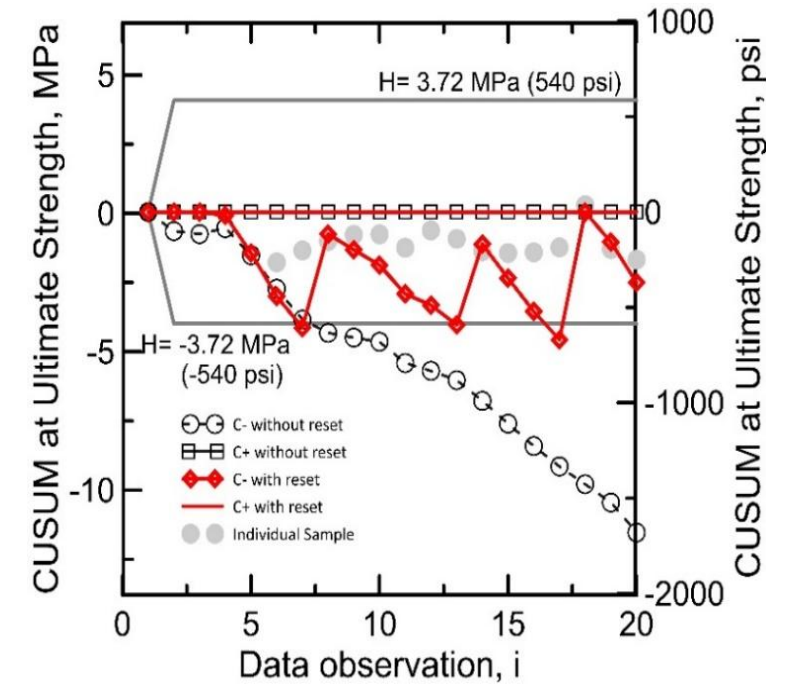
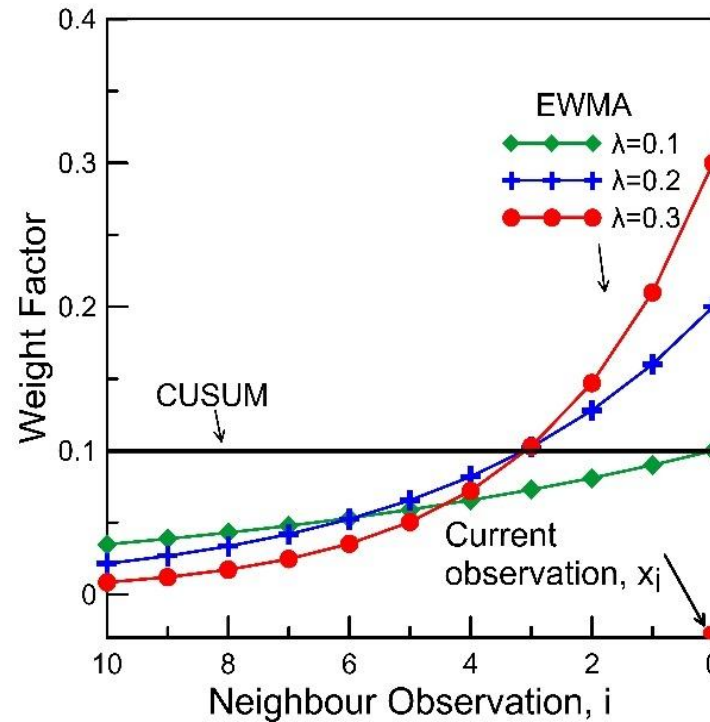
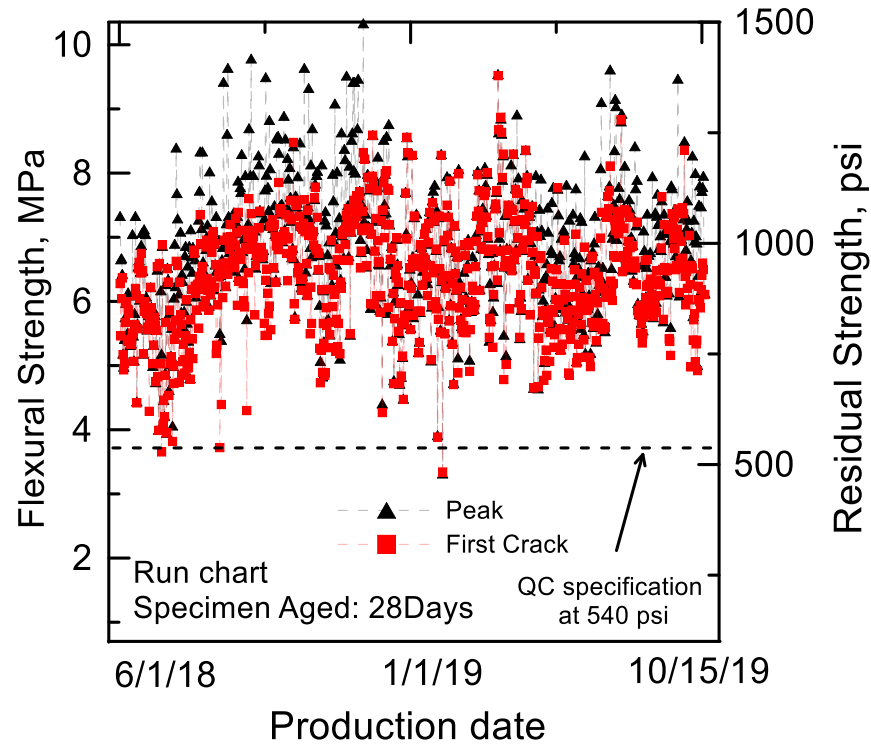


Control Charts and Machine Learning Applications for Production QC

- A control chart for individual observations monitors concrete residual strength by distinguishing between a **significant change** (real change in slope) and **non-significant change** (error of aberration).
- CUSUM, cumulative sum of errors as a function of time to hit the limits
- EWMA, Exponential Weighted Moving Average

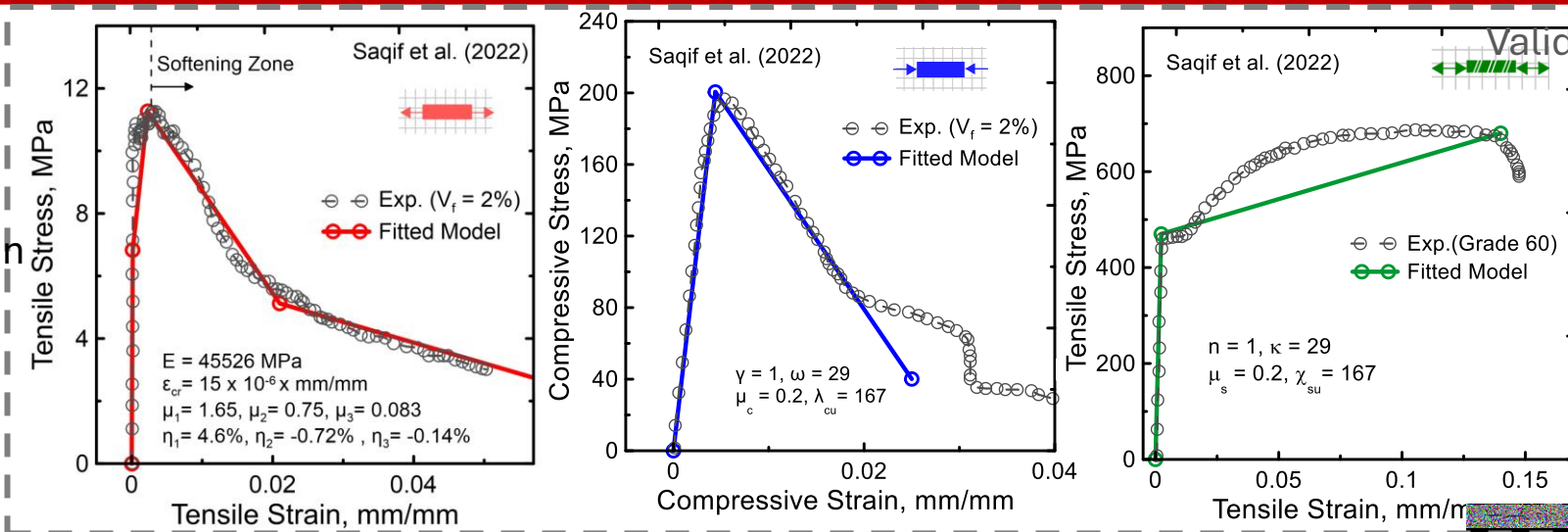


CUSUM and EWMA Run Charts to monitor and control Production problems and outlier analysis



Validation of closed-form approach for Hybrid Members:

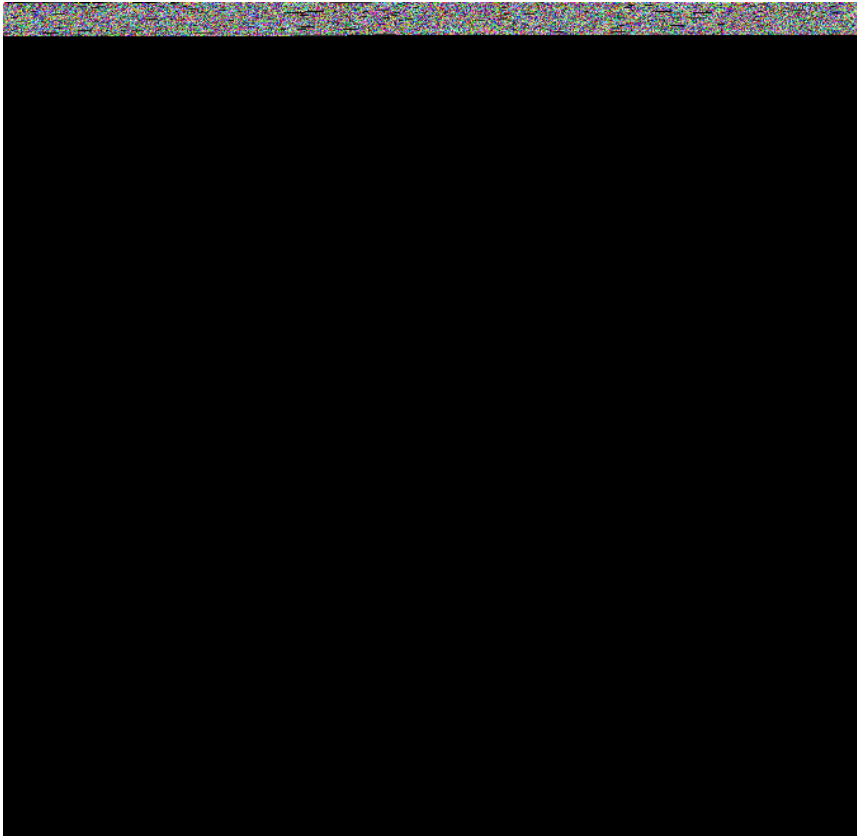
Material
Characterization



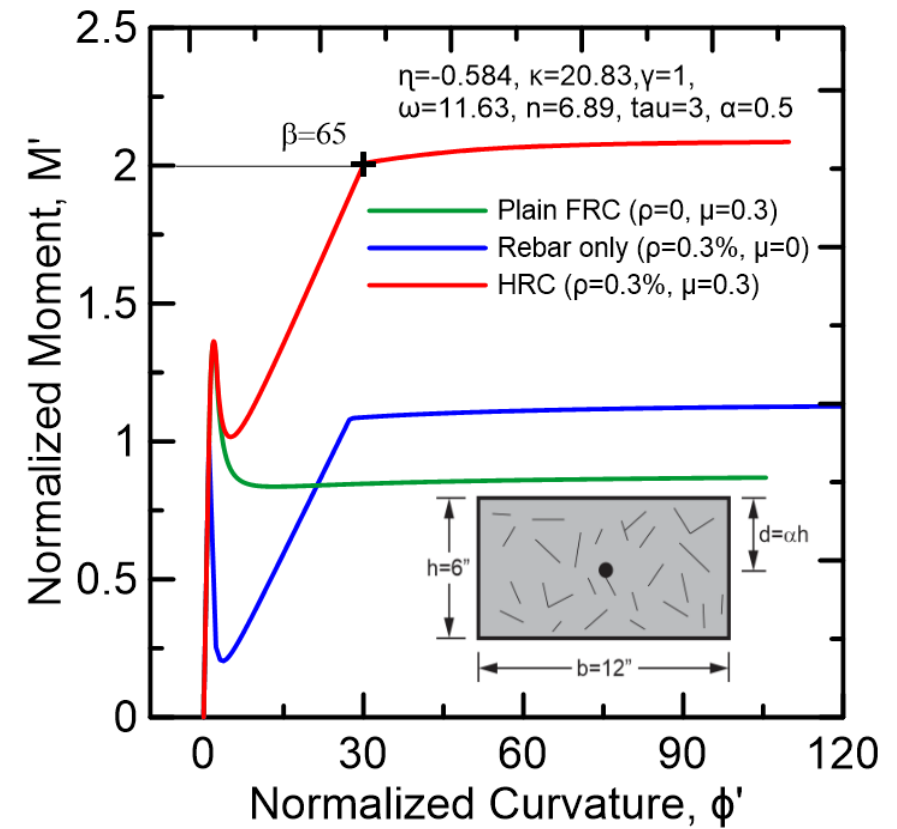
Stress
Distribution at
Limit States

The Hybrid Composite effect

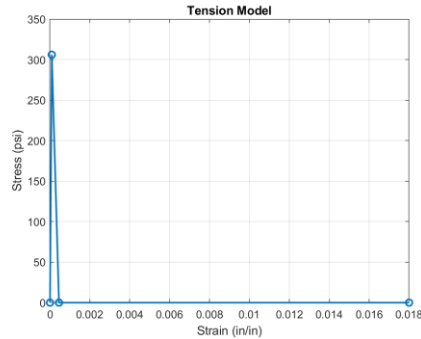
Reinforced concrete beams



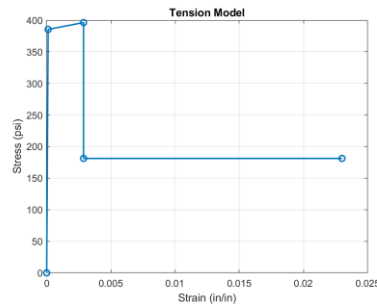
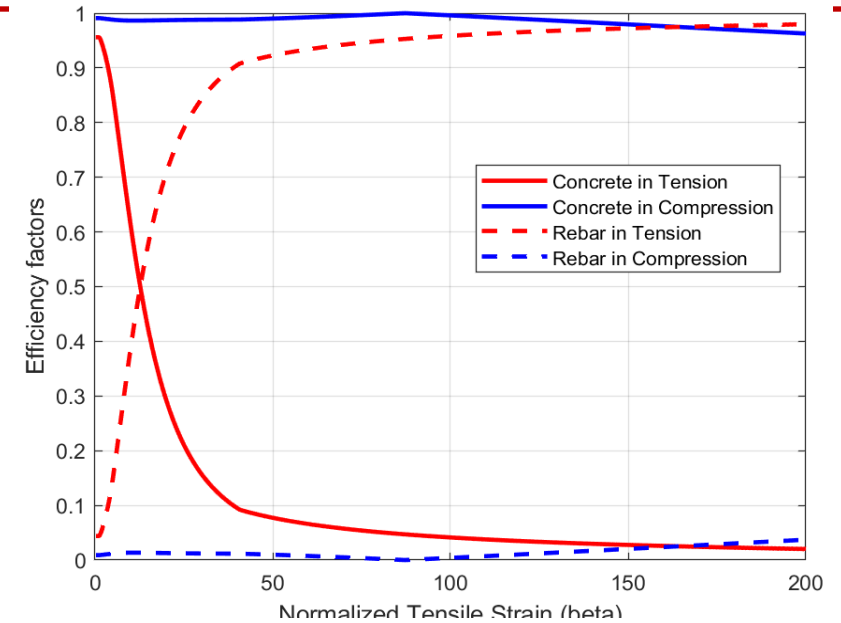
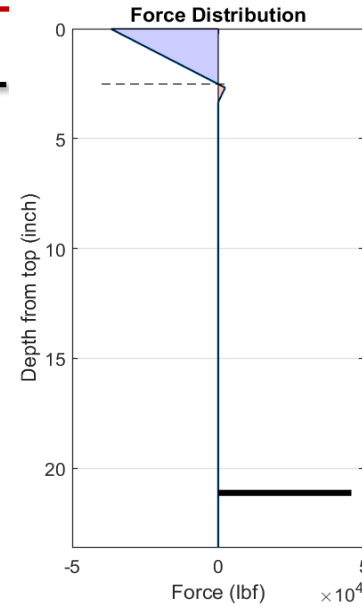
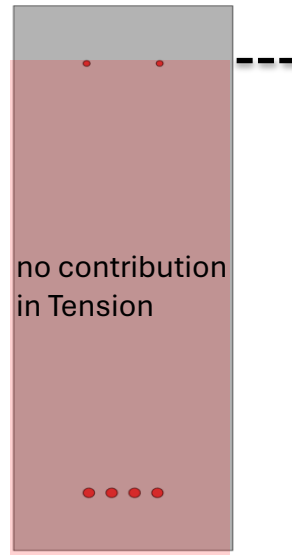
Reinforced Slabs



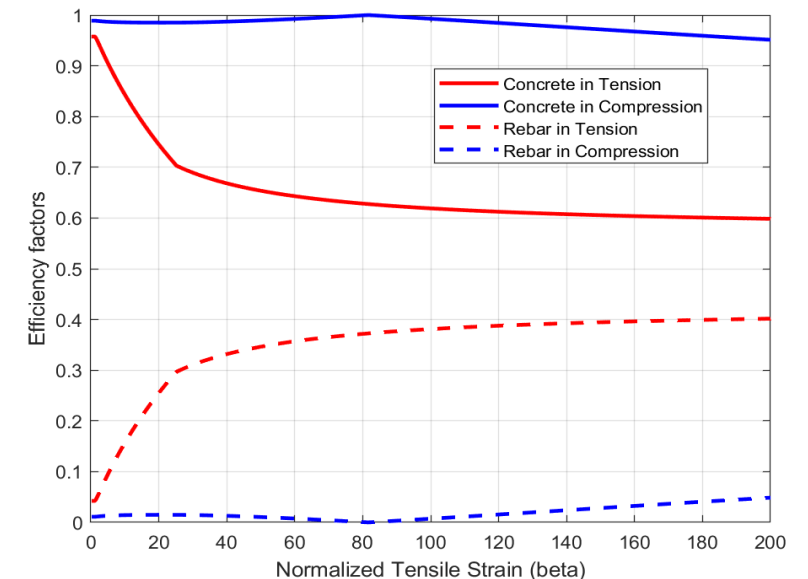
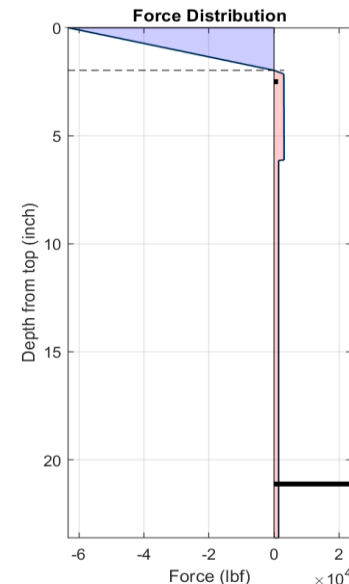
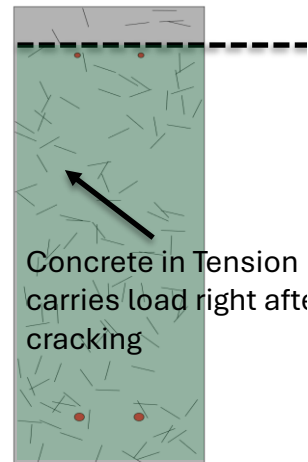
Contribution of Secondary Reinforcement in Conventional Design



$\rho = 0.28\%$
 $\mu = 0$ No residual strength
 Size = 7.5" x 23.5"



$\rho = 0.14\%$
 $\mu = 0.5$ residual strength
 (less than 1.4 MPa)
 Size = 7.5" x 23.5"



Hybrid
Reinforced
Section,
accounting for
Residual
Strength

Synergy of fiber and rebar combination as Hybrid Reinforcement

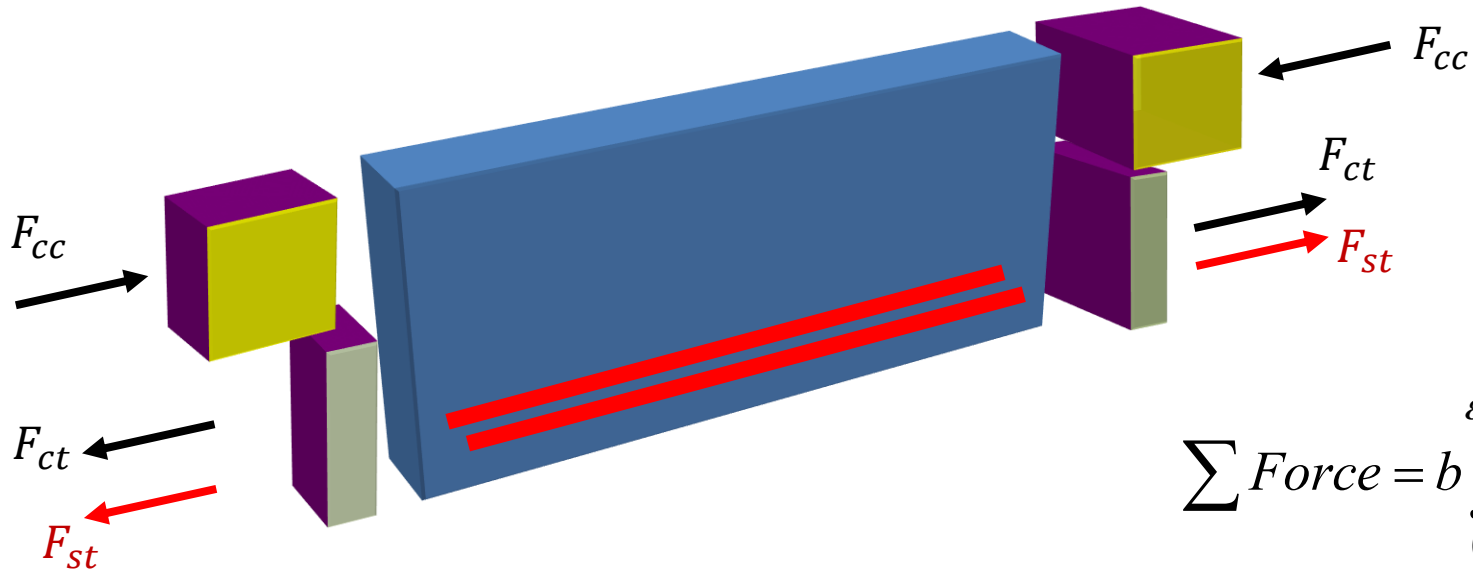
- As soon as post cracking starts, concrete contribution diminishes to less than 10% in plain RC concrete
- With only half the amount of steel in the hybrid system, the rebars in hybrid extend the serviceability range to a higher overall stiffness and extend to twice the deflection range
- Improving the efficiency of the reinforcement in delaying the yielding of steel
- Even after yielding of steel, Concrete in the hybrid system is carrying more than twice the steel



Conclusions

- Cost-effective tunnel lining sections can be developed using fiber reinforced concrete
- Presenting a robust, rational mixture and structural design
- Modeling of the compression and flexural Analysis for design using a closed form solution strategy
- Moment Curvature response can be used to isolate individual force components and obtain the materials efficiency parameters
- Statistical Process control can help QC/QA and material design to increase efficiency compared to Reinforced Concrete
- New and improved design models for Tunnel Section using Ssteel and Polymeric Fibers integration into codes an standards

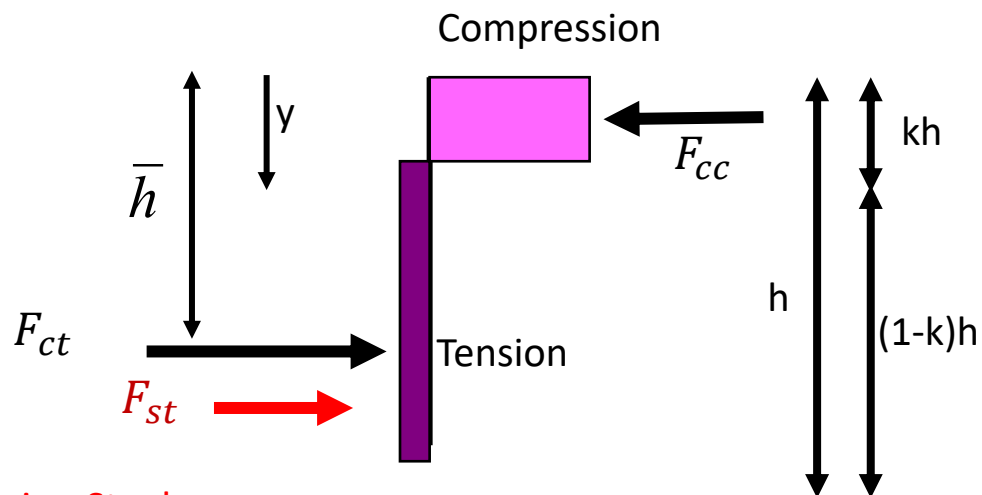
Mean Equivalent Stress distribution in an intermediate stage



$$\sum Force = b \int_0^{\varepsilon_i} \sigma(\varepsilon) \frac{(1-k)h_0}{\beta \varepsilon_{cr}} d\varepsilon = F_{cc} + F_{ct} + F_{st} + F_{sc} = 0$$

$$\bar{h} = \frac{b \int_0^{\varepsilon_i} h(\varepsilon) \sigma(x) dh}{\sum F_{ct}} = \frac{\int_0^{\varepsilon_i} h(\varepsilon) \sigma(\varepsilon) \frac{(1-k)h_0}{\beta \varepsilon_{cr}} d\varepsilon}{F_{ct}}$$

$$\sum M = b \int_0^{\varepsilon_i} \sigma(\varepsilon) \frac{y(1-k)h_0}{\beta \varepsilon_{cr}} d\varepsilon = M_{cc} + M_{ct} + M_{st} + M_{sc} = \bar{M}$$



Tension Steel